

Polynomial Witness Histories

Unique linear certificates with finite polynomial unknowns

Cellular histories over $\mathbb{N}[X, Y]$ and $\mathbb{N}[X]$, and linear boundary transport over every commutative ring

AI-assisted research report prepared for Vladimir Reshetnikov from three manuscripts (batch 79 of ProveIt's incoming reports, manuscripts 06, 15 and 01), 2 October 2026

Abstract

[write] Three manuscripts represent a whole terminating computation by a *fixed number of finite polynomials* that satisfy fixed linear identities (in one case quadratic identities that become linear once a single shared stride is fixed). In each, a clock forces the complete witness tuple to be unique whenever it exists, every coefficient of that witness is zero or one, and its degree or support is an exact running-time function for which no computable input-only bound exists.

Part I (source 06) compiles a deterministic radius-one cellular automaton with s symbols into affine-linear systems over $\mathbb{N}[X, Y]$ with $s^3 + s + 5$ polynomial unknowns and $3s + 3$ rows, or $3\lceil \log_2 s \rceil + 6$ unit-coefficient rows, or nine weighted rows, all with the same full solution tuples; a solution exists exactly at a first singleton acceptance. Part II (source 15) answers Part I's one-coordinate question: coefficient-bounded univariate linear systems are finite-state decidable, so no universal linear compiler over $\mathbb{N}[X]$ has computably bounded witness heights; eleven identities of degree two with one shared stride S , forced to equal X^{m+2h+2} , replace it, with witnesses in bijection with Part I's. Part III (source 01) compiles a two-counter machine with q states and d zero tests into $q + d$ linear identities in $q + 2d$ polynomial unknowns with omitted-variable sorts, valid over every nonzero commutative ring; it then derives an exact integrality gap (integer minimum 1, rational infimum 0) and a Hilbert-space operator with dense nonclosed range whose basis-vector membership is computably enumerable complete.

Polynomial unknowns are not ordinary integer witnesses: nothing here is a fixed-arity single-fold or finite-fold Diophantine representation over $\mathbb{N}$, and the repository's 75- and 87-operation bounds are unchanged.

Status. [write] AI-assisted and unrefereed; written proofs with exact finite computational checks; nothing in this report is formalized in Lean or Rocq, and historical priority is not established for any Part. Section 1 states the common spine, the notation, the relation to neighbouring reports and to the research programme's reviews of these manuscripts, and what is not claimed; Appendix A gives the provenance.

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1 Scope, common spine and conventions

[write] This section was written when the three manuscripts were merged. Parts I, II and III print the three sources in full; text there without a [write] marker is the source’s own.

1.1 The three Parts

This report is built from three research manuscripts delivered with batch 79 of ProveIt’s incoming reports (manuscripts 06, 15 and 01; Appendix A gives the full provenance). They are called *source 06*, *source 15* and *source 01* after their batch-79 manuscript numbers, which are also the file prefixes of their shipped programs and data. Each carries the author line “Research report prepared for Vladimir Reshetnikov” and names no assistant. Source 15 is a direct sequel of source 06: it cites source 06’s file by name and answers one of its research questions. Source 01 was written independently of both, from another snapshot of the repository, and shares no lemma with them.

- **Part I (source 06), *Unique histories from linear equations*.** For a deterministic radius-one cellular automaton with s symbols, a quiescent blank and an accepting alphabet, a matrix over $\mathbb{Z}[X, Y]$ independent of the input and of the running time has a solution $U \in \mathbb{N}[X, Y]^{s^3+s+5}$ exactly when the evolution has a first singleton acceptance; the whole tuple is then unique, Boolean-coefficient, with exact degrees h , $m + 2h + 1$ and $m + 3h + 1$ (Theorem 2.1). The direct form has $3s + 3$ rows; an exact marginal projection and injective features give $3\lceil \log_2 s \rceil + 6$ unit-coefficient rows or nine weighted rows with the same solution tuples (Theorem 8.1, Corollary 8.2). A fixed matrix is Σ_1^0 -complete on an unambiguous family (Corollary 2.2); a single quadratic over the same semiring has the same solutions (Theorem 11.2).
- **Part II (source 15), *The price of one coordinate*.** Bounded-height solvability of univariate affine-linear systems over $\mathbb{N}[X]$ is decidable by a finite-state argument (Theorem 21.1), so no universal univariate linear compiler can guarantee a computably bounded witness height (Corollary 21.3): Part I’s one-coordinate question (Section 16.3) has a negative answer. Eleven identities of degree at most two in $s^3 + s + 7$ unknowns over $\mathbb{N}[X]$, every nonlinear monomial containing one stride unknown S , replace the linear design; two of them force $S = X^W$ with $W = m + 2h + 2$ (Theorem 20.2), and the full solutions are in bijection with Part I’s (Theorem 25.2).
- **Part III (source 01), *Linear boundary transport*.** Over any nonzero commutative ring R , a deterministic transition graph has a finite-support transport equation whose fibre is the halting history plus the cycle-constant kernel (Theorem 38.1); a clock indeterminate Z makes it injective and the 0/1 history unique (Theorem 39.1). Two-counter machines with q states and d zero tests compile to $q + d$ linear identities in $q + 2d$ polynomial unknowns whose sorts omit the tested variable (Theorem 41.2), and a row tag W makes one equation with a computably enumerable complete accepted set (Theorem 41.3). The associated energy has integer minimum 1 but rational infimum 0 on nonhalting sources (Theorem 43.1), and the Hilbert-space operator has dense nonclosed range with c.e.-complete basis-vector membership (Theorem 44.2).

Each Part keeps its own definitions, letters and numbering of statements. Parts I and II share the cellular semantics, the ray-clock mechanism and the worked four-symbol example; Part III has a different substrate, a different witness ring and a different uniqueness mechanism, and is placed here because of the spine of Section 1.2.

1.2 The common spine

In each Part an entire computation is a tuple of a *fixed* number of polynomials whose degrees and supports grow with the running time, and the identities they satisfy are linear in them (Part II: of degree two, with every product through one stride unknown, and linear in the remaining unknowns once the stride is fixed, Proposition 26.4). Table 1 lines the three Parts up.

Part	Substrate and witness domain	Identities and unknowns	What forces the whole witness	Exact size; no computable bound
I (06)	radius-one cellular automaton, s symbols; $\mathbb{N}[X, Y]$	$3s + 3$ affine-linear rows (or $3\lceil \log_2 s \rceil + 6$, or nine), $s^3 + s + 5$ unknowns, one fixed matrix	nonnegative ray flows (Lemma 4.1), conservation of the light cone, first singleton acceptance	$\deg_Y = h$, $\deg_X = m + 2h + 1$; Theorem 10.2
II (15)	the same automata; $\mathbb{N}[X]$	eleven identities of degree ≤ 2 , $s^3 + s + 7$ unknowns, products only through S	univariate ray lemma, then $S + E = X + XE$ and $SB = XD$ force $S = X^{m+2h+2}$ (Lemma 24.3)	$D_{\max} = \max\{W, W(h + 1) - 1\}$; Corollary 26.3
III (01)	two-counter machines, q states, d tests; any nonzero commutative ring R , unknowns in $R[X, Y, Z]$ with omitted-variable sorts	$q + d$ linear rows, $q + 2d$ unknowns; one tagged equation in $R[W, X, Y, Z]$	clock Z : $I - ZU$ injective by coefficient recursion (Theorem 39.1); sorts make the boundary split unique (Lemma 41.1)	$\deg_Z H = h$, main support $h + 1$; Corollary 41.5

Table 1: [write] The common spine of the three Parts.

Three features recur. First, the no-bound results (Theorem 10.2, Corollaries 26.3 and 41.5) are all the bounded-search argument of Proposition `cdc:bd:prop:nobound` in [1], with polynomial degree, stride or support in place of an integer witness height; no novelty is claimed for the argument. Second, every Part separates its normal form from the older undecidability of linear equations over polynomial semirings (Narendran [8]; Dong [10] for a decidable single homogeneous equation) and does not claim it. Third, every Part warns that a fixed number of polynomial unknowns is not a fixed number of ordinary integer unknowns; the research programme’s reviews repeat the warning (Section 1.4).

One domain ledger

Source 15 compares witness domains in a table (Section 26.3). Table 2 extends that table by Part I’s signed counterexample and by rows for Part III, so that the three witness domains are compared once.

1.3 Notation across the Parts

Each Part keeps its source’s letters, and a letter means what its own Part says it means. One macro was renamed in the source text: source 01 wrote the blackboard $\mathbb{F}$ of finite fields with the same command that sources 06 and 15 use for the calligraphic accepting alphabet $\mathcal{F}$; source 01’s command is renamed, the printed glyphs are unchanged, so $\mathbb{F}_p$ (a finite field, Part III) and $\mathcal{F}$ (an accepting alphabet, Parts I–II) look as in the sources. No normalization was changed. Table 3 lists the letters that the Parts use differently.

Domain	Unknown-degree	Coefficient condition	Consequence
$\mathbb{N}[X]$	1	no height bound	general systems undecidable [8]
$\mathbb{N}[X]$	1	given computable height bound	decidable (Theorem 21.1); no universal compiler with that guarantee (Corollary 21.3)
$\mathbb{Z}[X]$, or $R[X]$ with R finite	1	computable height bound (automatic for finite R)	decidable by the same proof; editorial, Remark 49.1
$\mathbb{N}[X, Y]$	1	all full witnesses forced Boolean	canonical universal family (Theorem 2.1)
$\mathbb{Z}[X, Y]$	1	signed	infinitely many solutions whenever solvable (Proposition 12.1)
$\mathbb{N}[X, Y]$	2, one equation	same forced Boolean witnesses	Theorem 11.2
$\mathbb{N}[X]$	2, one shared stride	all full witnesses forced Boolean	eleven-row canonical universal family (Theorem 20.2)
$\mathbb{N}[X]$	4, one equation	same forced Boolean witnesses	Theorem 28.1
$R[X, Y, Z]$ with sorts $R[X, Y, Z]$, $R[X, Z]$, $R[Y, Z]$, any nonzero commutative R	1	none; uniqueness from sorts and clock	unique 0/1 witness exactly on halting (Theorems 41.2, 39.1); one tagged equation (Theorem 41.3)
$R[X, Y]$ with sorts $R[X, Y]$, $R[X]$, $R[Y]$ (unlocked)	1	none	same accepted set; fibre = history plus cycle-constant kernel (Theorems 38.1, 41.3)

Table 2: [write] Witness domains of the three Parts. Rows 1, 2, 4, 7 and 8 are source 15’s table; the others are added. No row asserts a fixed-arity ordinary single-fold representation over $\mathbb{N}$.

Table 3: [write] Letters used differently in the three Parts.

Letter	Part I (source 06)	Part II (source 15)	Part III (source 01)
s	alphabet size	alphabet size	source configuration
X, Y	space and time coordinates	X : the only coordinate	the two counters: multiplication by X or Y increments
Z	$Z_{\ell cr}$: source-triple masks	$Z_{\ell cr}$: source-triple masks	the clock indeterminate
W	— (w_t : row width)	$W = m + 2h + 2$: stride exponent	row-tagging indeterminate
S	$S = X^a Y^b$ in the ray lemma; $S = \mathbb{N}[X, Y]$	the stride unknown	set of configurations
U, V	witness vector; ray-lemma unknowns; V : vertical clock prefix	ray-lemma unknowns; V : vertical clock prefix	U : transport operator; $\mathcal{V}_R$: finite-support module
D, B	endpoints of the two clocks	endpoints of the clocks; B_0 , B_X : height bounds	D_r, B_r : split unknowns; D_X : boundary shift; B : indegree bound
Q	prefix of the right-boundary clock	prefix of the right-boundary clock	set of control states (and $\mathbb{Q}$, the rationals)

Letter	Part I (source 06)	Part II (source 15)	Part III (source 01)
H, K	derived total tile and terminal masks	total source and terminal masks	H_r : main unknowns; K_u : halting set
h	first acceptance time	first acceptance time	halting time; also a finite-support array in the energy
m	input length	input length	number of rows, $m = q + d$
q, d	$M = X^p Y^q$ in the ray lemma; $d = m + 2H_0 + 1$	d : matrix degree in the obstruction	q control states, d zero tests
E	—	positive-ray prefix of the stride	E_i : residual rows; E_{v_0} : energy
$\mathcal{F}, \mathbb{F}$	$\mathcal{F}$: accepting alphabet	$\mathcal{F}$: accepting alphabet	$\mathbb{F}_p$: finite field
A	$\mathcal{A}$: alphabet; $A_{f, \mathcal{F}}$: the compiler matrix	$\mathcal{A}$: alphabet; $A(X)$: a linear system's matrix	$A = I - ZU$: incidence operator
f	local cellular rule	local cellular rule	deterministic partial transition
N	—	$N = (B_0 + 1)^{nd}$: memory states	taper length

Watch for these readings:

- **Fixed arity.** “A fixed number of unknowns” counts *polynomials of unbounded support*, never integers. A polynomial unknown hides an unbounded list of exponents and coefficients.
- **Unique, single-fold.** Uniqueness is uniqueness of the complete polynomial tuple, including every clock, mask and boundary auxiliary. It is not an ordinary single-fold Diophantine representation: canonically coding the tuple as an integer and applying MRDP can introduce auxiliary integer witnesses of uncontrolled multiplicity (Sections 12 and 45.4).
- **Boolean coefficient.** Every coefficient is 0 or 1; arithmetic is ordinary semiring or ring arithmetic, not arithmetic in a Boolean ring. In Parts I and II Boolean coefficients are a *conclusion* of the shape lemmas, never a hypothesis.
- **Linear.** Parts I and III are linear in the unknowns. Part II is quadratic, every product passing through the stride S ; its fixed-stride fibres are linear and decidable (Proposition 26.4).
- **Degree.** The *witness degree* (coordinate degree of a solution) grows with the running time; the *unknown-degree* of the identities is 1, 2 or 4. Corollary 21.3 is a statement about unknown-degree 1 in one coordinate.
- **Clock.** In Parts I and II a clock is a pair of unknown polynomials (endpoint and prefix) forced by the ray lemma. In Part III the clock is an indeterminate Z of the coefficient ring, not an unknown.
- **Diophantine.** The title of source 01, *Linear Quasi-Diophantine Universality*, refers to polynomial-valued unknowns with prescribed sorts, not to ordinary integer unknowns (its abstract and Section 45.4 say so).

1.4 Relation to neighbouring reports, the formal project and the research programme's reviews

This report sits in the collection's `hilbert-tenth-problem` category. No earlier report of the collection treats polynomial-semiring or polynomial-sort witnesses; its neighbours have natural-number unknowns.

- *Canonical Diophantine certificates* [1], Part IX. The one-history relation (`cdc:eq:onehistory`), the equivalence of finite-fold representability with a finite-fibre universal halting polynomial (Theorem `cdc:bd:thm:universal`) and the absence of a computable witness-height bound (Proposition `cdc:bd:prop:nobound`) are the boundary that all three Parts stand outside. Part III's Section 45.4 restates that boundary without citing it, and the three no-bound results are the polynomial analogues of `cdc:bd:prop:nobound`; editorial notes in the Parts say so. Part IV of [1], *Polynomial trajectories*, shares only a name: its trajectories are integer-valued polynomial functions of time with natural-number witnesses.
- *Stochastic and thermal exactness* [2]. Batch-79 manuscripts 10 and 14 (archives `Coercive_Green_Diophantine.zip` and `Well_Conditioned_Diophantine_Computation.zip`) were placed there by commit `bbaf322e5`, for its Part V on coercive Green operators. They build fixed sparse operators from counter machines in the opposite regime to Part III's Section 44: a uniformly coercive operator instead of one with zero in its continuous spectrum. A note after Theorem 44.2 records this.
- *Smooth Diophantine finalizers* [3], opened by the same placement as this report, post-processes finite systems of degree at most two over $\mathbb{N}$ into one quartic with a hypersurface smooth over $\mathbb{Z}$. Such systems occur here only as the externally bounded exports of Proposition 11.3 and Theorem 28.2; neither source uses it, and it does not apply to equations whose unknowns are polynomials.

The research programme's reviews. The Hilbert's-tenth-problem research programme (`Computability/HilbertTenthProblem`, read-only for this report) reviewed all three manuscripts, byte for byte the archives placed here; this report does not present its placement as a first review.

- Source 06: [4], commit `e5497072e`. It accepts the written construction on its stated domain, replays the three author suites, independently checks 594 affine identities of the feature row maps, and finds that accepting symbols `1.0` or `True` pass validation and produce undeclared witness names and a `KeyError`; the tested repair is `unique_polynomial_histories_exact_domains.patch`. The theorem for exact alphabet symbols is unaffected.
- Source 01: [5], commit `49bc4c654`. It finds the mathematics sound within the stated polynomial and field witness domains, reproduces 32,502 checks, and finds that `compile_system`, `history` and `residual` accept non-Boolean clock modes (`clocked=0.5` loses uniqueness, `clocked=2` rejects a true history); the tested repair is `linear_boundary_exact_boolean_guards.patch`. The theorems, stated for the two Boolean modes, are unaffected.
- Both reviews are indexed in [7] (commit `c00ce825a`).
- Source 15 had only been inventoried as pending (commit `b927b2290`) when it was placed here; it was reviewed afterwards in [6] (commit `a21c86070`). The review finds no mathematical defect; it finds that the low-level `System` constructor of the compiler accepts float residual coefficients (a residual -1 next to 2^{60} rounds to zero, so a false witness is reported as a zero) and keeps caller-owned lists, and supplies the repair `one_coordinate_exact_system_inputs.patch`; the high-level compiler and the theorems are unaffected. It also proves a projection to ten identities, recorded in a note at Section 33.1.

The shipped programs are the delivered originals, unpatched; the patches live beside the reviews in the research programme.

The formal project. The report sits in the collection, not in `Computability/HilbertTenthProblem`, and placement beside a Lean/Rocq development confers no formal status. No statement of this report is formalized. The only project declaration a source names is `rePred_dioph` (`Computability/HilbertTenthProblem/Lean/Diophantine/Common/RecursivelyEnumerableDioph.lean`, line 32, built on `MRDP.mrdp`), which source 01 proposes as the endpoint of a future integer bridge; nothing here uses it.

1.5 Status and what is not claimed

The report claims conventional mathematical proofs, by its sources, of the results summarized in Section 1.1. Each source presents as its own contribution only a combined normal form: source 06 the explicitly counted fixed matrix with exact shape recovery, first-acceptance canonicalization and same-tuple feature compression; source 15 the negative answer to source 06’s one-coordinate design question, the self-calibrating stride and the full-witness bijection; source 01 the mixed-subring compiler with its characteristic-free fibre description, the clocked single-certificate form and the exact integer, rational and Hilbert-space comparison.

Every limitation stated by a source is kept in its Part. Collected here, the report does **not** claim:

- historical priority for any Part or for the combined normal forms (all three sources say so), refereeing, proof-assistant verification, or a rebuild of the repository’s Lean or Rocq developments; the finite checks illustrate and do not prove;
- that undecidability of linear equations over polynomial semirings, or monomial forcing, is new (Narendran, already in one indeterminate); the attribution to Narendran [8] is the sources’, and the research programme could not reread that paper behind its publisher page;
- any ordinary fixed-arity single-fold or finite-fold Diophantine representation over $\mathbb{N}$, or any change of the repository’s 75-operation complete-certificate and 87-operation universal-polynomial bounds;
- a numerical universal instance: no Part instantiates a universal machine table; the worked examples are not universal;
- *Part I*: that weighted marginals determine arbitrary multisets (the feature reduction uses the already forced one-hot occupancy); that fewer rows give a smaller expanded quadratic (they do not, in the worked example); an implementation of the nondeterministic extension;
- *Part II*: decidability of unrestricted univariate linear systems over $\mathbb{N}[X]$ (Narendran’s systems are undecidable); optimality of eleven rows or $s^3 + s + 7$ unknowns; a sharp degree for one equation (the quartic row is not claimed sharp); “sharp” refers only to the linear–quadratic boundary for computably bounded heights;
- *Part III*: a new ordinary universal-polynomial bound; undecidability of finite-dimensional convex least squares or a spectral-gap undecidability theorem; decidability beyond the machine-generated one-counter class; a production-strength MRDP encoding.

Part I

Unique histories from linear equations over $\mathbb{N}[X, Y]$ (source 06)

[write] Part I prints batch-79 manuscript 06, *Unique Histories from Linear Equations: A canonical cellular-computability compiler over $\mathbb{N}[X, Y]$* , under the kicker “Computability / Polynomial semirings”, “Research report prepared for Vladimir Reshetnikov”, 2 October 2026 (the manuscript names no assistant), pinned to repository commit e58b724c2. Its labels carry the prefix `pwh:uh:` and its bibliography keys the prefix `uh:`. Its abstract and status statement follow; its two appendices are Sections 18 and 19. Part II answers the question of Section 16.3. The research programme reviewed this manuscript before it was placed here [4] (Section 1.4).

Abstract (source 06). We give a complete affine-linear encoding of finite cellular computations in polynomials with nonnegative integer coefficients. For a deterministic radius-one cellular automaton with s symbols, a quiescent blank, and a designated accepting alphabet, the compiler has exactly $s^3 + s + 5$ polynomial unknowns. The direct symbol-marginal form uses $3s + 3$ equations; an exact binary-feature reduction uses $3\lceil \log_2 s \rceil + 6$ equations with unit coefficients. Allowing coefficients of magnitude at most $s - 1$ gives just nine equations for $s \geq 2$. Every matrix is independent of the input word and running time; its entries have at most two monomials and coordinate total degree at most three.

The solution set is empty unless the first accepting configuration has exactly one accepting cell. In that case the entire polynomial witness is unique, and all its coefficients are zero or one. Conservation equations force a canonical expanding space–time region, so neither occupancy inequalities nor an externally chosen time horizon is needed. A marginal projection removes $(s - 1)^2$ equations from a pair-colour compiler without changing any natural-polynomial solution tuple. Injective symbol features compress the same witnesses further without introducing new variables.

We prove Σ_1^0 -completeness for a fixed matrix, an exact running-time/degree correspondence, a non-computable degree-bound theorem, a parsimonious nondeterministic extension, and a single quadratic equation with the same solutions. We also give an executable compiler, exhaustive finite checks, and a fully expanded worked quadratic. The contribution is an explicit uniqueness-preserving normal form, not the already known undecidability of linear polynomial-semiring equations, and not an ordinary single-fold Diophantine representation over the natural numbers.

Status. Complete mathematical arguments and reproducible exact computations are supplied. This is an unrefereed research manuscript, not a Lean or Rocq formalization. Historical priority for the particular combined normal form has not been established. The distinction between polynomial unknowns and ordinary integer unknowns is essential throughout.

2 The question and the result

A finite computation can be certified by introducing variables for each successive state. The number of variables then grows with the chosen time horizon. A different question is whether an entire, arbitrarily long computation can be represented by a fixed number of algebraic objects while retaining a precise correspondence between computations and complete witnesses. Here the algebraic objects are finite polynomials: the number of unknown polynomials stays fixed, whereas their degrees and supports grow with the computation.

The main technical issue is not merely to encode local transition rules. It is to prevent additional tiles, detached components, multiplicities, extra blank padding, independently chosen clock lengths, and arbitrary accepting continuations from creating spurious or duplicate witnesses. The construction below resolves these issues within a particularly simple syntax: affine-linear equations over the semiring $\mathbb{N}[X, Y]$.

2.1 Main normal-form theorem

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. An accepting event means that at least one cell carries a symbol from a designated set $\mathcal{F}$. A *first singleton acceptance* is a first accepting event at which exactly one cell is accepting. This condition is automatic for the usual one-head simulation of a deterministic accepting Turing machine, but is not automatic for an arbitrary cellular automaton.

Theorem 2.1 (Canonical linear history compiler). *Let $\mathcal{A}$ be a finite alphabet of size s , with blank symbol 0 . Let $f : \mathcal{A}^3 \rightarrow \mathcal{A}$ satisfy $f(0, 0, 0) = 0$, and let $\mathcal{F} \subseteq \mathcal{A} \setminus \{0\}$. There is an explicitly constructible matrix*

$$A_{f, \mathcal{F}} \in \mathbb{Z}[X, Y]^{(3s+3) \times (s^3+s+5)}$$

and an effective encoding of each nonempty finite word w as a vector $\mathbf{b}(w)$ of zero–one coefficient polynomials such that:

- (i) $A_{f, \mathcal{F}}U = \mathbf{b}(w)$ has a solution $U \in \mathbb{N}[X, Y]^{s^3+s+5}$ precisely when the evolution from w , with blank exterior, has a first singleton acceptance.
- (ii) Whenever a solution exists, the complete tuple U is unique and every coefficient in every component of U is zero or one.
- (iii) If the first acceptance time is h and the input length is m , then

$$\max_i \deg_Y U_i = h, \quad \max_i \deg_X U_i = m + 2h + 1, \quad \max_i \deg U_i = m + 3h + 1.$$

- (iv) Every nonzero matrix entry has at most two monomials, with coefficients in $\{-1, 1\}$ and total degree at most three. The matrix depends only on f and $\mathcal{F}$, not on w , m , or h .

The theorem is proved in Sections 5–6. It is a statement about all solutions, not only about a specially constructed class of well-behaved solutions. No coefficient bounds, monomial predicates, support bounds, or time indices are added as hidden side conditions.

Corollary 2.2 (A fixed unambiguous hard instance family). *For one fixed choice of f and $\mathcal{F}$, deciding whether $A_{f, \mathcal{F}}U = \mathbf{b}(w)$ is solvable over $\mathbb{N}[X, Y]$ is Σ_1^0 -complete. Every input word in this family has either no solution or exactly one solution, and any solution has only zero–one coefficients.*

The direct form of Theorem 2.1 is convenient for understanding soundness. Section 8 proves two stronger row counts with *exactly the same complete solution tuples*: $3\lceil \log_2 s \rceil + 6$ rows retaining unit coefficients, or nine rows allowing coefficients of magnitude at most $s - 1$ for $s \geq 2$. All conclusions about uniqueness, Boolean coefficients, and degrees transfer unchanged.

The proof uses a direct one-head Turing-machine simulation, not an assumption that an arbitrary cellular rule is universal. The construction of that simulation is given in Section 9.

2.2 What is and is not new here

Narendran proved undecidability of systems of linear equations over polynomial semirings, already with one indeterminate, in 1996 [8]. Thus neither polynomial-semiring undecidability nor monomial forcing is claimed as a new idea. In particular, the elementary power-forcing equations used below have a predecessor in his construction.

Lohrey and Steinberg encode Turing machines by tilings to obtain undecidable subsemimodule membership for finite-rank free $(\mathbb{Z} \times \mathbb{Z})$ -modules, and related group-theoretic results [9]. Their zero-tiling-sum formulation permits multiple tiles at a location. It is a close conceptual precedent for interpreting linear equations as translated local constraints.

The contribution developed here is the combined, explicitly counted normal form: a fixed low-degree matrix, exact shape recovery from conservation, first-acceptance canonicalization, uniqueness of every auxiliary polynomial, forced zero–one coefficients, and an exact degree ledger. These properties are proved directly. We have not established that this precise combination is absent from all earlier literature, and do not present the theorem as a resolution of a named longstanding Diophantine conjecture.

Dong’s work provides a useful contrast: a single homogeneous linear equation over $\mathbb{N}[X]$ admits polynomial-time decision procedures for the nonzero-solution problem he studies [10]. This is not a decision procedure for arbitrary inhomogeneous systems. The contrast here is therefore not a claimed one-variable/two-variable undecidability threshold. Two coordinate indeterminates are used because they retain the actual space–time geometry transparently.

2.3 Connection with the repository

The inspected ProveIt snapshot contains both formalized MRDP material and many computational-substrate certificate projects. Its Hilbert-tenth README carefully distinguishes finite-horizon compilers from fixed universal arithmetic bounds and records a 75-operation certificate / 87-operation universal-polynomial boundary for a separate natural-number construction [12]. This report changes neither of those counts.

Instead, it contributes a different target domain and a potentially useful intermediate representation. The polynomial tuple is canonical before any attempt is made to pack it into ordinary integers. The transferable parts are the proof of shape, the marginal and feature projections, the witness reconstruction theorem, and the explicit conditions that a later packing stage must preserve. No repository theorem is used as an unexamined black-box lemma in the elementary compiler proof.

3 Semantics and notation

3.1 Polynomials are the unknown values

Let

$$\mathcal{S} = \mathbb{N}[X, Y] \subset \mathbb{Z}[X, Y].$$

An element of $\mathcal{S}$ is a finite sum $P = \sum_{i,j \geq 0} p_{ij} X^i Y^j$ with $p_{ij} \in \mathbb{N}$. The symbols X, Y are fixed coordinate indeterminates, not existentially quantified numbers. An equation is an identity of polynomials, equivalent to equality at every coefficient.

A linear system means

$$\sum_{j=1}^n a_{ij}(X, Y) U_j(X, Y) = b_i(X, Y), \quad 1 \leq i \leq r, \quad (3.1)$$

where the a_{ij}, b_i are supplied integer-coefficient polynomials and $U_j \in \mathcal{S}$ are unknown. There are no products $U_j U_k$ in a linear system. Negative coefficient terms in (3.1) may be moved to the opposite side, so this is equally an equality between two expressions formed with addition and multiplication by fixed elements of $\mathcal{S}$.

We call a representation *single-fold over $\mathcal{S}$* when every admitted input has exactly one full tuple of $\mathcal{S}$ -witnesses, and every other input has none. This terminology does not change the domain to $\mathbb{N}$. One polynomial variable contains an unbounded finite amount of data.

Write

$$[r]_X = 1 + X + \cdots + X^{r-1}, \quad [0]_X = 0.$$

For a nonnegative polynomial, its mass is $\text{wt}(P) = P(1, 1)$, the sum of all coefficients. A polynomial is called *Boolean-coefficient* when all its coefficients belong to $\{0, 1\}$; no Boolean-ring arithmetic is intended. We use $\deg 0 = -\infty$ for total and coordinate degrees, so maxima over tuples with zero components are unambiguous.

3.2 The cellular evolution

Let $w = w_0 \cdots w_{m-1} \in \mathcal{A}^m$, $m \geq 1$. The initial configuration is

$$c_0(i) = \begin{cases} w_i, & 0 \leq i < m, \\ 0, & \text{otherwise,} \end{cases} \quad i \in \mathbb{Z}.$$

Subsequent configurations satisfy

$$c_{t+1}(i) = f(c_t(i-1), c_t(i), c_t(i+1)). \quad (3.2)$$

Quiescence gives the elementary light-cone bound

$$c_t(i) = 0 \quad \text{if } i < -t \text{ or } i > m - 1 + t. \quad (3.3)$$

Indeed, a cell outside the larger interval at time $t + 1$ has three blank predecessors at time t . This proves (3.3) by induction.

We distinguish *configuration rows* from *transition rows*. Transition row t contains the triples whose centres are

$$i = -t - 1, -t, \dots, m + t.$$

There are

$$w_t = m + 2t + 2 \quad (3.4)$$

such triples. The two extreme centres are blank guard cells at time t ; their outputs are allowed to become nonblank at time $t + 1$. We encode the centre i in this row at horizontal exponent

$$j = i + t + 1, \quad 0 \leq j < w_t. \quad (3.5)$$

Thus a cell remaining at physical position i shifts one place to the right when moving to the next encoded row. This is the source of the factor XY in the vertical equations.

4 A finite nonnegative flow forces a clock

The following elementary lemma is the main mechanism for obtaining exact shape information from linear equations. Its power-forcing special cases are familiar from polynomial-semiring undecidability constructions [8]; the translated form is convenient for the present proof.

Lemma 4.1 (Ray-clock lemma). *Let $S = X^a Y^b$ and $M = X^p Y^q$, where $a, b, p, q \in \mathbb{N}$ and $p + q > 0$. For $U, V \in S$, the identity*

$$U + V = S + MV \quad (4.1)$$

holds if and only if, for some $h \in \mathbb{N}$,

$$U = SM^h, \quad V = S \sum_{t=0}^{h-1} M^t. \quad (4.2)$$

The integer h is uniquely determined, and equals $\text{wt}(V)$.

Proof. Evaluation at $(1, 1)$ gives $\text{wt}(U) = 1$. Hence U is a monomial with coefficient one. Regard the coefficients of V as nonnegative integer flows along directed edges

$$(i, j) \longrightarrow (i + p, j + q)$$

in $\mathbb{N}^2$. Equation (4.1) says that the unique source is at (a, b) and the unique sink is at the exponent of U . Every vertex has at most one predecessor and one successor, and total degree strictly increases along every edge.

For completeness, this flow description can be checked purely by coefficient induction. Before the source, and on every chain not meeting the source, the incoming coefficient is zero. A positive outgoing coefficient cannot appear there without a source; a sink there would force a negative coefficient. At the source the outgoing coefficient is one unless the sink is the source itself. That coefficient remains one along its unique successor chain until the sink is reached. Since V has finite support, this chain must terminate at the sink. There can be no other flow. It follows that the sink is SM^h and that precisely the first h edges of this ray carry coefficient one. This is (4.2). Conversely, those expressions satisfy (4.1) by telescoping. Their mass and uniqueness statements follow immediately. $\square$

Two aspects matter. First, finiteness of support excludes an infinite unterminated flow. Second, nonnegativity excludes a compensating signed flow on another chain. Ordinary scalar evaluation of the equation does not retain either of these coefficientwise arguments.

5 The complete compiler

We now list every unknown and every equation. The list itself is the compiler; no separate validity predicate is attached to its output.

5.1 Unknown polynomials and linear forms

For each triple $\tau = (\ell, c, r) \in \mathcal{A}^3$, introduce a polynomial $Z_{\ell cr}$. Its intended coefficient at $X^j Y^t$ is one when that triple occurs at transition site (j, t) , and zero otherwise. Introduce T_a for every $a \in \mathcal{A}$ to record the final configuration, with one blank guard at either end. Introduce five further polynomials

$$D, Q, B, V, J.$$

These are all the unknowns, giving $s^3 + s + 5$ in total.

For $a \in \mathcal{A}$, use the following abbreviations for *linear forms*, not additional unknowns:

$$\mathsf{L}_a = \sum_{\ell=a} Z_{\ell cr}, \quad \mathsf{C}_a = \sum_{c=a} Z_{\ell cr}, \quad (5.1)$$

$$\mathsf{R}_a = \sum_{r=a} Z_{\ell cr}, \quad \mathsf{O}_a = \sum_{f(\ell,c,r)=a} Z_{\ell cr}. \quad (5.2)$$

Indices not specified under a sum range independently over $\mathcal{A}$. Also put

$$\mathsf{I}_a(X) = \sum_{\substack{0 \leq j < m \\ w_j = a}} X^j, \quad G = V + B + Q + D. \quad (5.3)$$

The I_a are supplied data, and G is another linear form.

5.2 Equations

The two clocks are

$$D + Q = X^{m+1} + X^2 Y Q, \quad (C1)$$

$$B + V = 1 + Y V. \quad (C2)$$

The left-neighbour marginal equations, one for every $a \in \mathcal{A}$, are

$$X \mathsf{C}_a + \delta_{a,0} V = \mathsf{L}_a + \delta_{a,0} X Q. \quad (L_a)$$

The right-neighbour marginal equations are needed only for nonblank symbols:

$$X \mathsf{R}_a = \mathsf{C}_a \quad (a \in \mathcal{A} \setminus \{0\}). \quad (R_a)$$

The vertical equations, one for each symbol, are

$$\mathsf{C}_a + T_a = X \mathsf{I}_a + X Y \mathsf{O}_a + \delta_{a,0} G. \quad (V_a)$$

Finally, acceptance is enforced by

$$\sum_{a \in \mathcal{F}} \mathsf{C}_a = 0, \quad (F0)$$

$$\sum_{a \in \mathcal{F}} T_a + J = B + X J. \quad (F1)$$

All equalities are polynomial identities, and every listed unknown ranges over $\mathcal{S}$.

5.3 Why the matrix is fixed

Move the supplied terms X^{m+1} , 1, and $X \mathsf{I}_a$ to the right-hand vector, leaving every unknown on the left. The remaining coefficient polynomials depend only on the alphabet, the rule table, and the accepting set. Typical entries are $1 - X^2 Y$, $1 - Y$, $X - 1$, $1 - XY$, $1 - X$, X , and -1 . For an individual tile variable there is at most one positive and one negative occurrence in any row. Thus each entry has at most two monomials with coefficients of magnitude at most one, and total coordinate degree at most three.

The apparent input-dependent coefficient in a naive boundary construction would be $X^{m+1} Q$. It does not occur here. Instead, the right-boundary clock is *started* at X^{m+1} in the supplied constant term of (C1). All subsequent shifts are multiplication by the fixed monomial $X^2 Y$. This is the step that makes the matrix independent of the input length.

Component	Role	Number of rows
(C1)–(C2)	Right boundary and vertical clock	2
(L _a)	Left-neighbour matching and total shape	s
(R _a)	Right-neighbour matching, blank row omitted	$s - 1$
(V _a)	Input, transition, and terminal row	s
(F0)–(F1)	First singleton acceptance	2
Total	Complete affine-linear system	$3s + 3$

Table 4: No occupancy rows or coefficient-Booleanity equations are added.

6 Shape and soundness

6.1 The clocks synchronize without a separate equation

Lemma 6.1 (Forced boundary and transition region). *Every solution of the complete system has a uniquely determined $h \in \mathbb{N}$ for which*

$$D = X^{m+1+2h}Y^h, \quad Q = \sum_{t=0}^{h-1} X^{m+1+2t}Y^t, \quad (6.1)$$

$$B = Y^h, \quad V = \sum_{t=0}^{h-1} Y^t. \quad (6.2)$$

Moreover, the total tile polynomial $H = \sum_{\ell,c,r} Z_{\ell cr}$ is

$$H = \sum_{t=0}^{h-1} Y^t [m + 2t + 2]_X. \quad (6.3)$$

Consequently, at every site in this region exactly one tile polynomial has coefficient one, and all tile coefficients outside the region vanish.

Proof. Apply Lemma 4.1 separately to (C1) and (C2). This gives (6.1) for one integer h and gives $B = Y^k$, $V = \sum_{t < k} Y^t$ for a possibly different integer k .

Summing all equations (L_a) over a uses $\sum_a C_a = \sum_a L_a = H$ and yields

$$XH + V = H + XQ. \quad (6.4)$$

Evaluation at $(1, 1)$ gives $k = h$. Thus the two clocks have the same height without a separately paid synchronization variable or row.

Rearrange (6.4):

$$(1 - X)H = V - XQ = \sum_{t=0}^{h-1} Y^t (1 - X^{m+2t+2}).$$

Since $(1 - X)[r]_X = 1 - X^r$, the displayed expression for H satisfies this equation. It is the only such polynomial, because $1 - X$ is not a zero divisor in $\mathbb{Z}[X, Y]$.

Every coefficient of every $Z_{\ell cr}$ is nonnegative. Their sum is one at each site of (6.3), and zero elsewhere. Therefore exactly one tile is present at each permitted site, with coefficient one; no tile can occur elsewhere or with multiplicity greater than one. $\square$

This argument establishes occupancy rather than assuming it. In particular, it rules out disjoint extra computations and translated closed components: their nonnegative coefficients would alter the forced total H .

6.2 The terminal region is also forced

Lemma 6.2 (Forced top row). *Every solution satisfies*

$$K := \sum_{a \in \mathcal{A}} T_a = Y^h [m + 2h + 2]_X. \quad (6.5)$$

At each site of this row exactly one terminal symbol occurs with coefficient one; every terminal coefficient elsewhere is zero.

Proof. Each triple has exactly one central symbol and exactly one output under f . Hence $\sum_a C_a = \sum_a O_a = H$. Also, $\sum_a I_a = [m]_X$. Summing (V_a) gives

$$H + K = X[m]_X + XYH + G. \quad (6.6)$$

By (6.1)–(6.2),

$$G = \sum_{t=0}^h Y^t (1 + X^{m+2t+1}). \quad (6.7)$$

The coefficient row at Y^0 on the right of (6.6) is $X[m]_X + 1 + X^{m+1} = [m + 2]_X$. For $1 \leq t \leq h$, that row is

$$X[m + 2t]_X + 1 + X^{m+2t+1} = [m + 2t + 2]_X.$$

There are no other rows. Subtracting the already known H leaves exactly (6.5). This argument includes $h = 0$, when $H = 0$ and the initial guarded row is already the top row. Nonnegativity now proves the one-symbol-per-site assertion just as in Lemma 6.1. $\square$

6.3 Neighbour matching is exact

The omitted blank right-neighbour row is not a missing constraint. Summing (R_a) over $a \neq 0$ and subtracting that sum from (6.4) gives

$$XR_0 + V = C_0 + XQ. \quad (6.8)$$

Thus, with the blank boundary terms restored, right-neighbour matching holds for every symbol.

Lemma 6.3 (The tile triples are genuine consecutive windows). *Fix a transition row $t < h$, and write its unique tile at position j as (ℓ_j, c_j, r_j) , for $0 \leq j < w_t$. Then*

$$\ell_0 = c_0 = 0, \quad c_{w_t-1} = r_{w_t-1} = 0,$$

and for $1 \leq j < w_t$,

$$\ell_j = c_{j-1}, \quad c_j = r_{j-1}.$$

In particular, the tile at each site is the actual length-three window of the row of centre symbols, with blank exterior.

Proof. At an interior position j , coefficient comparison in (L_a) gives the equality of the one-symbol indicators of c_{j-1} and ℓ_j . The right-neighbour rows, including (6.8), give the equality of the indicators of r_{j-1} and c_j . Since the occupancy lemma has already proved that these are single symbols, the indicator equalities imply the asserted equalities of symbols.

At position zero, (L_a) forces $\ell_0 = 0$ and the right-neighbour rows force $c_0 = 0$. At the coefficient immediately after the last tile, the boundary monomial in XQ forces $c_{w_t-1} = 0$ in the left rows and $r_{w_t-1} = 0$ in the right rows. All boundary exponents agree with (3.4) by (6.1). Finally, the interior equalities also give $r_j = c_{j+1}$ wherever that expression is internal to the row. $\square$

It would not be sound to deduce equality of joint symbol pairs from marginal counts before proving occupancy. Here occupancy comes first; there is exactly one tile, not a distribution or a multiset, at a site.

6.4 Vertical matching recovers the actual evolution

Lemma 6.4 (Semantic reconstruction). *For each $t < h$, the coefficient of $X^j Y^t$ in C_a is one exactly when*

$$c_t(j - t - 1) = a, \quad 0 \leq j < w_t.$$

The coefficient of $X^j Y^h$ in T_a is one exactly when $c_h(j - h - 1) = a$, for $0 \leq j < w_h$.

Proof. When $h > 0$, the Y^0 part of (V_a) is

$$C_a|_{Y^0} = XI_a + \delta_{a,0}(1 + X^{m+1}).$$

This is precisely the input configuration with the two source guards. By Lemma 6.3, every tile in this row contains the true neighbourhood of its centre. Its assigned output is therefore the true cellular output. Multiplication by XY moves that output to the next encoded source row and shifts its exponent by one, as required by (3.5). The two new boundary terms in G supply the blank source guards for that row. Inducting in t recovers all transition rows.

At time h , there are no centre terms because H has no Y^h row. Equation (V_a) therefore places the last outputs, together with the guards, in the T_a . This proves the terminal assertion. If $h = 0$, the same equation directly reads $T_a = XI_a + \delta_{a,0}(1 + X^{m+1})$, which proves the assertion without a transition row. Bound (3.3) and quiescence ensure that the omitted infinite exterior is genuinely blank, not an unverified part of the computation. $\square$

6.5 Acceptance and all auxiliary witnesses

Lemma 6.5 (First acceptance and the horizontal marker). *Every solution encodes a first singleton acceptance at time h . If its accepting cell has encoded horizontal position j_* , then*

$$\sum_{a \in \mathcal{F}} T_a = X^{j_*} Y^h, \quad J = Y^h[j_*]_X, \quad 1 \leq j_* \leq m + 2h. \quad (6.9)$$

Proof. All summands in $(F0)$ are nonnegative. Thus no centre row before h has an accepting symbol. Lemma 6.4 identifies these with the genuine configurations, including their blank exterior.

Apply Lemma 4.1 to $(F1)$, with source $B = Y^h$, multiplier X , and sink $\sum_{a \in \mathcal{F}} T_a$. It yields exactly the first two formulas in (6.9). There is consequently one accepting cell at time h , not merely a selected cell among several. The two top boundary cells are blank by the vertical reconstruction and $0 \notin \mathcal{F}$, which gives the stated bounds on j_* . The marker J is uniquely forced; there is no slack in its choice. $\square$

6.6 Completeness and uniqueness

Completion of the proof of Theorem 2.1. Suppose first that the genuine evolution has a first singleton acceptance at time h . Define D, Q, B, V by (6.1)–(6.2). At every transition site (j, t) with $t < h$, put coefficient one in the unique $Z_{\ell cr}$ corresponding to the actual neighbourhood of physical centre $j - t - 1$. Put coefficient one in T_a at (j, h) according to the actual guarded final row. Define J by (6.9). All unspecified coefficients are zero.

The clock equations telescope. The spatial equations express exact adjacency and the blank source boundaries. The vertical equations express the input row, the deterministic rule, and the two new guards at each stage. The absence of earlier acceptance gives (F0), and the single terminal accepting cell gives (F1). Thus this tuple is a solution, proving completeness.

Conversely, the preceding lemmas reconstruct a genuine first singleton acceptance from every solution. The first accepting time of a deterministic evolution is unique. Hence every solution must have the same h . Its clocks are then fixed, every transition tile and terminal symbol is fixed by the actual evolution, and J is fixed by the location of the accepting cell. There is therefore exactly one full solution tuple. All its components are Boolean-coefficient by the construction and by the shape and clock lemmas. The matrix-size and coefficient assertions were checked in Section 5; the exact degree assertions are proved in Section 10 below. $\square$

Remark 6.6 (Two accepting cells are not one accepting certificate). If the first accepting configuration contains two or more accepting cells, the system has no solution. A later time with only one accepting cell cannot help, because (F0) detects the earlier event. For arbitrary cellular automata this restriction is part of the specified language, not an omitted case in the theorem. One-head machine encodings satisfy it automatically.

7 An exact marginal projection

A more direct tile formulation would use the ordered pair of symbols on each side of a tile as its horizontal colour. Define

$$W_{ab} = \sum_{(\ell, c)=(a, b)} Z_{\ell cr}, \quad E_{ab} = \sum_{(c, r)=(a, b)} Z_{\ell cr}.$$

The pair-colour spatial equations are

$$XE_{ab} + \delta_{a,0}\delta_{b,0}V = W_{ab} + \delta_{a,0}\delta_{b,0}XQ, \quad a, b \in \mathcal{A}. \quad (7.1)$$

Together with the same clocks, vertical equations, and acceptance rows, these give $s^2 + s + 4$ equations on exactly the same witness tuple.

Theorem 7.1 (Full-tuple preservation under marginal reduction). *The pair-colour system and the reduced system $\mathcal{E}_{f, \mathcal{F}(w)}$ have identical solution sets in $\mathcal{S}^{s^3+s+5}$. Replacing (7.1) by (L_a) – (R_a) removes exactly $(s-1)^2$ equations and changes neither the meaning nor the value of any witness coordinate.*

Proof. Sum (7.1) over b for fixed a . The resulting equation is (L_a) . Sum it over a for fixed b to obtain the complete right-neighbour equation for b ; discarding its blank instance gives (R_a) . Thus every pair-colour solution is a reduced solution.

For the reverse implication, Lemma 6.1 proves that each site contains one tile, and Lemma 6.3 proves equality of both components of the adjacent pair. Therefore the ordered pairs themselves agree. At the left and right boundaries the pair is $(0, 0)$, and there are no tiles outside the prescribed region. Comparing coefficients at every site gives (7.1). The equation-count difference is

$$(s^2 + s + 4) - (3s + 3) = (s - 1)^2.$$

No reconstruction map or fresh auxiliary variable is involved. $\square$

This theorem is a zero-set equivalence over nonnegative polynomials, not an off-zero identity between residual vectors. The proof of the reverse direction uses forced unit occupancy. In a signed or fractional setting, equality of two marginal distributions does not generally force equality of the joint pair distributions.

8 Injective features: logarithmically many rows, or nine

The pair-to-marginal reduction still assigns a channel to each symbol. Once the shape equations force exactly one label at each site, a full one-hot vector is unnecessary: any injective numerical description of a symbol suffices. The important order of argument is to force unit occupancy *before* using the injectivity of this description.

8.1 The complete feature system

Let $d \geq 0$ and let

$$\chi : \mathcal{A} \longrightarrow \{0, 1, \dots, B_0\}^d$$

be injective, with $\chi(0) = 0$. Here $B_0 \geq 1$ is an ordinary fixed integer, not the clock polynomial B . For $1 \leq k \leq d$, define the linear forms

$$\begin{aligned} L^{(k)} &= \sum_{\ell, c, r} \chi_k(\ell) Z_{\ell cr}, & C^{(k)} &= \sum_{\ell, c, r} \chi_k(c) Z_{\ell cr}, \\ R^{(k)} &= \sum_{\ell, c, r} \chi_k(r) Z_{\ell cr}, & O^{(k)} &= \sum_{\ell, c, r} \chi_k(f(\ell, c, r)) Z_{\ell cr}, \\ T^{(k)} &= \sum_{a \in \mathcal{A}} \chi_k(a) T_a, & I^{(k)} &= \sum_{j=0}^{m-1} \chi_k(w_j) X^j. \end{aligned}$$

As before, $H = \sum Z_{\ell cr}$, $K = \sum T_a$, and $G = V + B + Q + D$ are abbreviations, not additional unknowns.

Retain the two clock equations (C1)–(C2) and the two acceptance equations (F0)–(F1). Replace all the spatial and vertical symbol equations by the following two aggregate equations and $3d$ feature equations:

$$XH + V = H + XQ, \quad H + K = X[m]_X + XYH + G, \quad (8.1)$$

$$XC^{(k)} = L^{(k)}, \quad XR^{(k)} = C^{(k)}, \quad 1 \leq k \leq d, \quad (8.2)$$

$$C^{(k)} + T^{(k)} = XI^{(k)} + XYO^{(k)}, \quad 1 \leq k \leq d. \quad (8.3)$$

Thus the complete feature compiler has $s^3 + s + 5$ unknown polynomials and exactly $6 + 3d$ affine-linear equations. In particular, the aggregate shape equations are explicitly paid for; they are not free consequences of a family of feature rows whose total might have been lost.

Theorem 8.1 (Same-tuple injective-feature compression). *For every injective χ as above, the feature system and $\mathcal{E}_{f, \mathcal{T}}(w)$ have identical solution sets in $\mathbb{N}[X, Y]^{s^3+s+5}$. The feature matrix is independent of the input word and horizon. Every entry has at most two monomials, coefficient magnitude at most B_0 , and coordinate total degree at most three. The supplied right-hand side has coefficient magnitude at most B_0 as well.*

Proof. First suppose the original system holds. Summing its left-marginal rows gives the first aggregate equation; summing its vertical rows gives the second. Weighting the left-marginal rows by $\chi_k(a)$ eliminates

both blank boundary terms, since $\chi_k(0) = 0$, and gives the first feature spatial equation. Weighting the nonblank right-marginal rows gives the second. Weighted vertical rows give (8.3). The unchanged clock and acceptance rows require no argument. This proves inclusion of the original zero set into the feature zero set using no new witnesses.

Conversely, the clocks and the first aggregate equation are exactly the hypotheses used to derive synchronized clocks and the transition mask in Section 6. The second aggregate equation then forces the same terminal mask. Nonnegativity therefore gives exactly one tile at each source site and exactly one terminal symbol at each top site, with zero multiplicities elsewhere.

Write (ℓ_j, c_j, r_j) for the unique source triple at position j of one row. Coefficient comparison in the feature spatial equations gives, for every k , the boundary equalities

$$\chi_k(\ell_0) = \chi_k(c_0) = 0, \quad \chi_k(c_{\text{last}}) = \chi_k(r_{\text{last}}) = 0,$$

and the interior equalities

$$\chi_k(c_j) = \chi_k(\ell_{j+1}), \quad \chi_k(r_j) = \chi_k(c_{j+1}).$$

These equalities hold for the whole feature vector. Injectivity and $\chi(0) = 0$ recover both blank boundaries and the genuine overlapping windows. At each initial site, the feature vertical equations identify the central symbol with the corresponding input symbol, with zero features on the two guards. At each later source or terminal site, they identify that symbol with the deterministic output of the preceding row's unique tile. Again injectivity is applied only to a single symbol on each side, never to arbitrary sums of symbol codes. The recovered configuration history satisfies every original symbol equation. Since the clock and acceptance equations were retained verbatim, the complete original system holds on the very same tuple.

The matrix count and bounds follow by inspection. A source variable can occur in any one feature row only at its two specified coordinate shifts; its coefficients are feature values with signs. Aggregate rows have unit coefficients. The sole degree-three shift is the inherited X^2Y in the first clock. Input feature coefficients occur at distinct supplied positions. This proves all claims. $\square$

8.2 Three exact coefficient–row tradeoffs

Enumerate the alphabet as $0, 1, \dots, s-1$.

Corollary 8.2 (Binary and weighted forms). *The same complete natural-polynomial witness relation admits:*

- (i) $3\lceil \log_2 s \rceil + 6$ rows whose matrix coefficients lie in $\{-1, 0, 1\}$ and whose input vector has zero–one coefficients;
- (ii) nine rows with coefficient magnitude at most $s-1$, for $s \geq 2$;
- (iii) $3\lceil \log_b s \rceil + 6$ rows with coefficient magnitude at most $b-1$, for every fixed integer $b \geq 2$.

The number of unknown polynomials remains $s^3 + s + 5$. Every conclusion about complete-witness uniqueness, Boolean coefficients, exact degrees, and support size remains unchanged.

Proof. Use the fixed-length binary digits of each symbol for (i), the single feature $\chi(a) = a$ for (ii), and its fixed-length base- b digits for (iii). Each map is injective and sends blank to zero. Apply Theorem 8.1. When $s = 1$, the empty feature vector is injective on the singleton alphabet and the zero-feature form has six rows; its accepting set is empty, as required by the standing convention. $\square$

For a fixed universal automaton, part (ii) gives a fixed *nine-row* coefficient matrix with undecidable solvability on the effective word input family, while every solvable member has a unique Boolean-coefficient full witness. The constant number of rows is not a constant number of unknown scalar coefficients or an ordinary nine-variable Diophantine claim.

Within the stated feature alphabet $\{0, \dots, B_0\}^d$, the elementary counting bound

$$s \leq (B_0 + 1)^d$$

forces $d \geq \lceil \log_{B_0+1} s \rceil$. This is only a lower bound on the length of an injective bounded-coordinate code, not a lower bound on all possible algebraic compilers or even on all same-witness equation systems. Feature compression also need not reduce the size of a fully expanded sum of squares: fewer but denser rows can produce more cross terms. The worked example below exhibits this tradeoff explicitly.

9 Universality without a black-box cellular rule

9.1 An explicit machine-to-cellular construction

Fix a deterministic, one-tape, two-way-infinite Turing machine. Its tape alphabet is Γ with blank $\square$, its state set is Q , and it has a designated accepting state q_{acc} . For every nonaccepting state and tape symbol, let

$$\delta(q, a) = (q', b, d), \quad d \in \{L, R, S\}.$$

A rejecting halt may be replaced by a nonaccepting stationary loop. The accepting head is retained forever after its first appearance.

Take the cellular alphabet

$$\mathcal{A} = \Gamma \sqcup (Q \times \Gamma), \quad s = |\Gamma|(1 + |Q|). \quad (9.1)$$

An unmarked symbol $a \in \Gamma$ represents a tape cell without a head; (q, a) represents the same tape cell carrying the head in state q . Identify the unmarked blank with the cellular symbol 0.

Define the radius-one rule by the following ordered cases. If the centre is an accepting head, keep it. If the centre is a nonaccepting head with transition (q', b, d) , output (q', b) when $d = S$ and the unmarked b when $d = L$ or R . If the centre is unmarked and its left neighbour is a head instructed to move right, put that head's new state on the centre's existing tape symbol. Otherwise, if the centre is unmarked and its right neighbour is a head instructed to move left, do the analogous update. Otherwise retain the centre's unmarked symbol. Neighbourhoods with multiple inconsistent heads are resolved by the stated order of cases; they do not occur from a valid one-head encoding.

This description determines a finite total table $f : \mathcal{A}^3 \rightarrow \mathcal{A}$. In particular, an all-blank neighbourhood remains blank. Set

$$\mathcal{F} = \{(q_{\text{acc}}, a) : a \in \Gamma\}.$$

Proposition 9.1 (Step-for-step simulation). *From a finite encoding with exactly one head, the cellular automaton continues to have exactly one head and performs one machine transition per cellular step, until and including the first accepting step. Its first acceptance, if any, has exactly one accepting cell.*

Proof. The old head either remains in place with its new state or writes the prescribed unmarked symbol and leaves its position. In the moving case, exactly its indicated neighbour receives the new head state. All other unmarked cells remain unchanged. Since there is one old head, there is no competing incoming move. These statements prove the one-step claim and preserve the one-head invariant. Induction proves the simulation. An accepting state belongs to the unique head, which proves the singleton property. $\square$

An empty machine input is represented by the single blank cell carrying the initial head, so the compiler's condition $m \geq 1$ causes no loss. For a nonempty input, put the initial head on its first tape symbol and leave the remaining tape symbols unmarked.

9.2 Fixed-matrix hardness and effective input loading

A fixed universal machine can interpret a finite program followed by its input; the classical universality construction supplies such machines [11]. Apply Proposition 9.1 to one such fixed machine. This fixes $f, \mathcal{F}, s$, and hence the entire matrix $A_{f, \mathcal{F}}$. Only $b(w)$ changes with the encoded program and input. The input map forms the symbol-indicator polynomials I_a and the endpoint marker X^{m+1} ; it does not simulate the machine or compute its accepting time.

Proof of Corollary 2.2. The reduction from the universal machine's acceptance problem is the finite input encoding just described, followed by the compiler. Proposition 9.1 and Theorem 2.1 prove equivalence and uniqueness. Thus the problem is Σ_1^0 -hard. It is computably enumerable because finite tuples of nonnegative integer polynomials can be enumerated, and any candidate tuple can be checked by finitely many exact coefficient operations. This proves completeness. $\square$

Equivalently, one obtains a fixed finitely generated $\mathcal{S}$ -subsemimodule

$$\mathcal{M} = \{A_{f, \mathcal{F}}U : U \in \mathcal{S}^{s^3+s+5}\} \subseteq \mathbb{Z}[X, Y]^{3s+3} \quad (9.2)$$

with undecidable membership even on the structured right-hand-side family $b(w)$. On that family, the representation by the fixed generators is unique whenever it exists. Uniqueness is not asserted for arbitrary vectors outside this input family.

This is an effective general compiler, not a numerical smallest-universal cellular-automaton claim. The package instantiates and tests several finite rules, but does not supply an independently audited numerical transition table for a particular small universal machine. Once such a table is chosen, the construction above generates its cellular table and fixes the numerical matrix size without any further mathematical existence step.

10 Exact resources and a noncomputable degree bound

10.1 Degrees, support size, and what fixed arity means

Proposition 10.1 (Exact witness ledger). *Let the input length be m , the first singleton acceptance time be h , and the final accepting position in the sheared coordinates be j_* . For the unique complete witness,*

$$\max_i \deg_Y U_i = h, \quad (10.1)$$

$$\max_i \deg_X U_i = m + 2h + 1, \quad (10.2)$$

$$\max_i \deg U_i = m + 3h + 1. \quad (10.3)$$

The total number of nonzero coefficients across all witness polynomials is

$$h^2 + mh + 5h + m + 4 + j_*. \quad (10.4)$$

Proof. All tile supports lie at $0 \leq t < h$ and $0 \leq j \leq m + 2t + 1$. All terminal supports lie at $t = h$ and $0 \leq j \leq m + 2h + 1$. The clocks and the marker obey (6.1), (6.2), and (6.9). These give the upper bounds in (10.1)–(10.3). The monomial $D = X^{m+2h+1}Y^h$ attains all three maxima simultaneously.

The total tile mass is

$$\text{wt}(H) = \sum_{t=0}^{h-1} (m + 2t + 2) = h(m + h + 1).$$

The terminal mass is $m + 2h + 2$. The masses of (D, Q, B, V, J) are $(1, h, 1, h, j_*)$. Since every coefficient is zero or one, adding these masses counts all nonzero coefficients and gives (10.4). $\square$

The compiler therefore has fixed *polynomial arity*, not bounded certificate length. The canonical light cone contains quadratically many sites in h for fixed input length. Sparse storage still records those sites. No exponential acceleration of ordinary witness checking is claimed, and no computation disappears into a bounded-size integer tuple by a mere change of notation.

10.2 No computable global degree estimate

Theorem 10.2 (Degree-bound obstruction). *For the fixed universal compiler of Corollary 2.2, there is no total computable function of the input word that bounds the largest Y -degree of its solution whenever a solution exists. The same holds for the largest total degree, the largest X -degree, and the total number of nonzero witness coefficients.*

Proof. A computable bound on the Y -degree would, by (10.1), bound the first accepting time on every accepting input. Simulating the fixed machine for that many steps would then decide acceptance, including a negative answer when it did not accept by the bound. This contradicts Corollary 2.2.

A bound on either of the other degrees gives a time bound by (10.2) or (10.3). A support bound also bounds h : for $h > 0$, the polynomial Q alone has h nonzero coefficients, and $h = 0$ is harmless. These reductions are effective, so all the claimed bounds are impossible. $\square$

A bound depending only on the size of the encoded input would be an even stronger bound and is therefore impossible as well. This remains true although the number of equations, the number of polynomial variables, the matrix entries, and the witness coefficient alphabet are all fixed. The unbounded computational resource is the polynomial support and its degree.

[write] The proof is the bounded-search argument of Proposition `cdc:bd:prop:nobound` in [1], with the Y -degree in place of an integer witness height; Corollary 26.3 and Corollary 41.5 are its counterparts in Parts II and III.

11 From many linear equations to one quadratic

Let $L_1(U), \dots, L_{3s+3}(U)$ be the left-minus-right residuals of the complete reduced system. Define

$$\mathcal{P}_{f,\mathcal{F},w}(U) = \sum_{i=1}^{3s+3} L_i(U)^2. \quad (11.1)$$

This is a polynomial in the unknown polynomial values, with coefficients in $\mathbb{Z}[X, Y]$.

Lemma 11.1 (Sum-of-squares identity test). *For $p_1, \dots, p_r \in \mathbb{Z}[X, Y]$, $\sum_i p_i^2 = 0$ as a polynomial identity if and only if every $p_i = 0$.*

Proof. The reverse implication is immediate. In the forward direction, evaluate the identity at any real pair (x, y) . A sum of nonnegative real squares can vanish only if every $p_i(x, y)$ vanishes. A real polynomial vanishing at every point of $\mathbb{R}^2$ is the zero polynomial. Applying this to each p_i proves the result. Real evaluation is used only to prove an identity lemma; it does not change the domain of the witness variables. $\square$

Theorem 11.2 (Single quadratic polynomial-semiring representation). *The equation $\mathcal{P}_{f, \mathcal{F}, w}(U) = 0$, with $U \in \mathcal{S}^{s^3+s+5}$, has exactly the same complete solutions as $\mathcal{E}_{f, \mathcal{F}}(w)$. It is of degree two in the unknown polynomial values. Its solution, when it exists, is unique and Boolean-coefficient.*

Proof. Every residual is affine-linear in the unknowns. Lemma 11.1 shows that its sum of squares vanishes precisely when all residuals vanish. The uniqueness and coefficient assertions are therefore inherited without any additional variables. The degree is exactly two: for example, the coefficient of D^2 receives a nonzero square contribution and cannot vanish as a sum of coefficient-polynomial squares. $\square$

One may regard the input endpoint and the I_a as polynomial parameters. Then (11.1) is one fixed quadratic expression for fixed $f, \mathcal{F}$, with those parameters specialized to valid word encodings. This is the sense in which the construction gives a universal quadratic expression over a polynomial semiring. The parameter values are not ordinary scalar numerals, and validity of the word-encoding slice is part of the input specification.

11.1 A bounded-degree ordinary quadratic family

Proposition 11.3 (Coefficient extraction at an external bound). *Fix a nonnegative external bound H_0 and put $d = m + 2H_0 + 1$. Replace each unknown polynomial by*

$$U_i = \sum_{0 \leq p \leq d} \sum_{0 \leq q \leq H_0} u_{i,pq} X^p Y^q, \quad u_{i,pq} \in \mathbb{N}.$$

Coefficient comparison gives a finite affine-linear system in

$$(s^3 + s + 5)(d + 1)(H_0 + 1) \tag{11.2}$$

ordinary natural-number variables. Its sum of squared scalar residuals is an ordinary quadratic polynomial. It has a unique natural zero precisely when a first singleton acceptance occurs at some time $h \leq H_0$; otherwise it has no natural zero. Fixing $B = Y^{H_0}$ restricts this to first acceptance at exactly H_0 .

Proof. The matrix shifts increase X -degree by at most two and Y -degree by at most one. It suffices to compare all coefficients with $0 \leq p \leq d + 2$, $0 \leq q \leq H_0 + 1$ in each residual. These are ordinary linear equations in the supplied coefficient variables.

Any coefficient solution is a genuine polynomial solution and therefore has acceptance time $h = \deg_Y B \leq H_0$. Conversely, the unique witness for any first singleton acceptance at $h \leq H_0$ satisfies the imposed degree bounds by Proposition 10.1. Coefficients give a bijection between the bounded polynomial tuple and the scalar tuple, so uniqueness is preserved. The scalar sum-of-squares equivalence follows from nonnegativity of integer squares. The additional equation on B fixes the clock time and is again linear in its coefficients. $\square$

There are at most $(3s + 3)(d + 3)(H_0 + 2)$ coefficient residuals before optional clock fixing. These loose rectangular counts include coefficients that the shape proof forces to zero; they are not optimization claims. Crucially, the scalar arity in (11.2) depends on the external bound. Taking the union of all such bounded families does not produce one fixed-arity ordinary quadratic Diophantine equation.

12 Why domain changes do not preserve the theorem for free

12.1 Signed polynomial witnesses

Proposition 12.1 (Nonnegative kernel and signed ambiguity). *For the fixed compiler matrix, the homogeneous system $A_f \cdot U = 0$ has only the zero solution in S^{s^3+s+5} . In contrast, every solvable input system has infinitely many solutions when witnesses are allowed in $\mathbb{Z}[X, Y]$.*

Proof. In the homogeneous version of the first clock, $D + Q = X^2YQ$. Evaluation at $(1, 1)$ gives $D = 0$ because D is nonnegative; then the integral-domain property gives $Q = 0$. The second clock similarly gives $B = V = 0$. Summing the homogeneous left-neighbour rows yields $XH = H$, hence $H = 0$ and all nonnegative tile polynomials vanish. The vertical rows now give $T_a = 0$ for every a , and the final acceptance row gives $J = XJ$, hence $J = 0$.

For the signed assertion, the matrix has $n = s^3 + s + 5$ columns and $r = 3s + 3$ rows, with $n - r = s^3 - 2s + 2 > 0$ for every $s \geq 1$. Over the field $\mathbb{Q}(X, Y)$, it therefore has a nonzero kernel vector. Clearing polynomial and integer denominators gives a nonzero kernel vector $K \in \mathbb{Z}[X, Y]^n$. From any signed solution U , all vectors $U + tK$ with $t \in \mathbb{Z}$ are distinct signed solutions. They need not be nonnegative. $\square$

The absence of a nonzero nonnegative homogeneous kernel is not, by itself, a general uniqueness criterion for inhomogeneous semiring systems: the difference of two nonnegative solutions may be signed. The specific uniqueness theorem still needs semantic reconstruction and first acceptance.

[write] Part III restores a unique witness over every nonzero commutative ring, signed coefficients and zero divisors included, by a different mechanism: unknowns restricted to subrings that omit a tested variable, and a clock indeterminate (Theorems 39.1 and 41.2).

12.2 Numerical evaluation destroys the clock predicate

Consider the elementary clock

$$D + Q = 1 + X^2YQ.$$

As a polynomial equation over $\mathcal{S}$, Lemma 4.1 forces $D = (X^2Y)^h$. After specialization $X = b \geq 2, Y = c \geq 1$, the scalar natural values

$$Q = 2, \quad D = 2b^2c - 1$$

always satisfy the numerical equation. Yet

$$b^2c < 2b^2c - 1 < (b^2c)^2,$$

so this D is not a power of b^2c . Thus even the simplest clock loses its intended semantics under raw scalar evaluation. This is a counterexample to the clock specialization, not a claim that every full compiled system acquires a spurious solution in this way.

Likewise, $Y \mapsto X^k$ is not injective on all of $\mathbb{Z}[X, Y]$ for any fixed k : it identifies Y with X^k . It becomes injective on a suitably bounded strip of X -exponents, but our accepting degrees have no computable

input-dependent bound. This blocks a naive fixed-stride flattening argument. It does not rule out a more sophisticated univariate compiler; general univariate linear-semiring undecidability was already established by Narendran [8].

12.3 Ordinary Diophantine representations

A finite polynomial tuple has a canonical integer encoding, and its validity can be checked effectively. That observation explains why the represented input language is computably enumerable. It does not turn all coefficientwise equalities into a fixed-arity ordinary polynomial system with unique auxiliary integer witnesses.

Applying an MRDP theorem to the resulting enumerable language establishes ordinary Diophantine representability, but the theorem does not by itself preserve the uniqueness of a chosen computation certificate. The additional witnesses used to arithmetize coefficient access, support, exponentiation, bounds, and verification must all be accounted for in any single-fold claim. This report supplies none of those steps as a hidden assumption. It therefore makes no claim to settle ordinary single-fold or finite-fold Diophantine representation questions.

The distinction is especially important when comparing with native integer-packing compilers in ProveIt: their paid typing and bounding conditions cannot be discarded merely because the intended polynomial witnesses in this report happen to have Boolean coefficients.

13 A parsimonious nondeterministic extension

The proof separates local consistency from deterministic uniqueness. This permits a precise extension that is stronger than an existence-only translation.

Let $\Delta \subseteq \mathcal{A}^4$ be a finite allowed-transition relation. A quadruple (ℓ, c, r, d) means that a cell with neighbourhood (ℓ, c, r) may produce d . Assume every neighbourhood has at least one allowed output and that the all-blank neighbourhood has exactly the output zero. A finite history is legal when each of its cell updates belongs to Δ . Acceptance and the light-cone convention are unchanged.

Theorem 13.1 (Parsimonious history correspondence). *Replace triple variables by $Z_{\ell c r d}$ for $(\ell, c, r, d) \in \Delta$, and define the four marginal forms by their respective coordinates. Keep all the equations of Section 5. The resulting system has $|\Delta| + s + 5$ unknown polynomials and $3s + 3$ equations. Its solutions are in bijection with finite legal configuration histories whose final row is their first singleton acceptance. Every solution has Boolean coefficients. For a fixed such history every auxiliary polynomial is unique.*

Proof. Each allowed quadruple has one left, central, right, and output symbol, so all total-mass identities remain valid. The clock, shape, and occupancy proofs are unchanged. Neighbour matching reconstructs the actual source windows, and vertical matching reconstructs each succeeding configuration. At every site the unique tile certifies that its output is allowed by Δ . Acceptance has the same first-singleton meaning.

Conversely, a legal configuration history specifies each quadruple at each site, together with its length, boundaries, final row, and accepting position. The same canonical construction supplies its unique solution. The two constructions are inverse. Determinism is needed only to conclude that there is at most one legal accepting history for a fixed input. $\square$

The objects counted here are *configuration histories*. Distinct syntactic transition labels with identical source and destination are not counted twice, because Δ is a set of quadruples. To count such labels separately,

they must be represented in the local state or tile alphabet. The provided executable compiler implements the deterministic special case; the nondeterministic statement is proved in the article but is not claimed as implemented in that package.

14 A complete worked certificate

14.1 The illustrative automaton

Use four symbols: 0 is blank, 1 is a right-moving signal, 2 is a stationary target, and 3 is an accepting signal. Define the rule in the following ordered form:

$$f(\ell, c, r) = \begin{cases} 3, & c = 3, \\ 3, & c = 2 \text{ and } \ell = 1, \\ 2, & c = 2 \text{ and } \ell \neq 1, \\ 1, & c = 0 \text{ and } \ell = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (14.1)$$

Take $\mathcal{F} = \{3\}$ and $w = (1, 0, 0, 0, 2)$, so $m = 5$. This automaton is a small test case, not a claimed universal rule. On the displayed input the signal reaches the target after four steps and becomes accepting there.

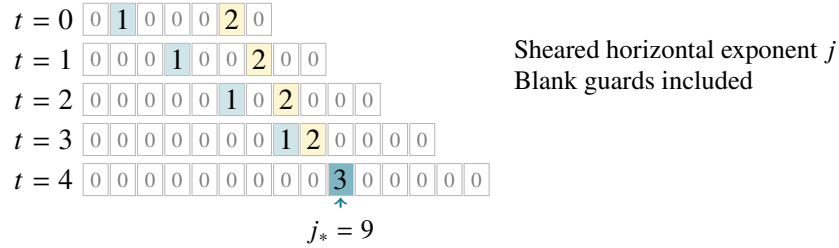


Figure 1: The source-centre rows for $t < 4$ and the terminal row at $t = 4$. The physical target remains at position 4; its encoded position is $4 + t + 1$, explaining the rightward shear.

14.2 All nonzero components

The five auxiliary polynomials are

$$\begin{aligned} D &= X^{14}Y^4, & Q &= X^6 + X^8Y + X^{10}Y^2 + X^{12}Y^3, \\ B &= Y^4, & V &= 1 + Y + Y^2 + Y^3, \\ J &= Y^4[9]_X. \end{aligned}$$

The final row is

$$T_3 = X^9Y^4, \quad T_0 = Y^4([15]_X - X^9), \quad T_1 = T_2 = 0.$$

The expression for T_0 is a polynomial with nonnegative coefficients: subtraction here simply removes the accepting position from the full zero–one row mask.

For compactness, the nonzero tile coefficients are listed by row as positions carrying a given triple. Any tile not listed is zero.

Time	Triple	Horizontal positions
0	001, 010, 100	0; 1; 2 respectively
	002, 020, 200	4; 5; 6 respectively
	000	3
1	001, 010, 100	2; 3; 4 respectively
	002, 020, 200	5; 6; 7 respectively
	000	0, 1, 8
2	001, 010, 102, 020, 200	4; 5; 6; 7; 8 respectively
	000	0, 1, 2, 3, 9, 10
3	001, 012, 120, 200	6; 7; 8; 9 respectively
	000	0, 1, 2, 3, 4, 5, 10, 11, 12

Each listed position j on row t contributes $X^j Y^t$ to that triple's polynomial. For example, the triple 120 at $(j, t) = (8, 3)$ outputs 3; its contribution is shifted by XY to the accepting terminal monomial $X^9 Y^4$. The boundary coefficients are included in this list, not suppressed as implicit blanks.

There are 73 polynomial unknowns and 15 reduced equations. The original pair-colour form has 24 equations, so the marginal projection removes nine rows. The unique witness has 74 nonzero coefficients across all components and degree maxima

$$\deg_X^{\max} = 14, \quad \deg_Y^{\max} = 4, \quad \deg^{\max} = 18.$$

These agree with Proposition 10.1, including $4^2 + 5 \cdot 4 + 5 \cdot 4 + 5 + 4 + 9 = 74$.

14.3 The supplied expanded quadratic

The file `results/example_system.json` gives every coefficient of every reduced linear residual for this example. The complete sum-of-squares polynomial has also been expanded exactly, not merely specified by a proposed expansion procedure. In `results/example_quadratic.json`, coefficients are polynomials in X, Y , and monomials in the unknown names are recorded explicitly. It contains 2,098 nonzero monomials in the unknowns, or 5,082 terms when their coordinate-polynomial coefficients are expanded as well. Its degree in the unknowns is two, and its maximum coefficient total degree in X, Y is twelve. Exact evaluation at the supplied certificate is zero.

14.4 Feature-compressed versions of the same example

For this four-symbol alphabet, choose either the two binary digits of a symbol or the single integer-valued feature $\chi(a) = a$. The complete counts are:

Form	Unknowns	Equations	Maximum coefficient magnitude
Pair colours	73	24	1
Symbol marginals	73	15	1
Binary features	73	12	1
One weighted feature	73	9	3

All four systems accept the identical tuple described above, and their *entire* natural-polynomial zero sets coincide by the proved reductions. The two feature systems and their fully expanded quadratics are supplied in the corresponding `example_binary_*` and `example_weighted_*` JSON files. Each feature quadratic has

2,704 nonzero unknown monomials and 12,364 terms after expanding coordinate coefficients. The coinciding term counts do not mean the two polynomials are equal: their coefficients differ. Both are exactly quadratic and vanish at the common certificate. Fewer linear rows have increased the expanded polynomial size in this example, so the row savings are not an arithmetic-operation optimum.

15 Reproducibility and validation

15.1 Executable scope

The package contains a dependency-free Python implementation using sparse integer-coefficient dictionaries. It accepts any total finite deterministic radius-one rule with quiescent blank, validates the alphabet and accepting set, and emits the entire system, including both clocks and both acceptance equations. It supports the reduced marginal form, the unreduced pair-colour form, and arbitrary validated injective-feature maps. Polynomial equality is checked coefficientwise; no floating-point arithmetic or numerical specialization is used.

The witness constructor takes a *supplied external horizon* for the purpose of generating or testing a candidate. The emitted mathematical system contains no such horizon. The constructor deliberately returns a candidate even when acceptance fails; the full residual check must then reject it. This separation tests the acceptance conditions rather than assuming them in the verifier.

Input rule tables are copied into immutable mappings. This prevents later mutation of a caller’s table from silently changing the meaning of an already compiled system.

15.2 Executed tests

Test family	Candidates or cases	Result
Ray-clock flows: coefficients 0, 1, 2 on a 3×3 box, three clock directions	59,049	11 admitted
All 128 quiescent binary rules, input lengths 1–4, horizons 0–4	19,200	1,280 accepted
Seeded rules on 3, 4, and 5 symbols, horizons 0–6	840	56 accepted
Independent triple assignments on one-row tableaux	16,384	32 structurally valid
Pair/marginal structural zero-set comparisons on those assignments	16,384	All agree
Worked-certificate coefficient changes, off-region insertions, and wrong horizons	373	All rejected
Expanded quadratic: canonical and altered tuples	8	Exact agreement

Table 5: Executed finite checks. “Accepted” means that the full residual system vanishes exactly when the independently evaluated first-singleton semantics holds. The pair/marginal line reuses the independent assignments, so it is not an additional disjoint sample.

The binary tests primarily stress the transition and boundary equations: with accepting alphabet $\{1\}$, a binary input already containing a one has an accepting event at time zero, while an all-zero input remains blank. The multistate tests and the four-step worked example provide the nontrivial later-acceptance cases. This limitation of the binary test set is recorded rather than hidden behind its case count.

For the independent tile enumeration, each of three sites chooses any of eight binary triples. Across

sixteen specified rules and two one-letter inputs, all $16 \cdot 2 \cdot 8^3 = 16,384$ assignments are checked. Exactly one structural tableau per rule/input pair is admitted; its triples are $(0, 0, w_0)$, $(0, w_0, 0)$, and $(w_0, 0, 0)$. The top row is formed from the chosen outputs, and the actual linear equations must recover the valid source and adjacency. These are not merely rechecks of the canonical constructor’s own tile choices.

The ray-clock enumeration allows coefficients equal to two, so it also checks that nonnegative multiplicities are rejected rather than assuming Boolean coefficients at input. The mutation tests delete or duplicate every nonzero coefficient of the worked witness, insert coefficients outside the region in every witness variable, and test six incorrect horizons. Matrix-invariance tests compare different input lengths and verify the stated coefficient sparsity and degree bounds.

The separate feature-verification run compares both binary and weighted forms with the direct symbol system on 7,168 binary rule/input/horizon cases and 288 seeded multistate cases. These are 7,456 cases with two feature-system checks per case, not an additional claim of exhaustive validation of all polynomials. For each feature form, all 367 coefficient mutations of the worked tuple are rejected; two exact expanded-quadratic evaluations agree with the linear residual sum of squares. The run also checks the zero-feature singleton alphabet, rejects three invalid feature maps, and verifies matrix invariance for two feature forms. Its multistate seed is 20261003; this is an arbitrary integer seed, not a report date. The exact counts are recorded in `results/feature_results.json`.

All these computations passed. They are finite validation, not a formal proof of the infinite claims and not an independent proof-assistant audit. The universal-machine reduction and the nondeterministic theorem rest on the written proofs, not on an assertion that the finite examples establish universality.

15.3 Files and commands

From the package root, with Python 3.10 or later, run

```
python code/verification.py
python code/export_quadratic.py
python code/feature_verification.py
pdflatex -interaction=nonstopmode -halt-on-error \
    unique_polynomial_histories.tex
pdflatex -interaction=nonstopmode -halt-on-error \
    unique_polynomial_histories.tex
```

The Python commands rewrite the JSON results and the complete worked example. The manuscript is a single self-contained TeX file with an embedded bibliography and diagram; it does not depend on downloaded data or on the repository being present locally. LaTeX requires the standard packages listed in its preamble.

[write] *Shipped names*. In this report the four programs are `code/06-polynomial-histories-*.py` and the recorded results data `data/06-polynomial-histories-*.json`; the manuscript is `Part I of article.tex`. Two delivered exports, `results/example_binary_quadratic.json` and `results/example_weighted_quadratic.json` (1,026,498 and 1,031,059 bytes), are not shipped: `feature_verification.py` regenerates them byte for byte, and the README says how. The commands above rewrite the recorded results in place; run them only on a copy with the delivered layout. The research programme’s review [4] reproduced all three commands and found an input-validation defect of the compiler (Section 1.4); the shipped `code/06-polynomial-histories-polynomial_histories.py` is the unpatched original.

16 Further research questions and concrete next steps

The following questions are proposed extensions of this construction. They are not all asserted to be previously published open problems; precise priority and existing special cases should be checked before a separate novelty claim is made.

16.1 A numerical universal matrix with a verified source table

Choose a particular universal Turing-machine table with documented tape, halt, and input conventions. Generate the cellular rule by Proposition 9.1, then emit its fixed matrix and input encoder. A useful deliverable would contain the table, the simulation theorem, all matrix entries, and a certificate checker, together with a numerical value for $s^3 + s + 5$ and the chosen row count (in particular nine in the weighted form). This would replace the generic universality instantiation by a fully auditable concrete object. Minimizing s is secondary to obtaining a correct ordinary-input interface.

[write] The same target is posed for Part II's eleven identities (Section 33.4) and for Part III's fixed tagged equation (Section 49.1, *A numerical universal table*); no source instantiates one.

16.2 Equation counts versus coefficient size

The feature theorem gives an explicit tradeoff between the number of rows and coefficient magnitude, without changing the witness tuple. Can the nine-row weighted form be reduced further while retaining the fixed matrix, canonical geometry, and full-tuple zero-set equivalence? For unit coefficients, is there a substantially better alternative to $6 + 3\lceil \log_2 s \rceil$ rows? Any lower-bound problem must specify the allowed fixed coordinate shifts and whether clock, geometry, and acceptance constraints may be combined. The counting bound on feature-vector length is not a lower bound for unrestricted linear systems. A finite-family Pareto analysis should also count matrix density and expanded quadratic size, which need not decrease with the number of rows.

16.3 A one-coordinate compiler with the same uniqueness guarantees

General linear-system undecidability over $\mathbb{N}[X]$ is known. The concrete question here is whether one can retain a fixed matrix, a transparent word-input map, Boolean-coefficient full witnesses, and first-history single-foldness using only one coordinate indeterminate. A successful construction must avoid the collisions of a fixed Kronecker substitution and must prove uniqueness of any width or stride auxiliaries. This is a sharpening target, not a proposed proof of a one-variable undecidability threshold.

[write] *Answered in Part II (note of 2 October 2026)*. In the form asked here the answer is negative: coefficient-bounded univariate affine-linear systems are decidable (Theorem 21.1), so no universal univariate linear compiler can guarantee even one Boolean-coefficient full witness per accepted input (Corollary 21.3, with bound $B_x = 1$). Eleven identities of degree two with one shared stride keep the full unique Boolean history (Theorem 20.2), in bijection with this Part's witnesses (Theorem 25.2). The research programme's review of source 15 adds a ten-identity projection (note at Section 33.1).

16.4 Packing with an exact auxiliary-fibre ledger

Can the canonical semiring witness be transferred to an ordinary integer or exponential-Diophantine representation with a proved bijection, or at least a controlled finite fibre, over the full auxiliary tuple? A

useful first milestone is a bounded version with explicit carry bounds and a proof that every integer solution has a unique digit decomposition and satisfies the polynomial coefficient equations. The unbounded extension must confront Theorem 10.2; it cannot assume a computable input-only degree limit.

[write] Part II asks the same for its stride witnesses (Section 33.6) and Part III for its sparse polynomial checker (Section 49.2, *An explicit integer bridge with controlled fibers*). The obstruction named here is, for all three, the bounded-search argument of Proposition `cdc:bd:prop:nobound` in [1].

16.5 Parsimonious branching compilers

Implement Theorem 13.1 with labelled transition choices and prove a root-counting correspondence for a carefully specified nondeterministic machine semantics. Distinguish configuration histories, labelled execution paths, and different schedules of independent actions. These choices determine whether the target should count traces, events, or equivalence classes. The existing equations already identify the point at which local multiplicity becomes genuine computational branching.

16.6 Smaller canonical regions

The fixed expanding light cone is simple but can be much wider than the actually visited tape interval. Can one use the minimal visited interval while retaining unique nonnegative linear witnesses and a fixed matrix? A naive freely chosen rectangle introduces padding ambiguity. Any improvement needs a canonical extremum condition, with all witnesses uniquely determined, rather than an existentially generous space bound. A successful version could substantially reduce support size even when it does not reduce the number of polynomial variables.

16.7 Output-preserving and compositional interfaces

The terminal polynomials already contain the whole final tape. A next step is to specify a canonical output projection and prove that composing two compilers preserves a bijection of complete witnesses. Input length, head position, blank padding, and return-to-origin conventions must be part of the interface. A composition theorem would make the normal form useful as a library component rather than only as a universal acceptance reduction.

16.8 Kernel-checked support and reconstruction proofs

A practical Lean or Rocq development should begin with finitely supported maps $\mathbb{N}^2 \rightarrow \mathbb{N}$, the coefficientwise ray-clock lemma, and the implication “sum of natural coefficients equals one $\Rightarrow$ unique occupied label.” Next come the two shape identities, the missing blank marginal row, and induction recovering the cellular evolution. Only after these parts should one formalize the Turing-machine bridge and the exact count ledger. The proof naturally separates algebraic identities, nonnegative coefficient reasoning, finite-state semantics, and the universality input reduction. No unbounded numerical search is required for these lemmas.

[write] Parts II and III propose the same formalization programme for their constructions (Sections 33.9 and 49.4, *A fully verified compiler pipeline*); nothing of it exists in the repository at this write.

17 Conclusion

The central construction answers a precise design question: a deterministic finite computation can have a fixed-arity, affine-linear certificate over $\mathbb{N}[X, Y]$ in which the *whole* witness, including clocks, boundaries, and accepting-position marker, is unique. Nonnegative conservation forces unit occupancy and a canonical light cone. Marginal matching then suffices to reconstruct the local windows, and first acceptance removes arbitrary post-acceptance extensions. Injective features compress the very same witness relation to $6 + 3\lceil \log_2 s \rceil$ unit-coefficient rows, or to nine rows when symbol-sized coefficients are allowed. The reduction is justified by already-forced one-hot occupancy, not by a generally false claim that weighted marginals determine arbitrary multisets.

The resulting fixed-matrix problem is Σ_1^0 -complete on an unambiguous input family, even though every actual witness coefficient is zero or one. The exact degree equals a linear function of the running time, and no computable input-only bound on that degree is possible. A single quadratic polynomial-semiring equation preserves all of these properties; a bounded ordinary quadratic family preserves them only with arity growing with the external bound.

The useful mathematical advance of this manuscript is the explicit normal form and its witness-preserving reductions. Its limitations are just as concrete: the unknowns are polynomials, the universal numerical rule is not instantiated in the package, the main theorem is not yet kernel-checked, and historical priority for the combined refinement remains to be investigated. These boundaries leave a well-defined path from the present proofs to a verified, compositional compiler and to carefully audited ordinary-arithmetic packing experiments.

18 Notation and proof dependency ledger (appendix of source 06)

Symbol	Meaning
S	The semiring $\mathbb{N}[X, Y]$ of finite nonnegative-coefficient polynomials.
X, Y	Fixed coordinate indeterminates, not numeric witness variables.
$\mathcal{A}, s, 0$	Cellular alphabet, its size, and its quiescent blank.
$f, \mathcal{F}$	Radius-one local rule and accepting alphabet, with $0 \notin \mathcal{F}$.
w, m	Supplied finite input word and its length, $m \geq 1$.
h	Acceptance time recovered from a solution, not supplied to the system.
w_t	Transition-row width $m + 2t + 2$.
$Z_{\ell cr}$	Unknown polynomial indicating sites carrying a source triple.
T_a	Unknown polynomial indicating terminal-row occurrences of symbol a .
D, Q	Endpoint and prefix of the sheared right-boundary clock.
B, V	Endpoint and prefix of the vertical clock.
J	Unique horizontal prefix ending immediately before the accepting site.
L_a, C_a, R_a, O_a	Linear forms for left, centre, right, and output symbols; not additional unknowns.
I_a	Supplied input symbol-indicator polynomial.
G	Linear form $V + B + Q + D$ for both blank source guards.
H, K	Derived total tile and terminal polynomials; not additional unknowns.
j_*	Encoded horizontal position of the unique accepting top cell.

The proof order is

$$\text{ray clocks} \implies \text{clock synchronization and shape} \implies \text{unit occupancy},$$

followed by

$$\text{marginal adjacency} \implies \text{true windows} \implies \text{true evolution} \implies \text{first acceptance and uniqueness}.$$

Boolean coefficient bounds are conclusions of this chain. They are never used to justify the preceding shape equations. This ordering is part of the soundness argument and should be preserved in any optimization or formalization.

19 Source and trust boundaries (appendix of source 06)

The repository review was tied to commit

e58b724c25bd34533b7a5834cfcbe873dfa01288.

The root and Hilbert-tenth README files were inspected, together with repository metadata and targeted searches. This is not a claim to have audited every proof or every imported report in the repository.

The external literature establishes the pre-existing context and supplies classical universality; it is not substituted for the displayed proofs of shape, uniqueness, projection, or exact resource bounds. The code checks finite instances and an explicit expanded polynomial. No Lean executable was run, and no theorem in this manuscript is described as machine-checked in a proof assistant.

The included code and results are original supporting artifacts for this report. The reference papers are linked bibliographically, not bundled as redistributed PDFs. Build intermediates and font files are not part of the archive.

Part II

The price of one coordinate: self-sizing quadratic certificates over $\mathbb{N}[X]$ (source 15)

[write] Part II prints batch-79 manuscript 15, *The Price of One Coordinate: Self-sizing quadratic certificates for universal computation*, under the kicker “Computability / Polynomial semirings”, “Research report prepared for Vladimir Reshetnikov”, 2 October 2026 (the manuscript names no assistant), pinned to repository commit 44983ed7e. Its labels carry the prefix pwh:oc: and its bibliography keys the prefix oc: . Its “preceding report” or “precursor” is source 06, printed as Part I; the two-coordinate system of its Section 22 is Part I’s nine-row weighted form (Corollary 8.2). Statements that re-prove Part I’s are marked where they occur, and the univariate counterparts of Part I’s results carry pointers. Its abstract and status statement follow; its two appendices are Sections 35 and 36. The research programme reviewed this manuscript after it was placed here [6] (Section 1.4).

Abstract (source 15). A preceding report asks whether a canonical cellular-computation compiler can be made affine-linear over $\mathbb{N}[X]$, while retaining Boolean coefficients in every witness and uniqueness of the complete accepting history. We prove that this proposed combination is impossible: coefficient-bounded univariate linear systems are decidable by a finite-state construction, even when the matrix varies with the input and uniqueness is not required.

We then give an explicit replacement. A deterministic radius-one cellular automaton with $s \geq 2$ symbols admits a compiler with $s^3 + s + 7$ polynomial unknowns and eleven equations of degree at most two. All nonlinear products share one stride variable. Two equations force that variable to be the monomial X^W and calibrate W to the exact terminal width $m + 2h + 2$, where m is the supplied input length and h is recovered from the solution. No time, width, degree, or coefficient bound is supplied.

There is exactly one full solution when the evolution has a first singleton acceptance, and none otherwise. Every solution coefficient is zero or one. We prove a bijection with the earlier two-coordinate witnesses, exact degree and support formulas, fixed-template Σ_1^0 -completeness, a same-witness quartic single equation, and a bounded ordinary-integer export. Executable compilers, an independent finite-state solver, negative examples, and an expanded worked quartic accompany the proofs.

Scope and status. This is an unrefereed mathematical manuscript with exact finite computational checks, not a Lean or Rocq formalization. The result concerns polynomial unknowns over $\mathbb{N}[X]$, not ordinary single-fold Diophantine representations over $\mathbb{N}$. Historical priority for the combined normal form has not been established.

20 The question, its obstruction, and its replacement

The computational-substrate program in ProveIt has developed many ways of turning executions into arithmetic certificates. A central issue is not merely whether a certificate exists. One also wants to know

what every solution means, whether the auxiliaries introduce artificial multiplicity, and whether an apparently fixed compiler hides an externally selected running-time or space bound. The repository's Hilbert-tenth overview makes these distinctions explicit and separately records its natural-number 75-operation certificate and 87-operation universal-polynomial bounds [20]. The present work changes neither bound.

The preceding manuscript *Unique Histories from Linear Equations* [19] gives a canonical compiler over $\mathbb{N}[X, Y]$. Its two indeterminates retain space and time. A proposed extension in its further research section asks for one coordinate while preserving a fixed *linear* matrix, a transparent word-input map, Boolean-coefficient full witnesses, and first-history uniqueness. This is a precise question about a representation, rather than a claim that undecidability over $\mathbb{N}[X]$ is unknown.

[write] The preceding manuscript is Part I of this report; the question is its Section 16.3.

This report answers that question negatively and supplies a degree-two replacement. The negative statement is stronger than required: even a computably varying matrix cannot represent an undecidable language if every solvable instance has some witness with a computably bounded coefficient height. The positive statement keeps the desirable semantic properties and uses only a single shared nonlinear parameter.

20.1 Two main theorems

All natural numbers in this report include zero. A polynomial is *Boolean-coefficient* when every coefficient is zero or one; arithmetic still takes place in the ordinary semiring, not in a Boolean ring.

Theorem 20.1 (The univariate linear obstruction). *Given an affine-linear system $A(X)u(X) = b(X)$ over $\mathbb{N}[X]$ and a natural number B_0 , it is decidable whether there is a solution all of whose coefficients are at most B_0 . If the system has n unknown polynomials, $\deg A \leq d$, and $\deg b \leq M$ with $M \geq 0$, then any solvable bounded instance has a bounded solution with*

$$\max_i \deg u_i \leq M + (B_0 + 1)^{nd}.$$

Consequently, no effective reduction of an undecidable language to univariate affine-linear systems can guarantee a computable bound on the coefficients of some solution to every solvable instance.

The proof is given in Section 21. It is an elementary finite-state argument and should not be confused with a decision procedure for unrestricted univariate polynomial-semiring linear systems.

For the positive theorem, a *first singleton acceptance* means that the first configuration containing any designated accepting symbol contains exactly one such cell. This holds automatically in the usual one-head simulation of a deterministic accepting Turing machine. It is a genuine restriction for an arbitrary cellular automaton.

Theorem 20.2 (Self-sizing univariate quadratic compiler). *Fix a deterministic radius-one cellular automaton with alphabet $\mathcal{A} = \{0, \dots, s-1\}$, $s \geq 2$, quiescent blank 0, total local rule $f : \mathcal{A}^3 \rightarrow \mathcal{A}$, and accepting set $\mathcal{F} \subseteq \mathcal{A} \setminus \{0\}$. For each nonempty finite input word w of length m , there is an effective system $\mathcal{E}_{f, \mathcal{F}}(w)$ of eleven polynomial identities with the following properties.*

- (i) *There are exactly $s^3 + s + 7$ unknowns in $\mathbb{N}[X]$. Every identity has degree at most two in those unknowns. Every nonlinear monomial contains one distinguished stride unknown S .*
- (ii) *The nonconstant-in-the-unknowns terms are fixed by f and $\mathcal{F}$. Their coefficients lie in $\mathbb{Z}[X]$, have coordinate degree at most two, and have coefficient magnitude at most $s-1$. Only the constant terms load the input word.*

(iii) *The system has a solution precisely when the evolution from w , with blank exterior, has a first singleton acceptance. In that case the complete polynomial witness tuple is unique, and all its coefficients are zero or one.*

(iv) *If the acceptance time is h , then*

$$S = X^W, \quad W = m + 2h + 2,$$

and the maximum degree among all witness components is

$$D_{\max} = \max\{W, W(h + 1) - 1\}.$$

The equations contain no externally supplied running time, stride, polynomial degree bound, occupancy condition, or coefficient bound.

We list the whole system in Section 23 and prove all-solution soundness, completeness, and uniqueness in Section 24. The theorem is not just a correct recipe for generating some valid certificates: it characterizes the entire natural-polynomial zero set.

20.2 What the degree boundary means

A fixed universal machine, translated to a one-head cellular automaton, gives a fixed instance of Theorem 20.2 whose word-input solvability problem is Σ_1^0 -complete. Theorem 20.1 rules out the analogous affine-linear compiler with computably bounded witness coefficients. Thus degree two is both sufficient and necessary *for systems in this coefficient-bounded representation class*. It is not a claim that unrestricted univariate linear systems are decidable, nor that eleven equations or $s^3 + s + 7$ variables are optimal.

The previously available two-coordinate linear construction and the present one-coordinate quadratic construction illustrate two different ways to retain unbounded computational geometry. A second coordinate permits fixed linear shifts in time. With one coordinate, the present construction pays for a variable shift by multiplication with an unknown monomial.

20.3 Literature and contribution ledger

Narendran proved undecidability of systems of linear equations over polynomial semirings, already over $\mathbb{N}[X]$, in 1996 [13]. The unbounded coefficient heights permitted there are essential to the contrast with our obstruction. Dong’s study gives a different positive result: polynomial-time decidability for a *single homogeneous* linear equation in which every unknown polynomial is required to be nonzero [14]. Neither result settles the canonical, automatically Boolean, full-witness formulation considered here.

Finite-state reasoning about finitely supported objects belongs to the classical automata tradition exemplified by Büchi [15]. Our coefficient-window construction is proved directly rather than attributed as a new general automata principle. Encoding computations by translated local constraints also has close precedents, including the Turing-machine tiling constructions of Lohrey and Steinberg [16]. Kronecker substitution is a standard way of encoding multivariate polynomial data by separated univariate exponent blocks; see, for example, Harvey [17]. None of these general techniques is claimed as an invention of this report.

The particular contribution is the following combination: an explicit negative answer to the precursor’s one-coordinate linear design question; a positive-stride polynomial equation; the exact calibration $SB = XD$; eleven complete low-degree identities; reconstruction of every solution; uniqueness of all stride, clock,

boundary, and position auxiliaries; and an exact resource ledger. The proofs below establish these claims without assuming an unexamined universal equation or an unproved repository lemma. A broader literature review would be needed to establish historical priority for the combined normal form.

21 Coefficient-bounded univariate linear systems are finite-state

21.1 Definitions and the finite-memory recurrence

Let $n, r \geq 1$ and consider

$$A(X)u(X) = b(X), \quad A(X) = \sum_{k=0}^d A_k X^k, \quad (21.1)$$

where $A_k \in \mathbb{Z}^{r \times n}$, $b \in \mathbb{Z}[X]^r$, and the unknown vector u lies in $\mathbb{N}[X]^n$. Zero leading matrices are allowed. Choose $M \geq 0$ with $\deg b \leq M$; the zero right-hand side may use $M = 0$. For a polynomial p , write

$$\text{ht}(p) = \max\{p_t : t \geq 0\}, \quad \text{ht}(0) = 0.$$

The bound B_0 is a bound on *all coefficients*, not on polynomial degrees or on values at selected numerical arguments.

Write $u_t \in \{0, \dots, B_0\}^n$ for the coefficient vector of X^t in u , put $u_t = 0$ for $t < 0$, and let b_t be the coefficient vector of b . Comparing coefficients in (21.1) gives

$$\sum_{k=0}^d A_k u_{t-k} = b_t \quad (t \geq 0). \quad (21.2)$$

Each test depends only on the next coefficient vector and the previous d coefficient vectors. There is no unbounded carry: these are identities of formal polynomials, not equalities after evaluation at an integer base.

21.2 Decision procedure and a uniform degree bound

Theorem 21.1 (Finite-state bounded-height theorem). *The bounded-height solvability problem for (21.1) is decidable. Set*

$$a = (B_0 + 1)^n, \quad N = a^d = (B_0 + 1)^{nd}.$$

If a bounded-height solution exists, one exists with $\max_i \deg u_i \leq M + N$. This includes $d = 0$ and $B_0 = 0$.

Proof. The coefficient alphabet is the finite set $C = \{0, \dots, B_0\}^n$, of size a . A memory state is the ordered tuple $(u_{t-1}, \dots, u_{t-d}) \in C^d$; there are N states. For $d = 0$, there is one empty state. The initial memory is all zero. A candidate symbol $v \in C$ is a legal next symbol at time t precisely when

$$A_0 v + \sum_{k=1}^d A_k u_{t-k} = b_t.$$

The new memory is obtained by prepending v and dropping the oldest vector. All tests use finite exact integer arithmetic.

First compute the set of memories reachable after reading positions $0, \dots, M$. During this finite prefix, the required right-hand vector may vary. After position M , the right-hand vector is always zero, so legal

transitions form a fixed directed graph on the same N memories. Call a memory *drainable* when a path in this zero-right-hand graph reaches the all-zero memory. Reachability in this finite graph is decidable. The original problem is solvable exactly when one of the reachable prefix memories is drainable.

For the forward direction, any genuine polynomial solution is eventually zero. After its last nonzero coefficient, d further zero coefficient vectors flush the memory to zero. Its coefficient sequence therefore supplies the required prefix and draining path. Conversely, a valid prefix followed by a draining path can be followed forever by zeros. All prefix and homogeneous equations then hold, and the resulting coefficient sequence has finite support. It is the desired bounded-height polynomial solution.

If a draining path exists, a shortest one has no repeated state and has length at most $N - 1$. Keeping a prefix that reaches its starting memory, then following that path and appending zeros, produces a solution with no nonzero coefficient beyond position $M + N - 1$; the stated bound $M + N$ is a harmless uniform relaxation. For $d = 0$, an admissible prefix can already be followed by zeros. For $B_0 = 0$, the only possible sequence is zero. Both cases are covered by the same argument. $\square$

This is a finite-state algorithm, not a polynomial-time claim for arbitrary encodings of the parameters. The state space may be enormous, and an exponent M given sparsely may itself be large. A direct implementation examines at most N memory states and a next symbols per state after the input prefix. The executable solver in the package follows precisely this construction and returns a witness, not only a Boolean decision.

Proposition 21.2 (Regularity refinement for a fixed matrix). *For fixed A and B_0 , the coefficient words of right-hand sides b that admit a bounded-height solution form a regular language, using least-degree coefficients first and allowing trailing zero vectors.*

Proof. Label a transition, specified by a memory and a next candidate coefficient vector, with the right-hand vector computed by the left side of (21.2). Only finitely many labels occur. Start at the zero memory and make the drainable memories accepting. A finite labelled path then recognizes exactly a right-hand prefix that can be completed by a zero-labelled tail and finally by zero coefficient vectors. The proof of Theorem 21.1 gives the equivalence with polynomial solvability. Right-hand letters outside the finite transition-label alphabet are rejected. $\square$

21.3 The impossibility result, with its exact quantifiers

Corollary 21.3 (No computably bounded-height linear universal compiler). *Let L be an undecidable language. There do not exist total computable maps $x \mapsto (A_x, b_x)$ and $x \mapsto B_x \in \mathbb{N}$ such that both*

$$\begin{aligned} x \in L &\iff \exists u \in \mathbb{N}[X]^{n_x} A_x u = b_x, \\ \exists u A_x u = b_x &\implies \exists v (A_x v = b_x \text{ and } \text{ht}(v_i) \leq B_x \text{ for all } i) \end{aligned}$$

hold for every input x .

Proof. Compute A_x, b_x, B_x and decide the bounded-height problem by Theorem 21.1. The second hypothesis makes bounded solvability equivalent to unrestricted solvability for this input family; the first then decides L , a contradiction. $\square$

In particular, choosing $B_x = 1$ rules out the precursor's proposed univariate affine-linear compiler with Boolean full witnesses. The proof does not need a fixed matrix, unique witnesses, Boolean coefficients in *every* solution, or a uniform number of unknowns. It only needs a computable bound for *some* solution whenever one exists. Thus a universal linear compiler over $\mathbb{N}[X]$, of the sort compatible with Narendran's

theorem, must have no total computable input-only upper bound on the minimum coefficient height of its accepting witnesses.

This establishes Theorem 20.1. It also identifies exactly what must change in the positive construction: either coefficient height must become unbounded in a noncomputably controlled way, a second coordinate must remain, or genuine nonlinearity must be introduced. We choose the last option.

22 Cellular semantics and the two-coordinate starting point

22.1 A guarded expanding computation region

Let $w = w_0 \cdots w_{m-1}$, with $m \geq 1$, and define the initial configuration on $\mathbb{Z}$ by

$$c_0(i) = \begin{cases} w_i, & 0 \leq i < m, \\ 0, & \text{otherwise.} \end{cases}$$

The evolution is

$$c_{t+1}(i) = f(c_t(i-1), c_t(i), c_t(i+1)). \quad (22.1)$$

Quiescence, $f(0, 0, 0) = 0$, implies by induction that

$$c_t(i) = 0 \quad \text{outside } [-t, m-1+t]. \quad (22.2)$$

At transition time t , include centres $i = -t-1, \dots, m+t$ and encode the centre by

$$j = i + t + 1, \quad 0 \leq j < w_t, \quad w_t = m + 2t + 2.$$

The first and last centre symbols are blank guards. Their outputs may become nonblank on the next step, while two new blank source guards are added. A cell at fixed physical position moves from encoded position j to position $j+1$ in the next row.

22.2 Polynomials and linear forms

For each triple $(\ell, c, r) \in \mathcal{A}^3$, use a tile polynomial $Z_{\ell cr}$. For each $a \in \mathcal{A}$, use a terminal polynomial T_a . We abbreviate the following linear forms:

$$H = \sum_{\ell, c, r} Z_{\ell cr}, \quad K = \sum_a T_a, \quad (22.3)$$

$$L = \sum_{\ell, c, r} \ell Z_{\ell cr}, \quad C = \sum_{\ell, c, r} c Z_{\ell cr}, \quad (22.4)$$

$$R = \sum_{\ell, c, r} r Z_{\ell cr}, \quad O = \sum_{\ell, c, r} f(\ell, c, r) Z_{\ell cr}, \quad (22.5)$$

$$\hat{T} = \sum_a a T_a, \quad I = \sum_{j=0}^{m-1} w_j X^j. \quad (22.6)$$

The alphabet labels themselves provide an injective numerical feature, with blank label zero. Also define

$$C_{\mathcal{F}} = \sum_{c \in \mathcal{F}} Z_{\ell cr}, \quad T_{\mathcal{F}} = \sum_{a \in \mathcal{F}} T_a, \quad (22.7)$$

where unspecified indices range freely. These expressions are abbreviations, not new unknowns. Write $[k]_X = 1 + X + \dots + X^{k-1}$ for $k \geq 1$, and $[0]_X = 0$.

The precursor [19] uses five further unknown polynomials D, Q, B, V, J and $G = V + B + Q + D$. Its weighted two-coordinate compiler is

$$\begin{aligned} D + Q &= X^{m+1} + X^2 Y Q, & B + V &= 1 + Y V, \\ XH + V &= H + XQ, & H + K &= X[m]_X + XYH + G, \\ XC &= L, & XR &= C, \\ C + \widehat{T} &= XI + XY O, & C_{\mathcal{F}} &= 0, \\ T_{\mathcal{F}} + J &= B + XJ. \end{aligned}$$

Here the unknowns lie in $\mathbb{N}[X, Y]$, and Y is a fixed indeterminate. The new system replaces Y by an unknown $S \in \mathbb{N}[X]$ and supplies two additional equations that make this replacement exact and canonical. We reprove the required reconstruction statements below; the precursor's conclusions are not treated as a black box.

22.3 Why a fixed substitution cannot be enough

For any fixed positive integer W , the homomorphism

$$\Phi_W : \mathbb{Z}[X, Y] \longrightarrow \mathbb{Z}[X], \quad \Phi_W(X) = X, \quad \Phi_W(Y) = X^W$$

is not injective: Y and X^W have the same image. It becomes injective on polynomials whose X -degree is strictly less than W , since then $j + Wt$ determines (j, t) by quotient and remainder.

The problem is that the required horizontal width depends on how long the computation runs. For a universal source, there cannot be a total computable input-only bound on that running time. Simply choosing a large computable stride does not solve the problem. Allowing an arbitrary sufficiently large stride would instead introduce infinitely many auxiliary witnesses. The new calibration chooses the exact stride from the certificate's own clocks.

23 The complete eleven-equation compiler

23.1 Unknowns, domain, and every residual

All unknowns now range over $\mathbb{N}[X]$. In addition to $Z_{\ell cr}$ and T_a , introduce

$$D, Q, B, V, J, S, E.$$

There are $s^3 + s + 7$ unknown polynomials. Continue to abbreviate $G = V + B + Q + D$, and retain the linear forms (22.3)–(22.7). The complete system is

$$D + Q = X^{m+1} + X^2SQ, \tag{1}$$

$$B + V = 1 + SV, \tag{2}$$

$$XH + V = H + XQ, \tag{3}$$

$$H + K = X[m]_X + XSH + G, \tag{4}$$

$$XC = L, \tag{5}$$

$$XR = C, \tag{6}$$

$$C + \widehat{T} = XI + XSO, \tag{7}$$

$$C_{\mathcal{F}} = 0, \tag{8}$$

$$T_{\mathcal{F}} + J = B + XJ, \tag{9}$$

$$S + E = X + XE, \tag{10}$$

$$SB = XD. \tag{11}$$

Equalities are identities at every polynomial coefficient. There are no implicit inequalities, no Booleanity equations, no support predicates, and no hidden constraints on the degrees of any unknowns.

The distinction between E and V is important. The former is a one-coordinate prefix that forces a positive stride. The latter is the clock prefix with one monomial per transition time. The calibration links the final points of two differently sloped clocks, not their numerical values at a chosen base.

23.2 Literal algebraic ledger

Proposition 23.1 (Fixed-template and degree bounds). *System (1)–(11) has eleven equations of unknown-degree at most two. Its quadratic dependence is confined to multiplication of S by the five linear forms*

$$Q, V, H, O, B.$$

After all abbreviations are expanded, every nonconstant-in-the-unknowns coefficient has X -degree at most two and integer coefficient magnitude at most $s - 1$. These terms depend only on f and $\mathcal{F}$. Input-dependent constant terms are exactly those formed from X^{m+1} , $X[m]_X$, and XI .

Proof. The only products of unknowns in the displayed system occur in (1), (2), (4), (7), and (11). They have the stated form. Each of H, O is a linear combination of tile unknowns, so expansion does not increase unknown-degree beyond two. The largest coordinate shift multiplying an unknown monomial is X^2 , in (1). All weights in $L, C, R, O, \widehat{T}, I$ lie between zero and $s - 1$; all other nonzero coefficients have magnitude one. Any two appearances of the same tile in a feature row have different coordinate powers or different unknown monomials, so the expansion does not accumulate larger integer coefficients. The input occurs only in the three supplied constant polynomials stated above. $\square$

This ledger counts polynomial identities and polynomial unknowns. The five factored products are not five ordinary-integer arithmetic operations: evaluating a polynomial product, or forming a dense linear form, has its own cost. No comparison with the repository's 87-operation ordinary universal polynomial is implicit.

23.3 A unit-coefficient variant

For an injective map $\chi : \mathcal{A} \rightarrow \{0, \dots, B_0\}^{d_0}$ with $\chi(0) = 0$, replace each numerical symbol weight in the three feature equations by each coordinate of χ . Retain all other equations unchanged. There are then

$$8 + 3d_0 \quad \text{equations on the same } s^3 + s + 7 \text{ unknowns.} \quad (23.1)$$

The proof in the next section applies coordinatewise after occupancy has been established. It proves that this variant has exactly the same full solution tuples as the weighted compiler.

Binary digits give $8 + 3\lceil \log_2 s \rceil$ equations with unit coefficients. Base- b digits give $8 + 3\lceil \log_b s \rceil$ equations with coefficient magnitude at most $b - 1$. The eleven-row form takes the single feature $\chi(a) = a$. No optimality assertion is made for these row counts; fewer, denser equations can have more terms after squaring.

24 All-solution soundness and canonical reconstruction

The proof order matters. We first derive monomial clocks, their common height, and the exact stride. Only then can disjoint blocks and unit occupancy be inferred. Weighted feature equalities are used only after there is a single symbol, rather than an arbitrary multiset, on each side.

24.1 Finite nonnegative flow on a ray

Lemma 24.1 (Univariate ray lemma). [write] *[The one-variable case ($S = X^a$, $M = X^b$) of Part I's Lemma 4.1; source 15's statement and proof are kept because the lemmas of this Part cite them.] Let $a \geq 0$, $b \geq 1$, and let $U, V \in \mathbb{N}[X]$. Then*

$$U + V = X^a + X^b V \quad (24.1)$$

holds if and only if, for a unique $k \in \mathbb{N}$,

$$U = X^{a+bk}, \quad V = X^a \sum_{t=0}^{k-1} X^{bt}.$$

In particular, $\text{wt}(U) = 1$ and $\text{wt}(V) = k$, where $\text{wt}(P) = P(1)$.

Proof. Evaluation at $X = 1$ gives $\text{wt}(U) = 1$, so U is a monomial with coefficient one. Regard V_e as a nonnegative integer flow along the directed edge $e \rightarrow e + b$ in $\mathbb{N}$. Equation (24.1) is conservation with one source at a and one sink at the exponent of U .

To make the flow argument explicit, partition $\mathbb{N}$ into the residue chains modulo b . On any chain not containing a , coefficient induction from its least exponent shows that no positive outgoing flow can arise; a sink on such a chain would demand a negative coefficient. On the chain containing a , the same induction gives zero flow before a , then unit flow after the source until the sink is reached. Finite support prevents this unit flow from continuing forever. After the sink the flow is zero. The sink therefore lies at $a + bk$ for some $k \geq 0$, with exactly the stated prefix. Conversely, these expressions telescope to (24.1). Their exponents and masses establish uniqueness. $\square$

The lemma uses both nonnegative coefficients and finite support. Neither is an optional side condition. It is a polynomial-semiring power-forcing mechanism of the kind already present in earlier work [13].

24.2 Positive stride and synchronization

Lemma 24.2 (Forced monomial stride and clock endpoints). *Every solution of the full system has integers $W \geq 1$ and $h, k \geq 0$ such that*

$$S = X^W, \quad E = X + X^2 + \cdots + X^{W-1}, \quad (24.2)$$

$$D = X^{m+1+(W+2)h}, \quad Q = \sum_{t=0}^{h-1} X^{m+1+(W+2)t}, \quad (24.3)$$

$$B = X^{Wk}, \quad V = \sum_{t=0}^{k-1} X^{Wt}. \quad (24.4)$$

Equation (3) then forces $h = k$.

Proof. Apply Lemma 24.1 to (10) with source X and multiplier X . Its endpoint is X^{1+q} for some $q \geq 0$; put $W = q + 1$. This gives (24.2), including the empty prefix at $W = 1$. In particular, S cannot be zero or the constant monomial one.

Now $X^2S = X^{W+2}$ and $S = X^W$ are positive-degree monomials. Apply the ray lemma to (1) and (2) to obtain (24.3)–(24.4). Finally evaluate (3) at $X = 1$. The two $H(1)$ terms cancel, giving $V(1) = Q(1)$, hence $k = h$. No separate synchronization equation is needed. $\square$

Lemma 24.3 (Exact self-sizing). *In every solution,*

$$W = m + 2h + 2. \quad (24.5)$$

Thus the stride equals the width of the guarded terminal row.

Proof. Substitute the monomial expressions from Lemma 24.2, with $k = h$, into (11). Equality of monomials gives

$$W(h + 1) = m + 2 + (W + 2)h.$$

Cancel Wh to obtain (24.5). Conversely, that equality makes the calibration identity hold for these clock expressions. $\square$

Notice that the cancellation is performed after S, B, D have been proved to be monomials. It is not an illicit replacement of arbitrary polynomial identities by degree identities. The positivity of W was obtained before either clock was decoded.

24.3 Forced transition and terminal masks

Lemma 24.4 (Canonical masks and unit occupancy). *Every solution satisfies*

$$H = \sum_{t=0}^{h-1} X^{Wt} [w_t]_X, \quad w_t = m + 2t + 2, \quad (24.6)$$

$$K = X^{Wh} [W]_X, \quad W = m + 2h + 2. \quad (24.7)$$

Each coefficient of these masks is zero or one. At each permitted source site exactly one tile polynomial has coefficient one; at each terminal site exactly one terminal-symbol polynomial has coefficient one. Every tile or terminal coefficient outside its respective mask vanishes.

Proof. Rearrange (3) as

$$(1 - X)H = V - XQ = \sum_{t=0}^{h-1} X^{Wt} (1 - X^{m+2t+2}).$$

Since $(1 - X)[r]_X = 1 - X^r$, the proposed H satisfies this identity. It is the unique polynomial solution because $1 - X$ is not a zero divisor in $\mathbb{Z}[X]$. Thus (24.6) holds.

The known clock expressions give

$$G = \sum_{t=0}^h X^{Wt} (1 + X^{m+2t+1}). \quad (24.8)$$

In (4), the row at time zero of $X[m]_X + XSH + G$ is $[m + 2]_X$. For each $1 \leq t \leq h$, its row is

$$X[m + 2t]_X + 1 + X^{m+2t+1} = [m + 2t + 2]_X.$$

Equivalently, summing these identities directly yields

$$X[m]_X + XSH + G = \sum_{t=0}^h X^{Wt} [w_t]_X.$$

Subtract the already determined H to obtain (24.7). This algebra also covers $h = 0$.

By (24.5), every w_t with $t \leq h$ is at most W . The exponent intervals $[Wt, Wt + w_t - 1]$ are therefore disjoint for distinct t . The displayed masks have only zero–one coefficients. Their definitions as sums of nonnegative tile or terminal polynomials then force exactly one coefficient-one label at each occupied site and zero coefficients elsewhere. No Booleanity hypothesis has been used. $\square$

24.4 Spatial feature equations recover actual windows

For a source row $t < h$, let $u = w_t$ and let (ℓ_j, c_j, r_j) be the unique triple occupying the coefficient X^{Wt+j} , $0 \leq j < u$. Here c_j is a local symbol label; the global configuration remains denoted $c_t(i)$.

Lemma 24.5 (Exact horizontal adjacency). *The feature equations imply*

$$\ell_0 = c_0 = 0, \quad c_{u-1} = r_{u-1} = 0,$$

and for $1 \leq j < u$,

$$\ell_j = c_{j-1}, \quad c_j = r_{j-1}.$$

Thus every source triple is the genuine consecutive window in its row of centre symbols, with blank exterior.

Proof. For source rows, $u \leq W - 2$. Multiplication by X moves their support only to positions at most $W - 2$ within a block, so it cannot mix distinct rows. Compare coefficients in $XC = L$. At position zero it gives $\ell_0 = 0$; at internal position j it gives $\ell_j = c_{j-1}$; at the position immediately following the source row it gives $c_{u-1} = 0$. Likewise $XR = C$ gives $c_0 = 0$, $c_j = r_{j-1}$ internally, and $r_{u-1} = 0$ at the position after the row. Unit occupancy ensures that each coefficient is the label of a single symbol, rather than an ambiguous sum of labels. Since the labels $0, \dots, s - 1$ are distinct, these numerical equalities are symbol equalities. $\square$

For vector-valued features, the same comparisons give equality of every coordinate of the two feature vectors. Injectivity and $\chi(0) = 0$ then recover exactly the same symbols and blanks. This proves the spatial part of the same-tuple feature variant stated earlier.

24.5 Vertical reconstruction, including guards and time zero

Lemma 24.6 (Recovery of the cellular evolution). *For $0 \leq t < h$, the tile at exponent $Wt + j$ is the actual triple*

$$(c_t(j - t - 2), c_t(j - t - 1), c_t(j - t)), \quad 0 \leq j < w_t.$$

The symbol in the terminal polynomial at exponent $Wh + j$ is $c_h(j - h - 1)$, for $0 \leq j < W$.

Proof. At time zero, the coefficient comparison in (7) sets the centre-symbol feature to w_{j-1} at positions $1 \leq j \leq m$ and to zero at the two guard positions. If $h > 0$, this is the first source row; if $h = 0$, it is already the terminal row. The mask lemma ensures that no terminal terms intrude on earlier source rows and no source terms remain on the terminal row.

Suppose a source row $t < h$ has been recovered. By Lemma 24.5, its triple at each site is the genuine neighbourhood of its centre. The corresponding coefficient of O is therefore the numerical feature of the actual local output. The factor $XS = X^{W+1}$ places that output in the next block one horizontal position to the right, exactly as required by the encoding $j = i + t + 1$. Equation (7) identifies each next-row centre or terminal symbol with that output.

The two new outer positions receive no contribution from XSO or XI . Their feature must be zero, and occupancy together with injectivity of the feature makes them blank. They are genuine blank source guards by the light-cone bound (22.2). This proves the next-row statement. Induction recovers every source row and the terminal row. For $h = 0$, the initial coefficient comparison already proves the whole assertion.

All involved shifts stay within the intended blocks: a source row has width at most $W - 2$, and after the horizontal shift its output lies at positions $1, \dots, W - 2$ at the last step. Thus no part of the induction uses a numerical equality produced by a block collision. $\square$

The vector-feature proof is identical except that equality of all feature coordinates is used to recover a symbol. We have therefore justified the full same-tuple equivalence of the feature variants: all reconstruct the same actual history with the same clock and marker polynomials.

24.6 First acceptance and uniqueness of the position marker

Lemma 24.7 (Canonical acceptance). *Every solution encodes a first singleton acceptance at time h . If its accepting cell has encoded position j_* in the terminal block, then*

$$T_{\mathcal{F}} = X^{Wh+j_*}, \quad J = X^{Wh}[j_*]_X, \quad 1 \leq j_* \leq W - 2. \quad (24.9)$$

Proof. Equation (8) is a sum of nonnegative tile polynomials equal to zero. Thus no source row before h has an accepting centre. The reconstruction lemma and blank exterior identify these with the actual earlier configurations.

Apply Lemma 24.1 to (9), with source $B = X^{Wh}$ and multiplier X . It gives a single monomial $T_{\mathcal{F}}$ and the unique prefix J in (24.9). The terminal mask restricts the monomial to the top row. Its two guards are blank, and $0 \notin \mathcal{F}$, so the accepting position lies strictly between them. There is one accepting cell, not a selected cell among several. $\square$

If the first accepting configuration contains two or more accepting cells, the system has no solution. A later singleton cannot repair this, because the earlier accepting event would violate (8). This qualification is part of the source language, not an omitted edge case. The one-head universal-machine instantiation avoids it automatically.

24.7 Completeness and full-tuple single-foldness

Proof of Theorem 20.2, except for the degree formula. Suppose the actual evolution has a first singleton acceptance at time h . Set $W = m + 2h + 2$ and define S, E, D, Q, B, V by the explicit formulas in Lemmas 24.2–24.3. At every source site (j, t) , put coefficient one at X^{Wt+j} in the polynomial for the actual neighbourhood triple. At each terminal site, put coefficient one in the polynomial for its actual symbol. Define J from the unique accepting position by (24.9). Every unspecified coefficient is zero.

The clocks and positive-stride equation telescope, and the calibration holds by the definition of W . The shape equations follow from the identities in Lemma 24.4. The feature equations express actual neighbours, the loaded input, the local update, and the blank guards. There is no earlier acceptance, and the terminal accepting set is a singleton. Thus all eleven equations hold.

Conversely, the preceding lemmas reconstruct a genuine first singleton acceptance from every solution. Determinism makes its time h unique. The calibration then makes W unique. All clocks and E are fixed by the ray lemma; every source tile and terminal symbol is fixed by the actual evolution; and J is fixed by the unique accepting position. Thus the entire witness tuple is unique, including all auxiliaries. Its coefficients are Boolean by the clock, mask, and marker formulas. The algebraic counts were proved in Proposition 23.1; Section 27 proves the exact degree formula. $\square$

25 An exact bijection with two-coordinate histories

The new system does more than reproduce the same accepted language. It translates the entire canonical witness relation of the precursor without creating stride multiplicity or losing information about any history component.

Proposition 25.1 (The two-coordinate witness shape). [write] *[The same as Part I, Theorem 2.1 in the nine-row weighted form of Corollary 8.2; source 15’s proof is kept because the bijection, Theorem 25.2, uses it.] For the nine two-coordinate identities in Section 22, every solution in $\mathbb{N}[X, Y]^{s^3+s+5}$ has a first singleton accepting time h and the forms*

$$\begin{aligned} D &= X^{m+1+2h}Y^h, & Q &= \sum_{t < h} X^{m+1+2t}Y^t, \\ B &= Y^h, & V &= \sum_{t < h} Y^t, \\ H &= \sum_{t < h} Y^t [m + 2t + 2]_X, & K &= Y^h [m + 2h + 2]_X, \\ T_{\mathcal{F}} &= X^{j_*}Y^h, & J &= Y^h [j_*]_X. \end{aligned}$$

All tile and terminal polynomials encode the actual history. The full witness is unique whenever it exists, and every component has X -degree at most $m + 2h + 1$.

Proof. The ray lemma extends verbatim to exponent pairs in $\mathbb{N}^2$ whenever the multiplier is a monomial of positive total degree. Its directed chains are rays parallel to that multiplier’s exponent vector; coefficient

induction on total degree excludes any source-free nonnegative flow. Apply it to the two clocks, with multipliers X^2Y and Y . Evaluation at $(1, 1)$ in the source-shape equation synchronizes their heights. Division by $1 - X$ gives the displayed H , and the terminal shape equation gives the displayed K .

Distinct powers of Y separate all rows, so these masks have unit coefficients without needing a stride inequality. Nonnegativity forces unit occupancy. The spatial and vertical feature comparisons now work exactly as in Lemmas 24.5 and 24.6, with Y recording time instead of a block offset. The acceptance equations and the horizontal ray lemma give the displayed $T_{\mathcal{F}}, J$ and first singleton acceptance. Determinism fixes every coordinate as before. The right endpoint and terminal guard attain X -degree $m + 2h + 1$; all remaining components have no larger horizontal degree. $\square$

Theorem 25.2 (Full-witness packing and unpacking). *For each input word, there is a bijection between the full solution set of the precursor's nine two-coordinate identities and the full solution set of the eleven univariate identities. On a two-coordinate solution of height h , set $W = m + 2h + 2$, apply $Y \mapsto X^W$ to every old component, and add*

$$S = X^W, \quad E = X + \cdots + X^{W-1}.$$

The inverse deletes these two added components and sends every old component's monomial X^e to

$$X^{e \bmod W} Y^{\lfloor e/W \rfloor}.$$

Both maps are effective on supplied finite witness tuples.

Proof. For a two-coordinate solution, Proposition 25.1 gives horizontal degree less than W in every component. Packing is injective on those components, and specialization preserves the nine inherited identities. The added ray equation telescopes; the calibration follows from $W = m + 2h + 2$. Thus the forward map is well defined.

For a univariate solution, Lemmas 24.2–24.7 put every old component into the stated canonical blocks. Quotient and remainder therefore recover its clock, tile, terminal, or marker data uniquely. One may verify the inherited equations either through the semantic reconstruction or directly through injectivity. For the latter, every term in each decoded residual has horizontal degree below W . The potentially largest shift is the term X^2YQ , whose final horizontal exponent is $W - 1$. Spatial shifts of source rows reach at most $W - 2$; vertical outputs reach at most $W - 2$ in the terminal block; XJ reaches at most $W - 2$; and terminal guards have exponent $W - 1$. The input endpoint has exponent $m + 1 \leq W - 1$.

Packing is injective on each such residual polynomial, so the vanishing univariate residual implies the vanishing two-coordinate residual. Finally, S and E are forced by the recovered height and cannot vary independently. The two maps therefore compose to the identity on both full solution sets. $\square$

[write] At the intake the bijection was checked on the shared worked example (input $(1, 0, 0, 0, 2)$, $h = 4$): packing all 73 components of Part I's witness by $Y \mapsto X^{15}$ gives exactly this Part's corresponding components, with $S = X^{15}$.

25.1 What is minimal about the stride

Proposition 25.3 (Minimal stride for the one-block-per-row convention). *Suppose a row-major encoder places each occupied site (j, t) at exponent $j + Wt$ and requires every row, including the entire terminal row, to have its within-block positions in $\{0, \dots, W - 1\}$. For the guarded history of height h , it must have $W \geq m + 2h + 2$. The calibration attains this lower bound.*

Proof. The terminal row contains exactly $m + 2h + 2$ consecutive occupied positions, starting at zero. Containment in one W -position block therefore requires the inequality. Equation (24.5) gives equality. $\square$

This is deliberately a restricted minimality statement. It is not a lower bound for all injective encodings of a finite cone, nor even for row-major encodings that allow the last row to spill beyond its nominal block. It says that the calibration wastes no extra width under the exact block-decoding convention used in our soundness proof.

26 Universality and the sharp representation boundary

26.1 A transparent one-head source

Take a deterministic one-tape Turing machine with tape alphabet Γ , blank $\square$, state set Q_0 , and accepting state q_{acc} . Its transition function is total on nonaccepting states; a rejecting halt can be replaced by a permanent nonaccepting stationary loop. After acceptance, keep the accepting head in place. Standard universal machines with a finite program-and-input word provide such a source [18].

Use the cellular alphabet

$$\mathcal{A} = \Gamma \sqcup (Q_0 \times \Gamma), \quad s = |\Gamma|(1 + |Q_0|), \quad (26.1)$$

identifying the unmarked blank with label zero. An unmarked symbol is a cell without the head; a marked pair (q, a) is a cell with the head in state q reading a . The radius-one update is defined by these ordered cases. A central accepting head stays fixed. A nonaccepting central head writes its new symbol and either stays marked with the new state or becomes unmarked when moving. An unmarked centre receives a head from its left neighbour when that head moves right; otherwise it receives a head from its right neighbour when that head moves left; otherwise it keeps its tape symbol. The order resolves invalid multiple-head neighbourhoods and makes the table total. Such neighbourhoods never occur from the one-head input encoding.

Lemma 26.1 (Step-for-step source simulation). *From a finite word carrying exactly one head, this cellular automaton simulates one machine transition per cellular step and preserves exactly one head. Its first acceptance, when present, is a singleton.*

Proof. The old head writes exactly the prescribed symbol. A stationary move retains exactly that head in its new state. For a moving instruction, exactly the indicated adjacent cell receives the new head state, while the old cell becomes unmarked. Every other cell retains its tape symbol. Since there was only one head, there are no competing incoming moves. Induction proves the simulation and the one-head invariant. Taking the accepting alphabet to be all (q_{acc}, a) gives the singleton claim. $\square$

The input encoder places the initial head on the first input symbol and uses the remaining cells unmarked. An empty machine input is represented by one blank cell carrying the head, so our condition $m \geq 1$ is harmless. For a fixed universal machine, both f and $\mathcal{F}$ are fixed. Loading a program and its input requires only forming the finite word and the three constant polynomials X^{m+1} , $X[m]_X$, and XI ; it does not require running the program.

26.2 Fixed-template completeness and noncomputable bounds

Corollary 26.2 (A fixed unambiguous hard family). *For one fixed cellular rule and accepting set, solvability of the eleven identities on the effective word-input family is Σ_1^0 -complete. Every input has either no solution or exactly one full solution, and all coefficients of any solution are zero or one.*

Proof. Apply Theorem 20.2 to the cellular simulation of a fixed universal machine. The computable input map just described reduces the machine's acceptance problem to solvability, proving Σ_1^0 -hardness. For membership, enumerate all finite tuples of nonnegative integer polynomials and check the eleven identities by exact coefficient arithmetic. The uniqueness and Boolean-coefficient assertions are those of the main theorem. $\square$

[write] The univariate counterpart of Part I's Corollary 2.2.

This is an explicit generic compiler theorem. The package does not claim to supply or formally verify a particular numerical universal-machine table. Its four-symbol worked rule is an illustrative terminating example, not a universal rule. The universal instance is obtained by the proved machine-to-cellular construction once a universal source table is chosen.

Corollary 26.3 (No computable stride or degree bound). *On the fixed universal word-input family, there is no total computable input-only function bounding the stride of the solution on every solvable input. There is also no such function bounding the maximum witness degree.*

Proof. A stride bound $g(w)$ would bound the possible accepting time through $W = m + 2h + 2$. Simulating up to that finite bound would decide acceptance. A degree bound would also bound W , because $S = X^W$ is one witness component. Either would contradict Corollary 26.2. Alternatively, a degree bound together with Boolean coefficients makes the set of candidate full tuples finite and effectively searchable. $\square$

[write] The univariate counterpart of Part I's Theorem 10.2, and like it the bounded-search argument of Proposition cdc:bd:prop:nobound in [1]; Part III's Corollary 41.5 is the counter-machine version.

Proposition 26.4 (Every fixed-stride fibre is decidable). *If $S = X^W$ is supplied as part of the input, the full system is solvable precisely when*

$$h = (W - m - 2)/2$$

is a nonnegative integer and the source evolution has first singleton acceptance at that time. Thus the fixed-stride problem is decidable. All remaining identities are affine-linear in the remaining unknowns.

Proof. Necessity is Lemma 24.3 followed by semantic soundness; sufficiency is the explicit witness construction. A finite simulation checks the condition. Substitution of a supplied S turns every listed quadratic product into multiplication by a fixed coefficient polynomial. $\square$

The undecidable family is therefore an existential union of decidable linear fibres over an unbounded monomial parameter. The ray equation makes that parameter Boolean-coefficient, but it does not put a computable bound on its exponent.

26.3 A precise comparison

Domain	Unknown-degree	Coefficient condition	Consequence
$\mathbb{N}[X]$	1	No height bound	General systems are undecidable; classical result [13].
$\mathbb{N}[X]$	1	Given computable height bound	Decidable; no universal compiler with the stated guarantee.
$\mathbb{N}[X, Y]$	1	All full witnesses forced Boolean	Canonical universal family in the precursor [19].
$\mathbb{N}[X]$	2	All full witnesses forced Boolean	Eleven-row canonical universal family here.
$\mathbb{N}[X]$	4, one equation	Same forced Boolean witnesses	Sum-of-squares form in Section 28.

The impossibility result is about *linear systems*; the single-equation quartic row is not claimed to be a sharp degree bound for one equation. No row of this table asserts a fixed-variable ordinary-natural-number single-fold representation.

[write] Table 2 of Section 1.2 extends this table by Part I's signed counterexample and by Part III's witness domains.

27 Exact degree, support, and coefficient ledgers

Proposition 27.1 (Exact maximum degree). *For an accepting history of height h and stride $W = m + 2h + 2$, the maximum witness degree is*

$$D_{\max} = \max\{W, Wh + W - 1\} = \begin{cases} W, & h = 0, \\ W(h + 1) - 1, & h \geq 1. \end{cases}$$

In particular, it grows quadratically in h for fixed input length.

Proof. The right-clock endpoint $D = X^{Wh+W-1}$ attains degree $Wh + W - 1$. The terminal row has the same final exponent. Every tile, terminal, clock-prefix, and position-marker exponent is at most this endpoint. The new polynomial E has degree $W - 1$, while S has degree W . These are all the components. For $h \geq 1$, $Wh + W - 1 \geq W$; for $h = 0$, the endpoint degree is only $W - 1$, so S is the largest component. $\square$

This completes the remaining assertion of Theorem 20.2. The special case $h = 0$ should not be hidden in an asymptotic expression: the stride polynomial then has degree one larger than the terminal row.

Proposition 27.2 (Exact full-witness mass). *If the unique accepting cell is at encoded horizontal position j_* , then the sum of all coefficients in the entire witness tuple is*

$$\mathcal{M} = h^2 + (m + 7)h + 2m + 6 + j_*. \quad (27.1)$$

Since every coefficient is Boolean, this is also the total number of nonzero coefficients across all components, counting components separately.

Proof. The source tiles contribute

$$H(1) = \sum_{t < h} (m + 2t + 2) = h^2 + (m + 1)h.$$

The terminal polynomials contribute W . The two clock endpoints D, B contribute 2; their prefixes Q, V contribute $2h$. The accepting-position marker contributes j_* . Finally S and E contribute $1 + (W - 1) = W$. Summing and substituting $W = m + 2h + 2$ gives (27.1). $\square$

The count includes blank guards, blank interior tiles, and all clock and stride auxiliaries. It does not silently count only nonblank computation cells. Compared with the precursor's full witness, packing preserves every existing coefficient and adds exactly W nonzero coefficients in S, E .

One may recover the acceptance time from the stride by the exact formula $h = (\deg S - m - 2)/2$. Every solution therefore satisfies $\deg S \geq m + 2$ and $\deg S \equiv m \pmod{2}$. These are consequences of the system, not separately imposed congruence tests.

28 A single quartic and an ordinary bounded export

28.1 Same-witness reduction to one equation

Let $P_1, \dots, P_{11}$ be the full left-minus-right residuals of the system, after expanding all linear-form abbreviations. Regard them as polynomials in the $n = s^3 + s + 7$ unknowns with coefficients in $\mathbb{Z}[X]$. Define

$$Q_w(U_1, \dots, U_n) = \sum_{i=1}^{11} P_i^2. \quad (28.1)$$

Theorem 28.1 (One quartic, identical full solutions). *Over $\mathbb{N}[X]^n$, the equation $Q_w = 0$ has exactly the same full solutions as the eleven-row system. Its unknown-degree is exactly four. Its coefficient polynomials have X -degree at most $2(m + 1)$.*

Proof. For any supplied tuple, the evaluated residuals are ordinary polynomials in $\mathbb{Z}[X]$. If some residual is nonzero, let e be the largest residual degree. The coefficient of X^{2e} in the sum of their squares is the sum of squares of the nonzero leading coefficients of precisely the residuals of degree e . It is a strictly positive integer. Thus the sum cannot vanish. Conversely, vanishing of every residual clearly makes the sum zero. This proves exact equality of the full solution sets.

Every residual has unknown-degree at most two, so the sum has degree at most four. In the first residual, the term $-X^2SQ$ contributes $X^4S^2Q^2$ after squaring. No other residual has a quadratic monomial SQ , and no other pair of terms can cancel this degree-four monomial. The degree is therefore exactly four. The largest coordinate coefficient degree in a residual is at most $\max\{2, m + 1\} = m + 1$, since $m \geq 1$. Squaring gives the stated coefficient-degree bound. $\square$

[write] The univariate counterpart of Part I's Theorem 11.2, one degree higher because the identities are quadratic.

This is a quartic equation over a polynomial semiring. Its root variables are themselves univariate polynomials. It is not an ordinary quartic with only n natural-number unknowns, and its existence does not settle an ordinary single-fold or finite-fold Diophantine conjecture.

28.2 External degree truncation gives ordinary integer equations

A useful, entirely explicit finite export is available when a cutoff is supplied externally. This makes the hidden amount of scalar data visible.

Theorem 28.2 (Bounded ordinary quartic export). *For each supplied integer $L \geq 0$, replace every polynomial unknown by*

$$U_i(X) = \sum_{t=0}^L u_{i,t} X^t, \quad u_{i,t} \in \mathbb{N}.$$

Coefficient comparison in the eleven residuals yields an ordinary quadratic system on $n(L+1)$ natural-number unknowns, with at most

$$11(K_L + 1) \text{ rows}, \quad K_L = \max\{2L + 2, m + 1\}.$$

The sum of squares of these scalar residuals is an ordinary polynomial of degree at most four with the same solutions. It has exactly one natural solution when the source has first singleton acceptance and its canonical witness has $D_{\max} \leq L$, and no solution otherwise. No Boolean side equations are required.

Proof. Under the substitution, the largest degree contributed by an unknown product and its fixed coordinate shift is at most $2L + 2$; input constants have degree at most $m + 1$. Comparing all coefficients through K_L therefore imposes the complete polynomial identities. Each coefficient is quadratic or lower in the scalar variables $u_{i,t}$. Natural scalar coefficients produce exactly polynomial tuples in $\mathbb{N}[X]^n$ of degree at most L , with a unique zero-padded coefficient representation. Theorem 20.2 now gives precisely the stated zero-or-one solution count and forces Boolean coefficients automatically. A sum of squares of integer residual values vanishes exactly when all are zero, proving the final scalar-polynomial assertion. $\square$

[write] The univariate counterpart of Part I's Proposition 11.3.

The cutoff is not part of the fixed univariate compiler. The scalar arity grows with it, and Corollary 26.3 prevents choosing a total computable adequate cutoff from the input alone for the universal family. Thus this theorem is a bounded export, not a shortcut from polynomial single-foldness to ordinary fixed-arity single-foldness.

29 A parsimonious extension to labelled branching

The clock and geometry proof does not require determinism. Determinism is used only to make the reconstructed accepting history unique. This separation allows a precise counting extension.

Let $\mathcal{R}$ be a finite indexed set of labelled local transitions. Each $\tau \in \mathcal{R}$ has a source triple $(\ell_\tau, c_\tau, r_\tau)$, an output a_τ , and its own identity label. Different labels may have identical source and output data. Assume each source triple has at least one allowed transition, and that there is exactly one transition with source triple $(0, 0, 0)$, with output zero. The latter condition makes the infinite blank exterior both quiescent and free of unrecorded choice multiplicity.

A labelled computation chooses a permitted labelled transition at every source site. Choices outside the finite guarded region are the unique all-blank transition. Its next configuration is given by the chosen outputs. First singleton acceptance is interpreted along that chosen history.

Theorem 29.1 (Labelled-history root correspondence). *Replace the s^3 deterministic tile unknowns by one Z_τ per labelled transition. Define $H, L, C, R, O, C_{\mathcal{F}}$ using the corresponding source, output, and accepting data. The same eleven identities, now on $|\mathcal{R}| + s + 7$ unknowns, have solutions in bijection with finite labelled computation histories ending at their first singleton acceptance. For each such history, every auxiliary polynomial is uniquely determined, all coefficients are Boolean, and the degree and mass formulas remain unchanged.*

Proof. The ray, synchronization, calibration, and mask arguments use only the sum of the tile unknowns, so they are unchanged. Unit occupancy now selects one labelled transition at every site. Feature adjacency recovers its source triple, and the vertical equation matches the next symbol to that transition's output. This reconstructs a valid labelled history. The acceptance equations give its first singleton accepting event.

Conversely, any such labelled history determines one coefficient-one label per source site, its terminal symbols, and all clocks and markers by the earlier formulas. They satisfy the system. Two different labelled histories differ in a source label, a configuration, or their stopping time, hence in a witness component. Conversely, no auxiliary can vary after the labelled history is fixed. The unique quiescent exterior prevents an unencoded infinite collection of exterior choices from being counted as distinct histories. The mask and clock counts establish the same resource formulas. $\square$

[write] The univariate counterpart of Part I's Theorem 13.1.

The counted objects are labelled local histories, not merely sequences of configurations and not equivalence classes under rescheduling of independent actions. Forgetting duplicate labels changes the counting problem. A probabilistic interpretation would likewise require explicit weights and a separate summation semantics; root correspondence alone is not a theorem about probabilities.

30 Two essential constraints and literal counterexamples

30.1 Without calibration, padding creates infinitely many witnesses

Fix any source evolution with first singleton acceptance at height h . Take its two-coordinate witness and substitute $Y = X^W$ for any $W \geq m + 2h + 2$. Add $S = X^W$ and $E = X + \dots + X^{W-1}$. Every inherited equation and the positive-stride equation hold. All components remain Boolean, because the blocks do not collide.

However, the calibration holds only when $W = m + 2h + 2$. Deleting it therefore introduces infinitely many full witnesses for the same computation, distinguished by S, E and the packed exponents. A mere lower bound on the stride cannot restore full-tuple uniqueness. This is a failure of canonicalization even among perfectly valid, nonoverlapping, Boolean histories; no malformed coefficients are needed.

30.2 Without the positive-stride equation, zero erases time

Consider alphabet $\{0, 1\}$, the all-zero local rule $f \equiv 0$, accepting set $\{1\}$, and input word $w = (0)$. The source never accepts. Delete only equation (10). The following natural-polynomial tuple satisfies every remaining equation:

$$\begin{aligned} S = E = D = B = J = 0, \quad Q = X^2, \quad V = 1, \\ Z_{000} = 1 + X + X^2, \quad Z_{\ell cr} = 0 \ ((\ell, c, r) \neq (0, 0, 0)), \quad T_0 = T_1 = 0. \end{aligned} \tag{30.1}$$

Indeed, the two clocks become $Q = X^2$ and $V = 1$; the source-shape equation is $(X - 1)(1 + X + X^2) = X^3 - 1$; and the terminal-shape equation is $1 + X + X^2 = X + (1 + X^2)$. All features and accepting forms vanish. The terminal marker equation and calibration are $0 = 0$. The deleted positive-stride residual, however, is $-X$.

Thus $S = 0$ can erase the entire future and leave a false certificate of acceptance. The equation forcing a positive-degree monomial must be proved before the clock equations are interpreted as finite runs. The implementation reproduces this literal counterexample coefficientwise.

31 A fully explicit four-step example

31.1 The source rule and its trajectory

Use alphabet $\{0, 1, 2, 3\}$, accepting set $\{3\}$, and the radius-one rule

$$f(\ell, c, r) = \begin{cases} 3, & c = 3, \\ 3, & c = 2 \text{ and } \ell = 1, \\ 2, & c = 2 \text{ and } \ell \neq 1, \\ 0, & c = 1, \\ 1, & c = 0 \text{ and } \ell = 1, \\ 0, & c = 0 \text{ and } \ell \neq 1. \end{cases} \quad (31.1)$$

The right neighbour is unused in this example, but the compiler still uses its complete radius-one interface. A pulse labelled 1 moves right; a stationary symbol 2 becomes accepting symbol 3 when the pulse reaches it. Input

$$w = (1, 0, 0, 0, 2)$$

has the following nonblank physical cells:

Time	Pulse	Stationary or accepting cell
0	$c_0(0) = 1$	$c_0(4) = 2$
1	$c_1(1) = 1$	$c_1(4) = 2$
2	$c_2(2) = 1$	$c_2(4) = 2$
3	$c_3(3) = 1$	$c_3(4) = 2$
4	none	$c_4(4) = 3$

Thus $m = 5$, $h = 4$, $j_* = 4 + 4 + 1 = 9$, and $W = 15$. There are $4^3 + 4 + 7 = 75$ polynomial unknowns and eleven equations.

31.2 All clocks and the terminal row

The unique clock and stride polynomials are

$$\begin{aligned} S &= X^{15}, & E &= X + X^2 + \cdots + X^{14}, \\ D &= X^{74}, & Q &= X^6 + X^{23} + X^{40} + X^{57}, \\ B &= X^{60}, & V &= 1 + X^{15} + X^{30} + X^{45}, \\ J &= X^{60} + X^{61} + \cdots + X^{68}. \end{aligned}$$

The terminal polynomials are

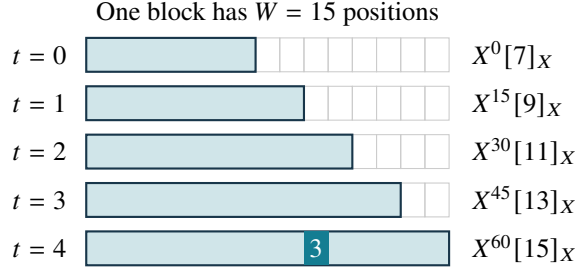
$$T_3 = X^{69}, \quad T_0 = X^{60}[15]_X - X^{69}, \quad T_1 = T_2 = 0.$$

The expression for T_0 has nonnegative coefficients: it means the terminal interval with its one accepting position removed. In particular, $SB = X^{75} = XD$ verifies the exact calibration, including its final horizontal factor X .

The source and terminal masks are

$$H = [7]_X + X^{15}[9]_X + X^{30}[11]_X + X^{45}[13]_X, \quad K = X^{60}[15]_X.$$

Figure 2 shows the one-block-per-row layout.



The top row is full; earlier rows leave unused block positions.

Figure 2: Canonical exponent blocks for the worked four-step history. The accepting cell is at exponent $60 + 9 = 69$. Blank cells inside each shaded region are still occupied and certified.

31.3 Every nonzero source polynomial

The following table completes the witness. An exponent list $e_1, \dots, e_k$ means the polynomial $X^{e_1} + \dots + X^{e_k}$. Every tile polynomial omitted from the table is zero.

Tile	Exponents with coefficient one
000	3, 15, 16, 23, 30, 31, 32, 33, 39, 40, 45, 46, 47, 48, 49, 50, 55, 56, 57
001	0, 17, 34, 51
002	4, 20
010	1, 18, 35
012	52
020	5, 21, 37
100	2, 19
102	36
120	53
200	6, 22, 38, 54

The last transition producing acceptance is tile 120 at exponent 53. Its output is shifted by $XS = X^{16}$ to the accepting exponent 69. The masks contain $7 + 9 + 11 + 13 = 40$ source sites. The full witness has

$$D_{\max} = 15 \cdot 5 - 1 = 74, \quad \mathcal{M} = 4^2 + (5 + 7)4 + 2 \cdot 5 + 6 + 9 = 89.$$

These are checked directly in the exported witness file.

31.4 The fully expanded quartic

The package supplies `results/example_system.json`, containing every term of each complete residual, and `results/example_quartic.json`, containing the fully expanded sum of their squares. No equation or input condition is left as an unexpanded external predicate.

[write] The first file is shipped as `data/15-one-coordinate-example_system.json`; the expanded quartic (1,641,389 bytes) is not shipped, because `code/15-one-coordinate-export_example.py` regenerates it byte for byte (README, *Reconstructing the excluded data*).

The expanded quartic has exactly 12,376 nonzero terms when the coordinate power of X and the unknown monomial are both expanded. Grouping terms by unknown monomial gives 7,455 monomials with polynomial coefficients. Its unknown-degree is four and its maximum coordinate coefficient degree is twelve. Exact

substitution of the canonical witness gives zero; three altered witnesses give nonzero polynomials, and all four evaluations agree with the separately evaluated sum of residual squares.

For comparison, the ordinary coefficient export with the external cutoff $L = 74$ has $75 \cdot 75 = 5,625$ scalar variables and at most $11 \cdot 151 = 1,661$ coefficient rows before removing identically zero rows. This is the actual scalar size of this simple bounded interface, not a claim that the 75 polynomial unknowns are 75 integer unknowns.

32 Executable validation and trust boundaries

32.1 Implementation interface

The implementation uses exact sparse dictionaries with nonnegative integer exponents. Residual polynomials may have signed integer coefficients, but the public witness interface validates every supplied witness coefficient as a natural number. It rejects negative exponents, negative coefficients, floating-point values, Boolean values masquerading as integers, and missing or extra witness coordinates. Rule tables are copied into immutable tuples; compiled residual mappings are deep-copied and frozen. Mutating a caller's original table does not alter the meaning of an already created compiler.

The source compiler accepts a rule, an accepting set, and a word. It does *not* accept a horizon. A separate candidate generator takes an external horizon so that tests can construct both valid and invalid proposed certificates. It deliberately generates candidates even at nonaccepting times or after a previous acceptance; the full residual checker must reject them. The independent truth check uses a separate dense cellular evolution, with no stride, clock, or polynomial packing.

The finite-state solver is separate from the cellular compiler. It accepts integer coefficient matrices, a coefficient-vector right-hand side, and a natural bound. It enumerates finite memories, solves the inhomogeneous prefix, and searches the homogeneous graph for a draining path. This independently exercises the negative theorem's algorithm.

32.2 Executed finite checks

Test family	Cases	Outcome
All 128 quiescent binary rules, input lengths 1–3, horizons 0–3	7,168	768 admitted; all agree with reference semantics.
Seeded rules on 3, 4, and 5 symbols, horizons 0–5	576	59 admitted; all agree.
Binary-feature checks of those same multistate candidates	576	Same full-tuple acceptance.
Stride-prefix coefficients in $\{0, 1, 2\}$ at degrees 0–7	6,561	Exactly 8 valid positive ray prefixes.
Clock parameters m, W, h, k in the declared finite ranges	432	108 synchronized; 10 also calibrated.
Independent one-row source-tile assignments	4,096	Exactly 8 structurally valid tableaux.
Worked-witness coefficient deletions and doublings	178	Every alteration rejected.

Test family	Cases	Outcome
Off-region insertion, one per worked-example variable	75	Every alteration rejected.
Worked-example candidate horizons 0–7	8	Only horizon 4 admitted.
Worked-example strides 15–22	8	All pass without calibration; only 15 passes it.
Scalar linear systems with coefficient data in $\{-1, 0, 1\}$	729	Automaton and exhaustive bounded-degree search agree.
Additional linear-solver edge cases	4	All pass, including $d = 0$ and $B_0 = 0$.
Invalid public-interface inputs	12	Every invalid input rejected.
Expanded quartic versus direct residual-square evaluation	4	Exact polynomial agreement.

The binary-rule family mostly stresses structural and boundary behavior: when symbol 1 is accepting, a word already containing a 1 accepts at time zero, and an all-zero word remains blank. It is not a large set of delayed acceptance examples. The multistate family and the explicit four-step trajectory address later acceptance. The multistate random seed is the fixed integer 20261002.

The independent source-tile test selects each of three source triples arbitrarily from eight binary triples, for four rules and two one-letter inputs. Thus it checks $4 \cdot 2 \cdot 8^3 = 4,096$ assignments rather than rebuilding only the canonical generator’s own choices. The structural rows, excluding the two acceptance rows, admit precisely the actual source triple sequence for each rule/input pair.

For the scalar linear check, both the degree-two coefficient polynomial $A(X)$ and the degree-two right side range over all triples in $\{-1, 0, 1\}^3$. There are $27^2 = 729$ systems. With $n = 1$, $d = 2$, $B_0 = 1$, and $M = 2$, the theorem’s bound is $2 + 2^2 = 6$. The automaton is compared against all $2^7 = 128$ Boolean polynomials through degree six. The two procedures agree on all systems; 75 are admitted. The exhaustive search is independent of the memory-graph implementation, although it uses the theorem’s specified cutoff.

The zero-stride ghost in (30.1) is also reproduced exactly, as are source-table immutability, fixed-template invariance under changing the input word, the degree bound of every emitted monomial, and the shared-stride form of every nonlinear monomial. Counts for each suite are recorded in separate JSON receipts and in `results/verification.json`.

All listed computations passed. They are finite tests, not an exhaustive search over all polynomial witnesses, a proof-assistant kernel check, an independent referee review, or a proof of universality from the toy example. The infinite statements rely on the written arguments.

32.3 Reproduction commands and package contents

With Python 3.10 or later, run from the package root:

```
python code/verify.py
```

```
python code/export_example.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
    one_coordinate_certificates.tex
```

Individual test suites can be selected with `python code/verify.py -part bounded_linear`, or the other names shown by the command's help. The tests and exports use only the Python standard library. The TeX source includes its bibliography and diagram; it does not fetch external data. Standard LaTeX packages named in its preamble are required.

File or directory	Purpose
<code>one_coordinate_certificates.tex/.pdf</code> <code>code/compiler.py</code>	Complete manuscript source and rendered article. Full exact eleven-row compiler, feature variants, witness generator.
<code>code/bounded_linear.py</code> <code>code/verify.py</code> <code>code/export_example.py</code>	Independent finite-state bounded-height solver. Finite correctness, mutation, and interface checks. Complete symbolic system, witness, and quartic expansion.
<code>results/</code>	Test receipts, expanded equations, explicit witness, and hashes.
<code>README.md</code>	Reproduction instructions and scope qualifications.

[write] *Shipped names.* In this report the four programs are `code/15-one-coordinate-*.py` and the receipts and exports `data/15-one-coordinate-*.json`; the manuscript is `Part II of article.tex`; the delivered `README` is not shipped. The delivered `results/example_quartic.json` is not shipped (see the note in Section 31), although the shipped `data/15-one-coordinate-export_hashes.json` still records its hash. `export_example.py` rewrites four shipped files and `verify.py` its receipts; run them only on a copy with the delivered layout. On Windows the rewritten files have CRLF line ends, so the hashes that `export_example.py` records differ from the delivered ones even though the files agree after LF normalization. The research programme's review [6] replayed both commands on the original and on a repaired copy and found a validation defect of the low-level `System` constructor (Section 1.4); the shipped `code/15-one-coordinate-compiler.py` is the unpatched original.

33 Further research questions and concrete next steps

These questions are proposed continuations of the proved construction. They are not all asserted to be previously published open problems. The first task in any priority claim is a focused literature comparison.

33.1 Can the eleven rows be reduced without changing the full fibre?

The three feature rows already use an injective one-coordinate code. The remaining rows pay separately for two clocks, two aggregate masks, first acceptance, a position marker, positive stride, and calibration. Which of these can be combined while retaining a bijection of all natural-polynomial witnesses? The counterexamples in Section 30 give immediate regression tests: any proposed merger must reject the zero-stride ghost and prevent arbitrary padding. A reduction should publish the full residual vector and inverse witness map, not only an implication between accepted inputs.

[write] *Answered in part by the research programme (note of 2 October 2026).* The review of this manuscript [6] proves, and checks by exact expansion on 25 emitted schemas, a projection to ten quadratic

identities on $s^3 + s + 6$ unknowns over $\mathbb{N}[X]$: put $S = XS_0$, $E = XE_0$, $D = S_0B$, replace the stride identity by $S_0 + E_0 = 1 + XE_0$ and delete the calibration identity; every product still contains S_0 , the largest coordinate shift rises from X^2 to X^3 , and the full natural-polynomial zero sets correspond bijectively. The review publishes the residuals and the witness map, as asked above. This was not rechecked at this write. Whether fewer than ten rows suffice remains open.

33.2 How much shared-stride nonlinearity is necessary?

The weighted compiler uses S times five specified linear forms. Can universality with automatically Boolean witnesses be achieved with fewer such products, or with only one explicit quadratic monomial after introducing linear auxiliaries? A useful classification would fix the coefficient ring, the allowed coordinate shifts, the variable count convention, and whether products of linear forms count as one gate or many expanded monomials. The linear obstruction gives an exact zero nonlinearity lower bound, but does not distinguish intermediate bilinear normal forms.

33.3 Can one equation have degree two or three?

The same-witness sum of squares gives a single quartic. Can one directly construct a single quadratic or cubic equation over $\mathbb{N}[X]$ with the same canonical universal behavior? The finite-state theorem rules out degree one, but it does not rule out two or three. Combining signed residuals by a simple sum is unsound because cancellation is possible; a lower-degree finalizer would need a rigorously proved noncancellation mechanism or a different certificate representation.

33.4 A fully numerical universal instance

Choose and document one actual universal machine table, including its tape alphabet, halt convention, ordinary program-input format, and empty input handling. Generate the cellular table by (26.1), then emit all eleven equations and their single quartic. The meaningful milestone is a verified complete input-to-certificate bridge with a numerical value of $s^3 + s + 7$, not merely a small table advertised as universal. A proof-assistant audit of the source simulation should precede any claim about a trusted numerical universal polynomial.

[write] The same target is posed in Part I (Section 16.1) and Part III (Section 49.1).

33.5 Tighter packing than constant-width blocks

Our stride is minimal for the stated one-block-per-row decoder, but early rows leave unused positions. A triangular layout using cumulative row widths would store the cone more densely. Can a fixed-degree polynomial system force such a variable-offset layout while keeping every auxiliary unique? A solution must account for the changing row-to-row displacement and must not replace it with a freely chosen sequence of offsets. Exact support and maximum-degree formulas would make the improvement measurable.

33.6 Ordinary arithmetic with a complete fibre ledger

The bounded coefficient export is explicit but grows in scalar arity. An ordinary integer packing would need a base, a carry bound, a finite support representation, and a proof that every integer solution decodes correctly. The new stride supplies a canonical polynomial size parameter, not automatically a canonical ordinary-integer base or an ordinary Diophantine monomial predicate. A first concrete goal is a bounded integer packing with

a proved zero-set bijection; an unbounded extension must keep the ordinary single-fold question separate from what has actually been proved.

[write] The same question is posed in Part I (Section 16.4) and Part III (Section 49.2).

33.7 Linear compilers with controlled but unbounded coefficient height

Corollary 21.3 rules out every total computable input-only bound on the minimum coefficient height for a universal univariate linear family. It does not forbid bounds in terms of the accepting running time, the degree of a solution, or another noncomputable complexity measure. For an explicit linear universal compiler, determine how coefficient height grows along genuine accepting histories, and whether that growth can be reduced while preserving full-witness uniqueness. This would quantify the alternative to our degree-two route.

33.8 Compositional output and labelled counting interfaces

The terminal polynomials contain the entire final tape, but the present interface is acceptance-oriented. A composition theorem should specify how one machine's canonical output becomes another's canonical input, including head position, blank padding, and word length. On the branching side, implement Theorem 29.1 and verify counts of labelled histories on finite examples. Distinguish duplicate labels, configuration histories, and schedules from the outset; these produce genuinely different root-counting questions.

33.9 Kernel verification of the obstruction and the positive compiler

The finite-state theorem is a useful first formalization target: bounded coefficient vectors, a memory graph, draining paths, and the polynomial reconstruction theorem are all finite combinatorial objects. For the positive compiler, begin with finitely supported maps $\mathbb{N} \rightarrow \mathbb{N}$, the ray lemma, and the cancellation identity for $1 - X$. Next prove clock synchronization and exact calibration, then the mask and unit-occupancy lemmas. Feature reconstruction and the source-machine bridge come last. This order ensures that no semantic assumption about well-formed tableaux is smuggled into the proof of their well-formedness.

[write] Parts I and III propose the same programme (Sections 16.8 and 49.4).

34 Conclusion

The precursor's requested one-coordinate affine-linear compiler cannot retain a computable bound on full-witness coefficient heights. This is not just a failure of a particular encoding: a finite-state algorithm rules out the whole proposed class, even with nonunique witnesses and input-dependent matrices.

The obstruction has a precise positive counterpart. Replacing the fixed time indeterminate by a shared stride unknown makes the system quadratic. The equations $S + E = X + XE$ and $SB = XD$ force a positive monomial stride and calibrate it to the terminal width. The resulting eleven identities encode a whole finite computation, force unit occupancy, reject earlier acceptance, and uniquely determine every auxiliary. Their full solutions are Boolean-coefficient and are in bijection with the earlier space-time witnesses.

The established advance is therefore a sharp linear-quadratic boundary for this coefficient-bounded representation problem, together with an explicit canonical construction and exact resource formulas. The univariate polynomial domain, the uninstantiated numerical universal source table, and the absence of

proof-assistant checking remain clear boundaries of the present result. They leave specific next steps rather than an unqualified claim to have solved an ordinary Diophantine representation conjecture.

35 Notation and proof-dependency ledger (appendix of source 15)

Symbol	Meaning
$\mathbb{N}[X]$	Finite polynomials in a fixed indeterminate, with natural coefficients.
$\deg, ht, wt$	Coordinate degree, maximum coefficient, and sum of coefficients. We set $\deg 0 = -\infty$.
$s, \mathcal{A}, 0, \mathcal{F}$	Alphabet size, alphabet, quiescent blank, accepting alphabet.
$f, c_t(i)$	Local cellular rule and the actual configuration at time t .
w, m	Supplied word and its positive length.
h	First singleton accepting time recovered from a solution.
w_t	Guarded source-row width $m + 2t + 2$.
W	Canonical stride $m + 2h + 2$, not supplied to the compiler.
$Z_{\ell cr}, T_a$	Source-triple masks and terminal-symbol masks.
D, Q	Endpoint and prefix of the right-boundary clock.
B, V	Endpoint and prefix of the vertical clock.
S, E	Stride monomial and its unique positive-ray prefix.
J, j_*	Terminal accepting-position prefix and the accepting position.
H, K, G	Total source mask, total terminal mask, and boundary linear form.
$L, C, R, O, \widehat{T}, I$	Weighted symbol-feature linear forms and supplied input feature.
$C_{\mathcal{F}}, T_{\mathcal{F}}$	Accepting source-centre and terminal linear forms.
B_0, d, M, N	Coefficient bound, matrix degree bound, right-side degree bound, and memory-state count in the linear obstruction.
L, K_L	External degree cutoff and residual coefficient-degree bound in the ordinary export; distinct from the spatial feature form L by context.

The positive proof has the following dependency order:

positive-stride ray $\implies$ two clock rays $\implies$ common height $\implies$ exact calibration,

followed by

canonical masks $\implies$ Boolean occupancy
 $\implies$ true neighbourhoods and evolution
 $\implies$ first acceptance and full uniqueness.

The finite-state obstruction is independent of this chain. The packing bijection uses the complete positive proof, rather than just a proposed Kronecker substitution. The single quartic and bounded scalar export are downstream of the full zero-set characterization.

36 Source, novelty, and reproducibility boundaries (appendix of source 15)

The repository inspection was tied to commit

44983ed7ebfd545de55b5fdb50e040c82f3d24295.

The root overview and the Hilbert-tenth README were inspected, together with selected computational-substrate research metadata and a detailed arithmetic-bound note. This is not a claim to have audited the

entire repository or any unrelated asserted result. The preceding *Unique Histories* source was inspected directly, including the compiler, its reconstruction proofs, its feature compression, and its one-coordinate further-research question.

External literature is used for historical context and classical universality. The obstruction, calibration, all-solution reconstruction, resource ledger, and extensions are supplied as explicit mathematical arguments here. The code verifies finite cases and exact expansions. No Lean or Rocq executable was run, and no theorem in this article is claimed to be kernel-checked.

The manuscript does not establish that every ingredient or their exact combination is absent from all prior literature. Relative to the inspected precursor, it establishes that one proposed conjunction of requirements is impossible and supplies an explicit construction after changing unknown-degree from one to two. The term “sharp” refers only to this degree boundary for systems with the specified computably bounded-height guarantee.

The archive contains this source, its PDF, original supporting Python code, and exact generated receipts. It contains no copies of external research papers, no font files, and no LaTeX build intermediates.

Part III

Linear boundary transport: unique polynomial certificates over every commutative ring (source 01)

[write] Part III prints batch-79 manuscript 01, *Linear Quasi-Diophantine Universality: Unique Polynomial Certificates, Boundary Tests, and an Exact Convex Integrality Gap*, “Research report prepared for Vladimir Reshetnikov”, 2 October 2026 (the manuscript names no assistant), pinned to repository commit 48ee077c7. Its title’s “Diophantine” refers to polynomial-valued unknowns (Section 1.3). Its labels carry the prefix `pwh:bt:` and its bibliography keys the prefix `bt:`; its three unlabelled numbered statements received labels at the merge (Remarks 38.3 and 41.4, Corollary 44.3), and its remarks are set in the same style as those of Parts I and II. It was written independently of Parts I and II. The two mechanisms are complementary: Part I’s uniqueness rests on nonnegativity, and its signed fibres are infinite (Proposition 12.1); here uniqueness rests on omitted-variable sorts and a clock indeterminate, and survives signed coefficients and zero divisors. Its abstract and keywords follow; its two appendices are Sections 51 and 52. The research programme reviewed this manuscript before it was placed here [5] (Section 1.4).

Abstract (source 01). We give an explicit, characteristic-independent representation of deterministic computation by linear polynomial identities with restricted polynomial sorts. The unknowns are finite polynomials, not a fixed tuple of integers. For a machine with q states and d zero tests, the compiler produces $q + d$ linear identities in $q + 2d$ polynomial unknowns. Its coefficient monomials have degree at most two, and each unknown has at most two signed monomial occurrences. A clock makes the complete certificate unique: a source has a solution if and only if its computation halts, and every coefficient of that solution is zero or one. The same equivalence holds over every nonzero commutative ring, including finite fields and rings with zero divisors. A row-tagging construction combines the identities into one linear equation while retaining the polynomial-sort restrictions.

We then turn the clocked compiler into a fixed, sparse, bounded linear operator and a strictly convex quadratic energy on finite-support arrays. For a nonhalting source the minimum over integer arrays is exactly one, whereas the infimum over rational or real arrays is zero and is not attained. An explicit tapered history has residual energy $1/(N + 2)$. The corresponding Hilbert-space operator has dense nonclosed range, with undecidable membership for a computable family of basis vectors. Full proofs, an executable

compiler, a worked certificate, and 32,502 exact validation checks are included. These are structural normal-form results; we do not claim a new ordinary integer universal-polynomial bound, a solution to finite-fold Diophantine representation, or established literature priority for the combined construction.

Keywords. Diophantine representation; two-counter machines; finite support; polynomial subrings; linear equations; convex relaxation; computable operators; certificate uniqueness.

Reading guide. The core representation is proved in Sections 38–41. Section 42 gives a complete numerical example. The analytic consequences are in Sections 43 and 44. Sections 45–49 explain the limitations, repository interface, tests, and further research program.

37 Scope, antecedents, and the repository boundary

37.1 The question addressed

The broad existence of an integer Diophantine representation for an effectively enumerable acceptance relation is supplied by MRDP. Merely applying that theorem to another universal programming language does not usually reveal what is special about the language. Here the target is a different question:

Can a Turing-complete transition system have an explicit *linear* algebraic certificate whose correctness survives arbitrary coefficient characteristic, whose full witness is unique, and whose failure can be compared exactly with continuous relaxation?

The answer is affirmative when the unknowns are finite-support polynomials in specified subrings. The representation exposes precisely where the computational power resides: in unbounded finite support and in two independent boundary restrictions, rather than in multiplication between unknowns.

The substrate used here is *linear boundary transport*. A monomial is a configuration. Multiplication by X or Y increments a counter; extracting the constant coefficient in one coordinate tests whether that counter is zero; shifting the remaining coefficients down decrements a positive counter. A finite control table combines these operations. The linear extension is not itself deterministic on arbitrary superpositions, but it is deterministic on every individual configuration monomial. The proofs exploit the latter fact and do not assume that arbitrary solutions are positive.

37.2 What is inherited and what is proved here

The universality of suitable deterministic two-counter machines is an external input. Instruction sets matter: Dudenhefner’s account explicitly distinguishes variants and establishes undecidability for its CM2 model [21]. Our instruction set allows arbitrary destinations for both branches and therefore contains that model. We do not infer universality from a reversible or differently restricted variant.

Undecidability of constrained linear polynomial equations is not new as a general phenomenon. Narendran proved undecidability when polynomial solutions are required to have nonnegative coefficients [23]. Dong proved a polynomial-time result for a particular single homogeneous univariate equation with all unknown polynomials nonzero and nonnegative [24]. Dong also proved undecidability for linear Laurent-polynomial systems with designated monomial unknowns [25]. Our restrictions are different: selected unknowns must omit a coordinate, and in the scalar presentation all unknowns must omit a row-tagging coordinate. No positivity restriction or monomial restriction is imposed on the unknowns.

[write] Two companion manuscripts of the same day, printed as Parts I and II, compile cellular automata into linear systems over $\mathbb{N}[X, Y]$ and quadratic stride systems over $\mathbb{N}[X]$ with unique Boolean histories. There uniqueness rests on nonnegativity (signed fibres are infinite, Proposition 12.1); here it rests on omitted-variable sorts and the clock, which is why the present theorem survives signed coefficients and zero divisors. Both credit Lohrey and Steinberg [9] for reading linear equations as translated local constraints, a precedent equally relevant here.

The contributions developed and proved in this report are the explicit mixed-subring compiler with its sparsity accounting; the complete characteristic-free description of its solution fibers; the clocked single-certificate normal form; and the exact comparison between finite integer feasibility, rational approximation, and Hilbert-space range membership. The graph lemmas and resolvent manipulations underlying these results are elementary and have standard incidence-operator antecedents. The report does not present their elementary ingredients as independently new discoveries. A focused source search did not establish priority for the combined theorem package, so no first-in-the-literature claim is made.

37.3 Pinned ProveIt audit

The repository snapshot used for the source comparison is commit

48ee077c7ab647d7121370e95a353169c89c79db.

Its commit timestamp is 2 October 2026, 19:11:29 UTC [28]. The audit inspected the Hilbert-tenth-problem overview, its directory structure and research summaries, and the source of `Common/RecursivelyEnumerableDioph.lean`. That file exposes `rePred_dioph`, adapting `MRDP.mrdp` to the repository’s Mathlib-facing predicate [30]. ([write] The full path is `Computability/HilbertTenthProblem/Lean/Diophantine/Common/RecursivelyEnumerableDioph.lean`; `rePred_dioph` is at its line 32, at the pin and at this write.)

The overview records a 75-operation complete certificate and an 87-operation universal polynomial, and distinguishes those fixed bounds from several externally bounded substrate compilers [29]. The present report does *not* change either operation bound. Its fixed number of unknowns counts *polynomials with unbounded support*, not scalar integer witnesses. Its equation degree is linear in those polynomial unknowns, not an ordinary universal-polynomial degree bound.

This was a focused source audit, not a rebuild of ProveIt or an exhaustive inspection of every incoming archive. The mathematical proofs below are written proofs, supplemented by executable tests. They are not claimed to be Lean-kernel-checked, independently peer-reviewed, or already merged into the repository.

[write] The manuscript is now in the repository as Part III of this report. It remains unformalized and unrefereed; the research programme has reviewed it [5] (Section 1.4).

38 The algebra of a deterministic transition graph

38.1 Finite support and terminal configurations

Fix a nonzero commutative ring R with identity. Thus $1_R \neq 0_R$; no assumption of characteristic zero, absence of zero divisors, or ordering is made. Let S be a set of configurations and let

$$f : S \longrightarrow S \cup \{\dagger\}$$

be a deterministic partial transition, where $f(s) = \dagger$ means that s is terminal. We count a terminal configuration as part of a halting history. A history that halts at time h is

$$c_0 = s, \quad c_{t+1} = f(c_t) \quad (0 \leq t < h), \quad f(c_h) = \dagger.$$

In particular, an initially terminal source has $h = 0$.

Let $\mathcal{V}_R = R^{(S)}$ be the free R -module of finitely supported coefficient functions on S , with basis e_s . Define the linear transport operator

$$Ue_s = \begin{cases} e_{f(s)}, & f(s) \neq \dagger, \\ 0, & f(s) = \dagger. \end{cases} \quad (38.1)$$

This definition gives a well-defined endomorphism of $\mathcal{V}_R$ even if a configuration has infinitely many predecessors: the image of a finite linear combination is still finite. We use bounded indegree only in the Hilbert-space section.

A *weak component* means a connected component after forgetting the directions of the edges $s \rightarrow f(s)$; terminal vertices have no outgoing edge. A weak component either contains one terminal vertex, contains one directed cycle, or contains neither. If a component contains a terminal vertex, every vertex in it eventually reaches that terminal vertex. Indeed, across each undirected edge the property of eventually reaching a specified terminal is equivalent at the two endpoints. The same argument shows that a component with no terminal contains only nonhalting configurations.

Theorem 38.1 (Finite-support transport and the full fiber). *For every $s \in S$, the equation*

$$(I - U)H = e_s, \quad H \in \mathcal{V}_R, \quad (38.2)$$

has a solution if and only if the computation from s halts. When the computation halts at time h , define

$$H_s = \sum_{t=0}^h e_{c_t}.$$

The complete solution set is

$$H_s + \bigoplus_{C \text{ a directed cycle}} R\mathbf{1}_C, \quad \mathbf{1}_C = \sum_{v \in C} e_v. \quad (38.3)$$

Here a direct sum means that only finitely many cycle coefficients are nonzero. All cycles are disjoint from the halting component of s .

Proof. For a halting history the sum telescopes:

$$(I - U) \sum_{t=0}^h e_{c_t} = e_{c_0}.$$

No configuration can repeat before a deterministic computation halts; a repetition would force an infinite periodic future. Hence the coefficients of H_s are zero or one.

Suppose instead that the computation from s does not halt, and let D be its weak component. Define the linear functional

$$\lambda_D(H) = \sum_{v \in D} H(v).$$

The sum is finite for each $H \in \mathcal{V}_R$. There is no terminal vertex in D , and no edge crosses a weak-component boundary, so $\lambda_D U = \lambda_D$. Applying λ_D to (38.2) would give $0 = 1_R$, a contradiction.

It remains to calculate the kernel. Let $K = UK$ have finite support, and let $K(v) \neq 0$. The coefficient equation at v implies that some predecessor w has $K(w) \neq 0$: otherwise the finite sum of nonzero-support predecessor coefficients would be empty. Repeatedly choosing such a predecessor stays in a finite set, so eventually produces a directed cycle. The selected predecessor chain gives a forward path from that cycle to v . A deterministic directed cycle cannot have an outgoing edge leaving the cycle, so v lies on it. Thus the support of K is a union of directed cycles. On a cycle the coefficient equations say that the coefficients are constant, because no noncycle predecessor has nonzero coefficient. Conversely each $\mathbf{1}_C$ belongs to the kernel. Subtracting H_s from any solution proves (38.3). $\square$

Corollary 38.2 (An exact sequence). *Let $\mathcal{C}$ be the set of directed cycles and $\mathcal{D}$ the set of nonhalting weak components. There is an exact sequence of R -modules*

$$0 \longrightarrow R^{(\mathcal{C})} \longrightarrow \mathcal{V}_R \xrightarrow{I-U} \mathcal{V}_R \longrightarrow R^{(\mathcal{D})} \longrightarrow 0. \quad (38.4)$$

The left injection sends a cycle to $\mathbf{1}_C$. The right map sends e_s to its component basis vector if its component has no terminal, and to zero otherwise.

Proof. The kernel statement is Theorem 38.1. In the cokernel the relations are $e_v = e_{f(v)}$ for a live vertex and $e_v = 0$ for a terminal vertex. Along any finite undirected path, adjacent basis vectors therefore represent the same class. Terminal components collapse to zero; every other component gives one generator. The component-sum maps show that these remaining generators have no nontrivial relations. All reductions of a finite combination use only finitely many paths. $\square$

Remark 38.3. Component membership is not being used as an algorithm. It is a mathematical separation functional proving nonsolvability. The theorem allows negative coefficients, modular cancellation, and zero divisors; none of those can turn the component sum 1_R into zero.

39 The clocked resolvent and uniqueness

Adjoin an indeterminate Z , and extend U coefficientwise to $\mathcal{V}_R[Z]$. The clocked equation is

$$(I - ZU)H = e_s, \quad H \in \mathcal{V}_R[Z]. \quad (39.1)$$

An element of $\mathcal{V}_R[Z]$ has both a finite time degree and finitely many configuration coefficients in each time layer. Equivalently, it has finite total support in $S \times \mathbb{N}$.

Theorem 39.1 (Characteristic-free unique certificate). *The operator $I - ZU$ is injective on $\mathcal{V}_R[Z]$. Equation (39.1) has a solution precisely when s halts. In that case its unique solution is*

$$H_s(Z) = \sum_{t=0}^h Z^t e_{c_t}. \quad (39.2)$$

Its degree in Z is h , its support size is $h + 1$, and all its nonzero coefficients equal 1_R . These conclusions hold for every nonzero commutative ring R .

Proof. Write $H = \sum_{t \geq 0} Z^t H_t$, with all but finitely many H_t zero. Comparison of coefficients in (39.1) gives

$$H_0 = e_s, \quad H_t = UH_{t-1} \quad (t \geq 1). \quad (39.3)$$

The same comparison for a homogeneous solution gives $H_0 = 0$ and then $H_t = 0$ for every t , proving injectivity. For the inhomogeneous equation we obtain $H_t = U^t e_s$. Along a deterministic trajectory this vector is exactly e_{c_t} until a terminal configuration is passed, and is zero thereafter. Since $1_R \neq 0_R$, a nonhalting trajectory has a nonzero coefficient in every time layer and cannot be a polynomial. The stated formula and counts follow. Different times remain distinct even when the unlocked configuration repeats. $\square$

39.1 What changes on completion

In the formal power-series module $\mathcal{V}_R[[Z]]$ there is always a unique solution,

$$H_\infty = \sum_{t=0}^{\infty} Z^t U^t e_s. \quad (39.4)$$

Multiplication by ZU is coefficientwise well-defined, and (39.3) verifies the equation directly. Thus existence in the power-series completion has no halting content. The relevant question is whether the formal solution belongs to the much smaller polynomial module. The clock enforces uniqueness; finite support enforces halting. These are separate mechanisms.

Corollary 39.2 (Exact base change). *For a fixed deterministic program and a single configuration source, solvability of the clocked equation is the same over $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}_p$, and $\mathbb{Z}/m\mathbb{Z}$ for every $m \geq 2$. If a solution exists, it is obtained over each ring from the same zero-one history.*

Remark 39.3 (The source restriction is essential). For a general linear combination of source configurations, finite-time cancellation may occur. For example, let two different states a, b both transition to a nonhalting state c . Over $\mathbb{F}_2$, $U(e_a + e_b) = 0$, so $H = e_a + e_b$ solves $(I - ZU)H = e_a + e_b$ although neither source halts. Over $\mathbb{Z}$ use $e_a - e_b$ instead. Our characteristic-independent halting statements always use one configuration monomial with coefficient one.

40 A universal substrate made of linear boundary operators

40.1 The machine convention

Let the control states be $Q = \{0, \dots, q-1\}$, with configurations $(r, a, b) \in Q \times \mathbb{N}^2$. Each state has one of the following instructions:

Instruction	Action
INC(j, p)	Increment counter j and go to p .
TEST(j, p_+, p_0)	If counter $j > 0$, decrement it and go to p_+ ; otherwise leave it at zero and go to p_0 .
HALT	Declare the current configuration terminal.

Here $j \in \{X, Y\}$. There are no implicit wraparound counters, negative counter values, or automatic jumps past the end of the table. Every destination is a valid state. A missing next instruction in a different machine convention can be translated to an explicit terminal state.

This model contains the undecidable CM2 instruction set specified in [21]. The standard universality construction also provides a fixed interpreter machine with computably encoded input in the counters. One may choose the usual interface with fixed initial state r_0 , second counter zero, and first counter containing an encoded input [22, 21]. Nothing here requires the physical counter value to equal an unencoded source-program number. All completeness statements concern computable many-one reductions; no polynomial-time size bound for that encoding is asserted.

40.2 Polynomial configurations

Identify

$$e_{(r,a,b)} \longleftrightarrow e_r X^a Y^b, \quad \mathcal{V}_R \cong R[X, Y]^Q.$$

For $F \in R[X, Y, Z]$ define

$$M_X F = XF, \quad P_X F = F(0, Y, Z), \quad D_X F = \frac{F - P_X F}{X}, \quad (40.1)$$

$$M_Y F = YF, \quad P_Y F = F(X, 0, Z), \quad D_Y F = \frac{F - P_Y F}{Y}. \quad (40.2)$$

The divisions in these definitions are exact polynomial shifts, not inversions in a field. They are valid over every coefficient ring. On a monomial,

$$P_X(X^a Y^b Z^t) = \begin{cases} Y^b Z^t, & a = 0, \\ 0, & a > 0, \end{cases}$$

and $D_X(X^a Y^b Z^t) = X^{a-1} Y^b Z^t$ for $a > 0$, while it is zero for $a = 0$. The analogous statements hold for Y .

Lemma 40.1 (Boundary algebra). *All six operators are $R[Z]$ -linear, and*

$$D_X M_X = I, \quad M_X D_X = I - P_X, \quad P_X^2 = P_X, \quad P_X M_X = 0, \quad D_X P_X = 0.$$

The Y identities are analogous. Every X -coordinate operator commutes with every Y -coordinate operator.

Proof. Each identity follows by evaluating it on $X^a Y^b Z^t$ and distinguishing $a = 0$ from $a > 0$ where necessary. The cross-coordinate statements follow because the two operations change or test different exponents. Linearity extends the identities to arbitrary polynomials. $\square$

For a vector $H = (H_r)_{r \in Q}$ define UH by adding the following contribution from each control state r :

instruction at r	contribution to UH
INC(X, p)	$e_p M_X H_r$
INC(Y, p)	$e_p M_Y H_r$
TEST(X, p_+, p_0)	$e_{p_+} D_X H_r + e_{p_0} P_X H_r$
TEST(Y, p_+, p_0)	$e_{p_+} D_Y H_r + e_{p_0} P_Y H_r$
HALT	0

(40.3)

On a single monomial configuration exactly the intended branch is nonzero. Thus this U is exactly the transport operator of (38.1), not an approximation to it.

Corollary 40.2 (Native boundary-transport representation). *For a source (r_0, a_0, b_0) , the linear operator equation*

$$H - ZUH = e_{r_0} X^{a_0} Y^{b_0}, \quad H \in R[X, Y, Z]^Q, \quad (40.4)$$

has a solution if and only if the counter machine halts on that source. Its solution, when present, is unique and binary. Removing Z preserves existence but introduces exactly the cycle fibers of Theorem 38.1.

This is already an explicit quasi-Diophantine representation. The next section removes P_X, D_X, P_Y, D_Y from the displayed language, at the cost of making the polynomial sorts visible.

41 Compilation to mixed-subring linear equations

41.1 Eliminating a boundary operator

Lemma 41.1 (Unique boundary split). *For $F \in R[X, Y, Z]$ there is a unique pair*

$$D \in R[X, Y, Z], \quad B \in R[Y, Z]$$

satisfying

$$F = XD + B. \quad (41.1)$$

It is $D = D_X F$ and $B = P_X F$. The analogous result holds with X and Y interchanged.

Proof. Group the monomials of F by their exponent of X . The zero-exponent terms give B , and shifting the other terms down gives D . For uniqueness, evaluate (41.1) at $X = 0$ to recover B , then use the injectivity of multiplication by the indeterminate X to recover D . That injectivity holds even when R has zero divisors. $\square$

Let $T \subseteq Q$ be the set of test states, and write $d = |T|$. Introduce $H_r \in R[X, Y, Z]$ for every state. For each $r \in T$ introduce $D_r \in R[X, Y, Z]$ and a boundary unknown B_r , whose sort is

$$B_r \in \begin{cases} R[Y, Z], & \text{the instruction tests } X, \\ R[X, Z], & \text{the instruction tests } Y. \end{cases} \quad (41.2)$$

Impose one split equation at each test state:

$$H_r - X_j D_r - B_r = 0, \quad X_X = X, \quad X_Y = Y. \quad (41.3)$$

For each destination state p , form the incoming expression I_p . An increment instruction at r contributes $X_j H_r$ to its destination; a positive test branch contributes D_r ; a zero branch contributes B_r ; a terminal instruction contributes nothing. Impose the balance rows

$$H_p - Z I_p = S_p \quad (p \in Q), \quad (41.4)$$

where $S_{r_0} = X^{a_0} Y^{b_0}$ and all other S_p are zero. If both branches of a test have the same destination, both contributions are added there. The split still makes them disjoint on a configuration monomial.

Theorem 41.2 (Explicit sparse linear compiler). *Equations (41.2)–(41.4) give a compiler with $q + 2d$ polynomial unknowns and $q + d$ linear equations. For every nonzero commutative coefficient ring, its solution set is in bijection with the solutions of (40.4), by forgetting the D_r, B_r unknowns. Consequently the complete*

compiled witness is unique when it exists, exists exactly for halting computations, and has only zero–one coefficients.

Every matrix-coefficient monomial is a signed member of

$$\{1, X, Y, Z, ZX, ZY\}.$$

Each unknown has at most two signed monomial occurrences in the entire matrix, with occurrences possibly combined in one entry. Thus matrix coefficient degree is at most two. If halting occurs at time h , the main H -support has size $h + 1$. The full witness support has size

$$h + 1 + \#\{t : 0 \leq t \leq h, \text{ the state of } c_t \text{ is a test state}\}, \quad (41.5)$$

and is at most $2h + 1$.

Proof. Lemma 41.1 forces $D_r = D_j H_r$ and $B_r = P_j H_r$. Substitution in the balance rows therefore gives precisely (40.4). Conversely a solution of that equation has exactly one extension through these boundary splits. This proves the bijection and the uniqueness claim.

There are q main unknowns and two new unknowns for each of the d tests. There are q balance rows and d split rows. An increment-state unknown occurs in its own balance row with coefficient 1 and in its destination balance row with coefficient $-ZX_j$. A test-state H_r occurs in its own balance row and its split row, each with coefficient 1. A terminal-state unknown occurs only in its own balance row. Each D_r occurs with $-X_j$ in the split row and $-Z$ in its positive destination row. Each B_r occurs with -1 in the split row and $-Z$ in its zero destination row. Self-destinations combine occurrences but do not increase their number or degree.

For a halting trajectory, each time layer of the main witness has exactly one monomial. At a test-state occurrence, that monomial contributes to exactly one of D_r and B_r , again with coefficient one. Time exponents prevent collisions. Hence (41.5) holds. A terminal state is not a test state, so there are at most h such additional monomials. $\square$

41.2 One scalar equation without sums of squares

Let $E_0, \dots, E_{m-1}$ be the residuals of the $m = q + d$ rows, including the subtraction of the right-hand side. Introduce an additional indeterminate W , and replace the system by

$$\sum_{i=0}^{m-1} W^i E_i = 0. \quad (41.6)$$

Every unknown retains its original sort and is independent of W . Thus the coefficient of W^i is exactly E_i , and the tagged equation is equivalent to the system. This uses no order and no sum-of-squares argument, so it works in characteristic two as well. After collecting each unknown, the result is one equation

$$\sum_{j=1}^{q+2d} p_j(W, X, Y, Z) F_j = W^{r_0} X^{a_0} Y^{b_0}. \quad (41.7)$$

The p_j are explicitly determined by the program, each has at most two signed monomials, and their total degree is at most $m + 1$. The degree in the unknowns F_j is one. The dependence restrictions on the F_j are part of the equation's specification, not informal advice.

Theorem 41.3 (A fixed single-equation universality normal form). *Choose a fixed universal deterministic two-counter interpreter $\mathcal{U}$ with an encoded-input interface $(r_0, n, 0)$. There are fixed integers q, d , fixed polynomials $p_j \in \mathbb{Z}[W, X, Y, Z]$, and fixed polynomial sorts chosen from*

$$R[X, Y, Z], \quad R[X, Z], \quad R[Y, Z]$$

such that the set of $n \in \mathbb{N}$ for which

$$\sum_j p_j F_j = W^{r_0} X^n \tag{41.8}$$

has a solution is computably enumerable complete. Its solution fiber is empty or a singleton. If it is a singleton, its coefficients are zero or one. The accepted set is identical over every nonzero commutative ring R , in particular over each fixed finite field.

There is also an unlocked version: one equation in $R[W, X, Y]$ with unknown sorts $R[X, Y], R[X], R[Y]$ has the same accepted set, but need not have singleton fibers.

Proof. Compile the fixed interpreter using Theorem 41.2, then apply (41.6). The source monomial is computable from n . Theorem 39.1 identifies its accepted set with the interpreter's halting set, giving completeness and all fiber statements. The unlocked variant follows from Theorem 38.1 and the same unique boundary splitting without Z .

Over $\mathbb{Z}, \mathbb{Q}$, or a finite field, one can also see semidecidability by enumerating finite polynomial tuples and checking the identities and sorts. For an arbitrary ring no effective presentation is presumed: the algebraic equivalence with the same halting set remains true, but an enumeration of arbitrary ring elements is not asserted. $\square$

Remark 41.4 (What is and is not numerically instantiated). The compiler accepts every concrete valid finite program, and the worked example below is completely instantiated. The existence of a fixed universal table in Theorem 41.3 uses counter-machine universality; this report does not supply a particular numerical universal interpreter table or optimize its q, d . Thus the theorem is an effective structural construction, not a claimed small numerical universal equation or an arithmetic-circuit record.

Corollary 41.5 (No computable complete degree bound). *For the fixed universal source family there is no total computable function bounding $\deg_Z H$ on all solvable instances. There is also no total computable bound on the main-witness support size, or on the full witness support size, on all solvable instances.*

Proof. If $B(n)$ bounded the time degree, simulate the interpreter for $B(n)$ transitions; all halting cases would have been detected. This would decide its halting set. A support bound gives a time bound because the main support has size $h + 1$ and is contained in the full support. $\square$

[write] The bounded-search argument of Proposition `cdc:bd:prop:nobound` in [1], with $\deg_Z H$ and support in place of an integer witness height; its counterparts are Theorem 10.2 (Part I) and Corollary 26.3 (Part II).

42 A complete five-polynomial certificate

Consider the program that transfers the first counter to the second:

State	Instruction
0	TEST($X, 1, 2$)
1	INC($Y, 0$)
2	HALT

For input $(0, 2, 0)$ the exact trajectory is

$$(0, 2, 0) \rightarrow (1, 1, 0) \rightarrow (0, 1, 1) \rightarrow (1, 0, 1) \rightarrow (0, 0, 2) \rightarrow (2, 0, 2).$$

It halts at time $h = 5$. The five unknowns have sorts

$$H_0, H_1, H_2, D_0 \in R[X, Y, Z], \quad B_0 \in R[Y, Z].$$

The complete system is

$$H_0 - ZYH_1 = X^2, \tag{42.1}$$

$$H_1 - ZD_0 = 0, \tag{42.2}$$

$$H_2 - ZB_0 = 0, \tag{42.3}$$

$$H_0 - XD_0 - B_0 = 0. \tag{42.4}$$

Its unique solution is

$$H_0 = X^2 + Z^2XY + Z^4Y^2, \quad H_1 = ZX + Z^3Y, \tag{42.5}$$

$$H_2 = Z^5Y^2, \quad D_0 = X + Z^2Y, \quad B_0 = Z^4Y^2.$$

Direct substitution verifies every row, including the final terminal state. The main support has six monomials and the full witness has nine. The three additional monomials correspond exactly to the three visits to state zero.

Tagging the four rows by $1, W, W^2, W^3$ yields the single equation

$$(1 + W^3)H_0 + (W - ZY)H_1 + W^2H_2 - (WZ + W^3X)D_0 - (W^2Z + W^3)B_0 = X^2. \tag{42.6}$$

All unknowns are independent of W , and B_0 is additionally independent of X . Formula (42.5) is the unique solution over the integers, over the rationals, and over every finite field. In characteristic two the minus signs become plus signs; the uniqueness and halting equivalence do not change.

The distribution contains the exact trajectory and sparse coefficient lists in `examples/transfer_certificate.json`, the human-readable witness in `examples/transfer_witness.txt`, and the compiler's uncollected tagged equation in `examples/transfer_equation.txt`. ([write] Shipped as `data/01-boundary-transport-transfer_certificate.json`, `data/01-boundary-transport-transfer_witness.txt` and `data/01-boundary-transport-transfer_equation.txt`.) These files are generated by the same verification command that runs the tests, not manually transcribed numerical evidence.

43 A strictly convex energy with an exact integrality gap

43.1 Clocked configurations and the incidence operator

Let

$$\Omega = Q \times \mathbb{N}^2 \times \mathbb{N}$$

be the clocked configuration set. For $v = (r, a, b, t)$ define $F(v)$ by executing the machine step and increasing t by one; if the instruction is terminal, leave $F(v)$ undefined. Write v live for the first case. On finite-support arrays define

$$Ae_v = \begin{cases} e_v - e_{F(v)}, & v \text{ live,} \\ e_v, & v \text{ terminal.} \end{cases} \quad (43.1)$$

This is $I - ZU$ in coefficient notation. Every column has one or two nonzero entries, all equal to 1 or -1 . A row has at most $2q + 1$ nonzero entries: at most two predecessors can arise from each control state, in addition to its diagonal entry. The matrix is fixed by the finite program, although its row and column sets are infinite.

Fix the initial clocked source $v_0 = (r_0, a_0, b_0, 0)$. For finitely supported real h define

$$E_{v_0}(h) = \|Ah - e_{v_0}\|_2^2. \quad (43.2)$$

All sums are finite at such an h . The zero set of this energy is exactly the clocked certificate set.

Theorem 43.1 (Exact integer gap and nonattained real infimum). *The functional E_{v_0} is strictly convex on the real vector space of finite-support arrays. Its integer minimum is*

$$\min_{h \in \mathbb{Z}^{(\Omega)}} E_{v_0}(h) = \begin{cases} 0, & \text{the source halts,} \\ 1, & \text{the source does not halt.} \end{cases} \quad (43.3)$$

Over finite-support rational or real arrays the infimum is always zero. That infimum is attained if and only if the source halts; if attained, its minimizer is the unique binary history.

For a nonhalting trajectory $v_0, v_1, \dots$, set

$$h^{(N)} = \sum_{t=0}^N \frac{N+1-t}{N+2} e_{v_t} \quad (N \geq 0). \quad (43.4)$$

Then

$$Ah^{(N)} - e_{v_0} = -\frac{1}{N+2} \sum_{t=0}^{N+1} e_{v_t}, \quad (43.5)$$

$$E_{v_0}(h^{(N)}) = \frac{1}{N+2}, \quad (43.6)$$

$$\|h^{(N)}\|_2^2 = \frac{(N+1)(2N+3)}{6(N+2)}. \quad (43.7)$$

Proof. The clocked operator is injective by Theorem 39.1. For $h \neq k$ and $0 < \theta < 1$, the parallelogram expansion gives

$$\begin{aligned} E_{v_0}(\theta h + (1-\theta)k) &= \theta E_{v_0}(h) + (1-\theta)E_{v_0}(k) \\ &\quad - \theta(1-\theta)\|A(h-k)\|_2^2. \end{aligned}$$

The last term is strictly negative, proving strict convexity.

If a finite integer array has nonzero residual, the residual contains a nonzero integer coefficient, so its squared norm is at least one. A nonhalting source has no zero residual by Theorem 39.1, and $h = 0$ has energy one. This proves the nonhalting case of (43.3). A halting source has its integer binary history of energy zero.

For the nonhalting taper, the coefficient at v_0 in the residual is $(N+1)/(N+2) - 1 = -1/(N+2)$. At each v_t with $1 \leq t \leq N$, the coefficient is the difference of consecutive taper values, also $-1/(N+2)$. At v_{N+1} it is

the negative of the final taper value. All clocked vertices are distinct, even for an unlocked periodic run. This proves (43.5) and (43.6). Summing $1^2 + \dots + (N+1)^2$ and dividing by $(N+2)^2$ gives (43.7). Thus rational arrays already make the energy arbitrarily small. Attainment of the infimum would be a finite-support zero residual, which is equivalent to halting and has the unique witness already described. $\square$

43.2 The taper is optimal on its trajectory segment

Proposition 43.2 (Exact segment minimum). *For a nonhalting source, the minimum of E_{v_0} on $\text{span}_{\mathbb{R}}\{e_{v_0}, \dots, e_{v_N}\}$ is $1/(N+2)$, attained uniquely at (43.4). The same minimum holds over the rational span.*

Proof. Write $h = \sum_{t=0}^N a_t e_{v_t}$. Its residual coefficients on $v_0, \dots, v_{N+1}$ are

$$a_0 - 1, \quad a_1 - a_0, \quad \dots, \quad a_N - a_{N-1}, \quad -a_N.$$

Their sum is -1 . By Cauchy–Schwarz the sum of their squares is at least $1/(N+2)$, with equality if and only if all are $-1/(N+2)$. Solving those differences gives exactly (43.4). $\square$

This is an optimum on the indicated *trajectory-supported* subspace, not a claim about every finite spatial box. Allowing branches outside that trajectory can alter finite-box approximation rates. The global statements of Theorem 43.1 do not require a formula for those other finite-dimensional minima.

43.3 The complete quadratic and its locality

For distinct vertices v, w with a common successor write $\{v, w\} \in \mathcal{S}$. Expanding the square gives

$$\begin{aligned} E_{v_0}(h) &= 1 - 2h_{v_0} + \sum_v (1 + \mathbf{1}_{v \text{ live}}) h_v^2 \\ &\quad - 2 \sum_{v \text{ live}} h_v h_{F(v)} + 2 \sum_{\{v, w\} \in \mathcal{S}} h_v h_w. \end{aligned} \tag{43.8}$$

To verify this, expand $\|h\|^2 + \|ZUh\|^2 - 2\langle h, ZUh \rangle$ and the source terms. No transition enters time zero, so the linear term is exactly $-2h_{v_0}$. The diagonal contribution of $\|ZUh\|^2$ is one for each live column; its off-diagonal terms are precisely pairs with a common successor. Clocking prevents a pair from being both a parent–child pair and a sibling pair.

Thus the scalar coefficients of this infinite quadratic expression lie in $\{0, \pm 1, \pm 2\}$. Interactions occur only between a configuration and its next configuration or between two configurations with a common successor. In the three coordinates (a, b, t) their ℓ^1 separation is at most two; the finite state labels are additional fixed types. The local rules differ at the coordinate boundaries, which is exactly where the zero tests act. This is a finite-rule description on $\mathbb{N}^3$, not a claim of a translation-invariant system on all of $\mathbb{Z}^3$.

For a fixed universal program, deciding whether the integer minimum is zero or one is undecidable. Deciding whether the real infimum is attained is also undecidable, even though the infimum value itself is always zero. This is an *additive* integrality gap of one in the nonhalting case, not a finite multiplicative gap ratio. There is no contradiction with solvability of finite-dimensional least squares: no computable finite support bound is available for all instances.

44 Dense nonclosed range in a fixed Hilbert space

The finite-support statements admit a particularly sharp analytic completion. Let $\ell^2(\Omega)$ be the real or complex Hilbert space of square-summable arrays on clocked configurations.

Lemma 44.1 (Bounded extension). *Suppose every unlocked configuration has at most B predecessors. Then U extends to a bounded operator on $\ell^2(S)$ with $\|U\| \leq \sqrt{B}$. Its clocked shift ZU is bounded on $\ell^2(S \times \mathbb{N})$ with the same bound, and*

$$\|A\| = \|I - ZU\| \leq 1 + \sqrt{B}.$$

For the counter machines above one may take $B = 2q$.

Proof. For a finite-support vector x ,

$$\|Ux\|_2^2 = \sum_w \left| \sum_{v: f(v)=w} x_v \right|^2 \leq B \sum_w \sum_{v: f(v)=w} |x_v|^2 \leq B \|x\|_2^2.$$

Each live predecessor occurs in exactly one inner sum. Density extends the operator to ℓ^2 . The time shift preserves the norm when applied separately from U , giving the clocked bound. For each source control state an increment gives at most one predecessor of a specified configuration, and a test gives at most two, one from each branch. $\square$

Theorem 44.2 (Exact range encodes halting). *The bounded operator A on $\ell^2(\Omega)$ is injective and has dense range. For a clocked basis vector $e_{(s,t_0)}$,*

$$e_{(s,t_0)} \in \text{ran } A \iff \text{the unlocked computation from } s \text{ halts.} \quad (44.1)$$

If at least one source does not halt, the range is not closed. In particular a fixed universal counter machine yields a fixed computably specified sparse bounded operator whose range membership on a computable family of basis vectors is computably enumerable complete.

Proof. Decompose $h \in \ell^2(\Omega)$ into time layers $h_t \in \ell^2(S)$. For a homogeneous equation $Ah = 0$ the time-zero equation gives $h_0 = 0$, and the successive equations give $h_t = Uh_{t-1} = 0$. This proves injectivity without any finite-support assumption.

For $Ah = e_{(s,t_0)}$, the same recursion forces zero layers below t_0 , followed by

$$h_{t_0+t} = U^t e_s \quad (t \geq 0).$$

If the source halts this is the finite binary history and belongs to ℓ^2 . If it does not halt, every displayed layer has squared norm one, so no square-summable solution exists. This proves (44.1).

For density, it is enough to approximate every basis vector. A halting source basis vector already lies in the finite-support range. For a nonhalting source the shifted tapers of (43.4) satisfy $\|Ah^{(N)} - e_{(s,t_0)}\|_2 \rightarrow 0$. Finite combinations of basis vectors are dense, so the range is dense. A nonhalting source is in its closure but not in the range, proving nonclosedness. The fixed-machine computability statement follows from the finite transition table and its effective monomial input encoding. $\square$

Corollary 44.3 (Spectral location, without a spectral-gap claim). *Over the complex Hilbert space, if a nonhalting source exists then zero belongs to the continuous spectrum of A : the operator is injective with dense proper range. It is not an eigenvalue of A .*

This conclusion is about a generally non-self-adjoint incidence operator. It is not an assertion about a finite-dimensional quantum Hamiltonian or a new many-body spectral-gap undecidability theorem. The distinctive statement here is the exact range membership of the specified computable basis-source family.

[write] The opposite regime is treated in batch-79 manuscripts 10 and 14, placed in the report *stochastic-and-thermal-exactness* [2] for its Part V (Section 1.4): there a fixed sparse operator built from a counter machine is uniformly coercive, hence boundedly invertible, and halting is read off its Green data, whereas here zero lies in the continuous spectrum (Corollary 44.3) and the range is not closed.

The nonhalting tapers also illustrate why one should not mistake $\|Ah^{(N)} - e_{v_0}\| \rightarrow 0$ for convergence of the witnesses. Their norms diverge by (43.7). Coefficientwise they converge to the infinite binary computation history, which lies in the formal time completion but not in ℓ^2 . The three domains therefore give different answers: polynomial witnesses detect halting; formal-power-series witnesses always exist; square-summable witnesses again detect halting, despite the dense approximate range.

45 Failure modes and the ordinary Diophantine boundary

45.1 Dropping a boundary sort creates an actual false positive

Consider the three-state program

$$0 : \text{TEST}(X, 1, 2), \quad 1 : \text{INC}(X, 0), \quad 2 : \text{HALT}.$$

From $(0, 1, 0)$ it repeats $(0, 1, 0) \rightarrow (1, 0, 0) \rightarrow (0, 1, 0)$ and never halts. Its clocked rows are

$$\begin{aligned} H_0 - ZXH_1 &= X, & H_1 - ZD_0 &= 0, \\ H_2 - ZB_0 &= 0, & H_0 - XD_0 - B_0 &= 0, \end{aligned}$$

with the essential type $B_0 \in R[Y, Z]$. If that type is relaxed to $R[X, Y, Z]$, the tuple

$$H_0 = X, \quad H_1 = 0, \quad H_2 = ZX, \quad D_0 = 0, \quad B_0 = X \quad (45.1)$$

satisfies every displayed identity. It illegitimately routes the positive counter value through the zero branch. The supplied checker accepts this tuple when sort enforcement is disabled and rejects it when the declared sorts are enforced. This is an explicit counterexample to that relaxed compiler, not merely a warning about a hypothetical bug.

45.2 Discarding a final spill row creates another false positive

For a nonhalting source, the truncated true history $H_{\leq N} = \sum_{t=0}^N Z^t e_{c_t}$ satisfies

$$(I - ZU)H_{\leq N} - e_s = -Z^{N+1}e_{c_{N+1}}.$$

If a bounded coefficient checker retains only rows of time degree at most N , it incorrectly accepts every truncated nonhalting run. A correct finite-box checker must include *all output rows generated by the allowed columns*, even when those rows lie just outside the candidate box. The independent finite-field checker in this package uses precisely that rule. Truncation modulo Z^{N+1} is a different problem from a polynomial identity.

45.3 Clocking, signs, and several sources

The unclocked equation is sound for a single source even with signed coefficients, but its global witness need not be unique because unrelated directed cycles may be added. The clock eliminates those cycles from the kernel, rather than relying on a minimality convention. Nonnegative coefficients alone do not remove disconnected positive cycle additions. Conversely, permitting several sources can introduce finite-time cancellation even in the clocked construction, as in Remark 39.3. Positivity, clocking, and a one-monomial source are different assumptions and should not be interchanged.

45.4 Why this is not fixed-arity integer MRDP

The sorts $R[X, Y, Z]$ and $R[Y, Z]$ quantify over objects with arbitrarily large finite support. Calling them $q + 2d$ unknowns does not make them a fixed tuple of $q + 2d$ integers in an ordinary Diophantine equation. A support list contains an unbounded number of exponents and coefficients. The subring restriction is also indispensable, as (45.1) demonstrates.

For $R = \mathbb{Z}$ one can canonically encode each finite polynomial tuple as a natural number. A checker can decode its lists, reject malformed or noncanonical encodings, verify the omitted-variable restrictions, compute every residual coefficient, and test that all vanish. With standard finite-list and integer encodings this is a primitive recursive verification procedure. The accepted input relation is then

$$n \in K_{\mathcal{U}} \iff \exists w \in \mathbb{N} \text{ Check}(n, w) = 1.$$

MRDP converts this enumerable relation to an ordinary integer polynomial with finitely many natural witnesses. ProveIt's `rePred_dioph` provides an appropriate existing endpoint for such an integration [30].

But that conversion can introduce new auxiliary witnesses. The uniqueness of the canonical polynomial tuple gives no proof that those integer auxiliaries are unique or even finite in number. Thus the present theorem does not settle single-fold or finite-fold ordinary Diophantine representation. The distinction is substantial; it is the subject of dedicated work on Diophantine univocity [26, 27]. Nor is the operation count of polynomial-list verification the operation count of an expanded universal integer polynomial.

[write] [The same boundary is recorded in Part IX of the collection's report on canonical Diophantine certificates [1]: the one-history relation (`cdc:eq:onehistory`), Theorem `cdc:bd:thm:universal` and Proposition `cdc:bd:prop:nobound`, of which Corollary 41.5 is the polynomial-degree analogue. Both labels existed at this manuscript's pin.]

45.5 An explicit hierarchy of certificate languages

Language	Unknown domain	Conclusion here
Boundary-operator equation	Finite polynomials; P_j, D_j allowed	Explicit, unique halting certificate.
Mixed-subring equation	Finite polynomials with declared omitted variables	Explicit sparse linear compiler; same unique fiber.
Formal-time equation	Infinite series with finite configuration support per layer	Always soluble; not a halting certificate.
Hilbert-space equation	Square-summable clocked arrays	Exact solvability detects halting; approximate range is dense.
Ordinary integer equation	Fixed finite scalar tuple	Existence follows after an MRDP bridge; no explicit size or uniqueness gain.

46 A decidable one-counter boundary

The second independently testable exponent is not a cosmetic feature. The corresponding deterministic one-counter machine class has a direct halting algorithm. We give the proof to delimit the exact scope of this comparison.

Proposition 46.1 (One-counter machine-generated certificates). *For the same instruction convention with only one nonnegative counter, halting from a specified configuration is decidable. Consequently the polynomial-certificate existence problem is decidable for the subclass compiled from such one-counter programs.*

Proof. While the counter is strictly positive, every instruction has a fixed successor control state: an increment contributes $+1$, a test takes its positive branch and contributes -1 , and a terminal state halts. Thus a positive excursion follows a deterministic finite control graph with edge weights ± 1 , until it reaches zero or halts.

Follow the excursion until one of those events occurs or a control state repeats. A repeated state identifies a control cycle that has already been traversed with all intermediate tested counter values positive. Let Δ be its net counter change. If $\Delta \geq 0$, each successive traversal starts with at least as large a counter as the previous one, so the same positive cycle repeats forever. If $\Delta < 0$, repeated traversals decrease the starting counter by the positive integer $-\Delta$; after finitely many traversals the counter reaches zero during a traversal. One can compute the number of full safe traversals from the cycle's partial sums, or simply iterate these decreasing cycles until zero is reached. Either method terminates in this case. Hence each positive excursion can be classified as halting, returning to zero, or infinite.

At counter zero there are only q configurations, one per control state. Record each zero configuration encountered. Repetition of one forces an infinite deterministic future. Otherwise after at most q distinct zero configurations the procedure either halts, reaches an infinite positive excursion, or repeats a zero configuration. A possible initial positive excursion is handled first by the same procedure. This is a terminating decision algorithm. $\square$

This proposition concerns the *machine-generated* class. It is not a theorem that every linear equation with one restricted polynomial sort is decidable. General linear systems may have branching, coefficients, several sources, or dependence restrictions not realizable by this one-counter control discipline. The distinction between a syntactic subclass and the whole ambient equation problem must be kept explicit.

47 Executable compiler and validation

47.1 What the implementation does

The accompanying Python implementation uses sparse polynomials keyed by triples of nonnegative exponents (a, b, t) . Coefficients are exact integers or rational numbers, and optional modular arithmetic is implemented by reduction modulo a supplied integer. No floating-point arithmetic is used in any certificate check.

`code/boundary_transport.py` implements validated finite program tables; exact transitions; sparse addition and shifts; boundary splitting; the linear transport action; construction of histories and residuals; the mixed-subring compiler; sort-aware certificate checking; and emission of the tagged scalar equation. It also contains an independent finite-box solver over $\mathbb{F}_2$.

The latter solver constructs a coefficient matrix directly from the configuration successor relation, rather than from the generated polynomial rows. For each allowed column e_v it inserts its diagonal row and, when live, its successor row. It retains every such row, including spatial and temporal spill rows outside the candidate box. Bitset Gaussian elimination then checks column rank and consistency. This tests the compiled semantics against an independent finite linear-algebra route instead of merely asking the certificate generator to confirm its own output.

The implementation is a research artifact, not a certified prover or a high-performance universal-machine simulator. Its input validation rejects empty programs, invalid destinations, invalid counter indices, and negative exponent or counter values in the exercised cases. It also rejects nonintegral indices and moduli, and snapshots a caller-supplied instruction sequence as an immutable tuple. Canonical polynomial sorting is used in the mathematical discussion of integer coding; no claim is made that the current Python API is already a production-strength MRDP encoding format.

47.2 The executed checks

Running

```
python code/verify.py
```

from the package root regenerates the example files and `validation/test_results.json`. The supplied run reports **32,502 passing exact checks**. The main coverage is as follows.

Coverage item	Count
All two-state programs over the specified instruction options	169
Program/source simulations through at most 32 transitions	3,042
Those simulations halting within that horizon	366
Those not halting within that horizon	2,676
Independent finite-box $\mathbb{F}_2$ systems	1,352
Random exact polynomial inputs for boundary identities	400
Exact rational taper lengths, $N = 0, \dots, 30$	31
Random checks of the expanded quadratic formula	360

For two states there are thirteen allowed instructions per state: one terminal instruction, four increment instructions, and eight test instructions. Thus $13^2 = 169$ exhausts that finite program family. The simulation inputs have either state and both counters in $\{0, 1, 2\}$. For every input the tests compare direct transport

residuals with the compiled row residuals, including exact terminal or spill behavior. Halting witnesses are checked for binary coefficients, exact main support, the unlocked variant, and rejection of a coefficient mutation. The same acceptance equivalence is tested modulo 2, 3, 5, 6; modulus six is included to exercise a coefficient ring with zero divisors.

The finite-box tests use both initial states and counter values in $\{0, 1\}$, allow both counter coordinates at most one and time at most two, and compare consistency with an independently simulated halting trajectory contained in that box. Every tested matrix has full column rank, as clocked injectivity predicts. Additional tests enumerate small signed integer arrays on a nonhalting trajectory, check the rational taper formulas, verify (43.8), and reproduce both the boundary-sort false positive and an unlocked cycle-kernel addition.

These tests are evidence about the implementation and small instances. They do not replace the all-input proofs above. In particular, “not halting within 32 transitions” is *not* reported as “divergent.” The examples that are labeled periodic are verified by an actual repeated configuration. No numerical fixed universal machine was instantiated or tested in this run.

[write] *Shipped names.* In this report the programs are `code/01-boundary-transport-boundary_transport.py`, `code/01-boundary-transport-verify.py` and `code/01-boundary-transport-Makefile`; `examples/*` and `validation/test_results.json` are shipped as `data/01-boundary-transport-*`, and `SOURCE_AUDIT.md` as `01-boundary-transport-SOURCE_AUDIT.md`. The command above rewrites the example files and the validation record in place (with CRLF line ends on Windows); run it only on a copy with the delivered layout. The research programme’s review [5] reproduced the 32,502 checks and found that non-Boolean clock modes are accepted (Section 1.4); the shipped `code/01-boundary-transport-boundary_transport.py` is the unpatched original, and the theorems, stated for the Boolean modes, are unaffected.

47.3 A practical verifier contract

A robust consumer of these certificates should treat the following as part of the trusted interface: the validated instruction table; the source configuration and its coefficient-one convention; the declared polynomial sorts; exact coefficient arithmetic; the complete set of residual monomials, including spill; and the independent status of the clock variable and row tag. Omitting any of these changes the theorem being checked. A certificate need not carry a claimed runtime: its degree in Z reveals the runtime after the identities have been verified.

48 Integration with ProveIt

A useful formalization would separate the generic algebra from the machine compiler. The first layer is a finite-support module over an arbitrary nontrivial commutative ring, a deterministic partial successor, and its linear extension. The proofs of Theorems 38.1 and 39.1 can then be stated without counter-machine syntax. The clocked proof is especially small: it is coefficient induction, followed by evaluation of the iterates on one basis vector.

The second layer identifies two-counter configurations with monomials and proves the boundary split. It should make omitted-coordinate conditions explicit in the types, for example by storing B_r as an actual polynomial over the smaller variable index set rather than as a large-ring polynomial with an unchecked promise. The compiler-correctness theorem is then a bijection of full solution sets, not just a forward construction of a witness.

A third layer could specify a canonical natural-number encoding of finite polynomial tuples and prove that

its verifier is primitive recursive. Composing that result with the existing `rePred_dioph` interface would yield an ordinary Diophantine existence theorem for the accepted language. The primitive-recursive verifier and the MRDP-generated polynomial should remain separate objects with separate complexity ledgers. The repository's existing source bridge supplies the latter logical endpoint, but none of these new layers has been implemented in Lean in this package [30].

Finally, the convex-energy theorem can be formalized over finitely supported real arrays before bringing in any Hilbert-space completion. The integer lower bound is just the fact that a nonzero integer residual has squared norm at least one; the approximation formula is a finite telescoping sum. This separation would keep the most concrete new certificate results independent of analytic-library overhead.

49 Further research questions

49.1 Concrete universal instances and smaller interfaces

A numerical universal table. Choose and validate a published universal two-counter interpreter, including its ordinary encoded-input loader, then emit the complete fixed p_j of Theorem 41.3. The measurable targets are the number of states q , tests d , atomic coefficient terms, input encoding cost, and total witness support. This would upgrade the structural fixed-program theorem to a directly inspectable numerical instance; it would not by itself improve ordinary integer universal bounds.

[write] The same target is posed in Part I (Section 16.1) and Part II (Section 33.4).

Uniqueness without an extra clock coordinate. Can a universal counter-machine compilation eliminate all directed cycles globally while preserving a small two-counter instruction table and a manageable input interface? Theorem 38.1 shows exactly why this would produce unique unlocked certificates. The question is not merely whether a selected run has no repetition: every unrelated cycle contributes a kernel vector.

Minimal polynomial sorts. Determine the smallest number of independent indeterminates and omitted-coordinate sorts needed for a universal *linear* certificate with this sparsity discipline. The scalar construction uses a row tag solely to replace a vector equation. Can that coordinate be removed without hiding vector-valued equality or introducing nonlinear witness products? A lower bound must specify a precise syntactic class; Proposition 46.1 gives only one restricted decidable boundary.

Remark 49.1 (Editorial; not stated by source 01 or source 15). [write] Part II's finite-state theorem bounds this question from below. Its proof (Theorem 21.1) uses only that the coefficient vectors range over a finite alphabet and that the window identities (21.2) are checked by exact arithmetic in the coefficient ring. With the alphabet $\{-B_0, \dots, B_0\}^n$ it decides solvability of a univariate linear system over $\mathbb{Z}[X]$ by a solution of coefficient height at most B_0 , and over a finite ring R (for example $\mathbb{F}_p$ or $\mathbb{Z}/6\mathbb{Z}$) with the alphabet R^n it decides solvability of univariate linear systems outright. As in Corollary 21.3, no computable reduction of a computably enumerable complete set to univariate linear systems over $\mathbb{Z}[X]$ can guarantee a solution with coefficients in $\{0, 1\}$ on every solvable instance, and none to univariate linear systems over a finite ring exists at all. So the 0/1 certificates of Theorem 41.2, and its validity over finite fields, cannot be kept with a single indeterminate (clock and row tag included): at least two are necessary in this linear syntax. This is a partial answer only; the gap between two indeterminates and the four (W, X, Y, Z) of Theorem 41.3 remains open. The extension of the proof to $\mathbb{Z}[X]$ and to finite rings was observed at the intake and checked by reading the proof; source 15 states it only over $\mathbb{N}[X]$.

49.2 Arithmetic compilation and quantitative certificates

An explicit integer bridge with controlled fibers. Develop a concrete arithmetization of the sparse polynomial checker, track every scalar auxiliary, and determine which steps preserve a bijection of witnesses. The hard objective is not the existence statement already supplied by MRDP, but an audited operation count and controlled auxiliary multiplicity. Any claimed single-fold conclusion must account for all newly introduced scalar witnesses.

[write] The same question is posed in Part I (Section 16.4) and Part II (Section 33.6).

Sharp support-growth calibration. For fixed universal machines, compare maximal halting time at bounded input description length with maximal minimal certificate support. The clocked correspondence gives the exact identity $|\text{supp } H| = h + 1$ before arithmetic compression. Which natural changes of machine or native substrate preserve this identity up to affine, polynomial, or only computable distortion?

Decidable fragments. Extend the one-counter analysis to carefully delimited classes: restricted zero-test placement, bounded changes in counter direction, acyclic branch-control patterns, or fixed bounds on one counter. The goal is a theorem about the *generated equation class*, with an effective bound or structural algorithm, not an assertion that all similar-looking mixed-subring equations are decidable.

49.3 Native computational substrates

Direct encodings beyond counters. For a deterministic evaluator for combinatory logic, interaction nets, queue rewriting, or another esoteric substrate, construct a native monomial transition operator and prove the relevant boundary or rewriting decomposition. The generic clocked theorem then applies immediately, but the native compiler must still pay for syntax, matching, wiring, or queue legality. The research target is reducing that overhead while retaining a full solution-set bijection.

Nondeterminism and cancellation. A linear extension that sends a configuration to the sum of all its successors encodes a branching computation, not existential acceptance. Over finite characteristic, distinct branches can cancel. Find exact conditions or additional certificate structure that represent existence of a halting branch while preserving a useful characteristic-independent semantics. The one-source deterministic proof cannot simply be reused.

49.4 Approximation and functional analysis

Optimal finite-box approximation. The trajectory-segment optimum is exactly $1/(N + 2)$. What is the optimum when all configurations in a counter/time box are allowed? Incoming branches may spread the residual and change the rate. An exact combinatorial characterization could connect least-squares residuals with finite network potentials while retaining the terminal-column convention and spill rows.

Effective range criteria. The fixed operator has dense nonclosed range, and membership is hard already on a selected family of basis vectors. Identify natural restricted source subspaces where membership becomes decidable, or where an effective bound on the inverse norm is possible. Multiple source vectors bring genuine cancellation and require a different classification from the single-basis theorem.

Rational generating functions for nonhalting histories. An eventually periodic unlocked run has a rational formal clocked history. More complicated nonhalting runs need not have such a simple form. Characterize those machine fragments for which a rational representation of the formal resolvent is effectively recognizable, and determine what termination information survives this larger certificate language. Rationality must not be confused with polynomiality.

A fully verified compiler pipeline. Formalize the sparse compiler, sort checker, row-tagging map, spill-aware finite-field checker, and integer verifier bridge as separate verified components. The deliverable should include an explicit theorem that the complete compiled witness is unique, not only soundness of the forward history construction. This is the most direct route for converting the present research package into a reusable ProveIt component.

[write] Parts I and II propose the same programme (Sections 16.8 and 33.9).

50 Conclusion

A deterministic computation admits an exact linear certificate before any conversion to an ordinary integer Diophantine equation. Two boundary projections supply the zero tests; a clock makes the entire certificate unique; finite support determines whether the computation terminates. The resulting mixed-subring system is sparse, linear in all its unknowns, and valid over every nonzero commutative coefficient ring. A row-tagging variable turns it into one equation without invoking an order or a sum-of-squares test.

The same construction also gives a precise obstruction to treating approximate feasibility as exact feasibility. On a nonhalting input, integer residual energy is at least one, yet rational finite-support arrays approach zero energy with the explicit rate $1/(N + 2)$. In Hilbert space the operator is injective with dense nonclosed range, and exact basis-source membership retains the halting problem.

The research value is a transparent normal form linking computation, polynomial sorts, unique witnesses, and convex relaxation. Its limits are equally explicit: unrestricted polynomial sorts are unsound, truncated spill rows are unsound, infinite power-series existence is uninformative, and scalar MRDP uniqueness does not follow. Those boundaries identify concrete next steps rather than concealing them inside a generic universality assertion.

51 Compiler specification in pseudocode (appendix of source 01)

The following specification is independent of the sparse-polynomial storage format. Each balance row initially contains only its diagonal main unknown. The source term is subtracted from the selected row.

```
compile(program, source, clocked=True):
    validate(program, source)
    clock = Z if clocked else 1
    create H[r] for each control state r
    balance[r] = H[r] - source_polynomial_in_state(r)
    splits = []
    for r, instruction in program:
        if instruction is INC(j, p):
            balance[p] -= clock * X[j] * H[r]
        elif instruction is TEST(j, positive, zero):
            create D[r] in the full polynomial ring
```

```

        create B[r] in the subring omitting X[j]
        splits.append(H[r] - X[j]*D[r] - B[r])
        balance[positive] -= clock * D[r]
        balance[zero]      -= clock * B[r]
    elif instruction is HALT:
        pass
    rows = balance + splits
    scalar_equation = sum(W**i * rows[i] for i in range(len(rows)))
    require every unknown to omit W
    return sorts, rows, scalar_equation

```

The compiler itself does not simulate the source and does not need a halting bound. Simulation is used only to construct candidate witnesses in tests and examples. Checking a supplied tuple is a terminating finite algebraic operation regardless of whether the source halts.

52 Artifact manifest and reproducibility (appendix of source 01)

[write] The table lists the delivered layout; the shipped names are given in Section 47. `article.pdf` and `README.md` of the delivery are not shipped; `article.tex` is Part III of this report.

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Complete source and compiled report.
<code>README.md</code> , <code>Makefile</code>	Build and verification instructions.
<code>SOURCE_AUDIT.md</code>	Pinned repository paths, external sources, and audit limits.
<code>code/boundary_transport.py</code>	Exact sparse compiler and certificate tools.
<code>code/verify.py</code>	Reproducible exact tests and example generation.
<code>examples/transfer_certificate.json</code>	Complete transfer-machine instance and witness.
<code>examples/transfer_equation.txt</code>	Emitted tagged scalar equation.
<code>examples/transfer_witness.txt</code>	Human-readable witness polynomials.
<code>validation/test_results.json</code>	Check counts, parameters, and limitations.

The Python code uses only the standard library and modern type-annotation syntax. It was executed with Python 3.13. The PDF is built with pdfLaTeX; all references are embedded below, so BibTeX and network access are not needed to rebuild it. Two pdfLaTeX passes resolve cross-references and the table of contents. No external article PDFs, font files, repository archives, or unexecuted Lean proof claims are included.

A Provenance of this report

[write] This appendix was written when the three manuscripts were merged.

The report is built from three manuscripts of batch 79 of ProveIt’s incoming reports, cluster J1, placed by commit 224ca41df:

Source	Archive and main file	Pin	Arrival	Printed as
06 (base)	unique_polynomial_histories.zip, unique_polynomial_histories/unique_polynomial_histories.tex, 28-page PDF	e58b724c25bd 34533b7a5834 cfcbe873dfa01288	060e08a07	Part I
15	one_coordinate_certificates.zip, one_coordinate_certificates/one_coordinate_certificates.tex, 30-page PDF	44983ed7ebfd 545de55bfdb5 0e040c82f3d24295	ef2fc7990	Part II
01	Linear_Boundary_Transport_Research.zip, linear_boundary_transport/article.tex, 22-page PDF	48ee077c7ab6 47d7121370e9 5a353169c89c79db	060e08a07	Part III

Source 15’s pin is later than source 06’s arrival; it read source 06 as a “user-provided manuscript” (its bibliography entry `oc:Precursor`).

Contributions: source 06 the canonical cellular compiler over $\mathbb{N}[X, Y]$, its marginal projection and feature compression, universality, degree ledger, quadratic form, bounded export, domain barriers, nondeterministic extension and worked example; source 15 the univariate finite-state obstruction, the eleven-identity stride compiler, the full-witness bijection with source 06, its resource ledgers, quartic and bounded export, labelled branching and counterexamples; source 01 the ring-uniform transport theorems, the mixed-subring compiler with row tagging, the energy gap, the Hilbert-space range theorem, the failure modes and the one-counter boundary. Every result, proof, example, remark, limitation, research question and appendix of the three manuscripts is printed.

Where the merge had to choose:

- *Base and order.* Source 06 is the base and Part I because source 15 continues it and answers its question; source 01, whose witness ring is the most general, could have opened the report, but it shares no lemma with the other two and is Part III.
- *Duplicated statements.* Source 15’s Proposition 25.1 restates and re-proves the two-coordinate witness shape of Part I (Theorem 2.1 in the nine-row form of Corollary 8.2), and its Lemma 24.1 is the one-variable case of Lemma 4.1; both are printed where they stand, with a bracket, because Part II’s bijection uses them. No other statement is shared; the univariate counterparts in Part II and the polynomial analogues of `cdc:bd:prop:nobound [1]` carry pointers.
- *Research questions.* The three question lists are printed in full. Three themes recur in all three lists (a numerical universal instance; packing into ordinary integers with an exact fibre ledger; kernel-checked proofs); each occurrence carries a note naming the others. Part I’s one-coordinate question is answered by Part II (dated note); source 15’s question on reducing the eleven rows is answered in part by the research programme’s review (dated note); source 01’s question on minimal polynomial sorts receives an editorial remark derived from Part II’s finite-state theorem (Remark 49.1).
- *Layout.* The three title pages are replaced by the Part openings, which print each abstract and status statement. Section numbers run through the report, so each source’s appendices are printed as numbered

sections at the end of its Part, titled “(appendix of source NN)”. Source 01’s remarks, set in the remark style there, are set in the definition style of the other two sources. The three bibliographies are printed together at the end, with their keys prefixed by `uh:`, `oc:` and `bt:`; entries cited by more than one source are printed once per source, as delivered.

- *Macros.* Source 01’s command for the blackboard $\mathbb{F}$ was renamed (Section 1.3); its remaining macros, and those of sources 06 and 15, are unchanged.
- *File names.* The Parts’ texts name the delivered layout (`code/`, `results/`, `examples/`, `validation/`). Notes in the validation sections map them to the shipped names; three expanded-polynomial exports (two of source 06, one of source 15) are not shipped because the shipped programs regenerate them exactly, and the README gives the commands.
- *Labels.* Every label carries the prefix `pwh:`. The 76 delivered labels of source 06, 75 of source 15 and 65 of source 01 are printed as `pwh:uh:`, `pwh:oc:` and `pwh:bt:` plus their delivered names; no delivered label was dropped or renamed apart from the prefix. Added labels: 34 in all, listed in the README.

Every sentence written in the merge is marked [write]: Section 1, the openings of the three Parts, the notes in the Parts, the editorial Remark 49.1, the bracket notes in the bibliography, and this appendix.

References

[write] Cited in the merge

- [1] *Canonical Diophantine Certificates*, ProveIt research report, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/canonical-diophantine-certificates`. Cited by label names (a separate document); Part IX is *The boundary of fixed-arity compression*.
- [2] *Stochastic and Thermal Exactness*, ProveIt research report, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/stochastic-and-thermal-exactness`.
- [3] *Smooth Diophantine Finalizers*, ProveIt research report, `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/smooth-diophantine-finalizers`.
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